A Machine Learning-based Mobile Application with Augmented Reality for Rental Housing Prediction and Visualisation in Nairobi

By

Baraka Mitigoa

100185

Submitted in Partial fulfilment of the Requirements for the Degree of Master of Science in Computing and Information Systems at Strathmore University

School of Computing & Engineering Sciences

Strathmore University

Nairobi, Kenya

June 2026

Declaration and Approval

Declaration

I declare that this work has not been previously submitted and approved for the award of a degree by this or any other University. To the best of my knowledge and belief, the dissertation contains no material previously published or written by another person except where due reference is made in the dissertation itself.

Student's Name: Baraka Mitigoa

Sign: ___ ______________________ Date: _ 23/03/2026

Approval

The dissertation of Baraka Mitigoa was reviewed and approved for examination by the following:

Sign: B.K.S Date: March 26th, 2026

Prof. Bernard Shibwabo
Associate Professor,
School of Computing & Engineering Sciences,
Strathmore University

Abstract

The rapid growth of Nairobi's rental housing sector has intensified challenges related to affordability, pricing transparency and limited access to reliable housing information. Tenants often face difficulties in determining fair rental prices and visualising properties before making decisions. This study developed a mobile application that integrates machine learning (ML) for rental price prediction with augmented reality (AR) for property visualisation. A review of existing literature revealed that while ML algorithms such as LightGBM achieve high predictive accuracy in real estate markets, their application in Nairobi remains limited and although AR enhances spatial understanding, its integration within Kenya's rental platforms is minimal. To address this gap, the study adopted a Design Science Research (DSR) approach and utilised a dataset of 580 rental listings collected across Nairobi. The LightGBM model was trained and evaluated using standard metrics, achieving low prediction errors with Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) within acceptable bounds ($\leq$ KSh 2,000) and a high coefficient of determination ($R^2 \approx 0.85$–0.90), indicating strong predictive performance. The system was implemented as an Android-based mobile application integrated with Firebase for real-time data management and ARCore for augmented visualisation. Usability evaluation using the System Usability Scale (SUS) yielded a score above 70, indicating good to excellent usability. The AR module successfully enabled interactive, real-time visualisation of property information, enhancing users' spatial understanding and decision-making experience. Overall, the study presents a practical and innovative solution that improves transparency, supports informed decision-making and enhances access to rental housing in Nairobi by combining predictive analytics and immersive visualisation within a single mobile platform.

Keywords: Machine Learning, Augmented Reality, Rental Housing, Mobile Application, Nairobi.

Table of Contents

List of Figures

List of Tables

List of Abbreviations

AR	Augmented Reality
CPU	Central Processing Unit
DSR	Design Science Research
GIS	Geographic Information System
GPS	Global Positioning System
GUI	Graphical User Interface
HCD	Human-Centered Design
ICT	Information and Communication Technology
IoT	Internet of Things
KMRC	Kenya Mortgage Refinance Company
KNBS	Kenya National Bureau of Statistics
MAE	Mean Absolute Error
ML	Machine Learning
PEoU	Perceived Ease of Use
PU	Perceived Usefulness
RMSE	Root Mean Squared Error
$\mathbf{R^2}$	Coefficient of Determination
SDK	Software Development Kit
SQL	Structured Query Language
SUS	System Usability Scale
TAM	Technology Acceptance Model
TF-IDF	Term Frequency-Inverse Document Frequency
UI	User Interface
UML	Unified Modeling Language
UX	User Experience
VIO	Visual Inertial Odometry

Definition of Terms

Algorithm	A systematic procedure or set of rules followed in problem-solving operations, particularly by a computer, used to train and optimize machine learning models (Rahaman, 2024).
Augmented Reality (AR)	An immersive technology that overlays digital content such as 3D models and interactive information onto the physical environment through mobile devices (Pamungkas & Astrianty, 2024).
Artificial Intelligence (AI)	A branch of computer science concerned with building systems capable of simulating human intelligence processes like learning, reasoning, and decision-making (Cacciamani et al., 2024).
Dataset	A structured collection of data points, such as rental prices, locations, and property features, used for training and testing predictive models (Zhang, 2023).
Light Gradient Boosting Machine (LightGBM)	A machine learning algorithm based on gradient boosting decision trees, optimized for speed, memory efficiency, and high predictive accuracy (Zhang, 2023).
Machine Learning (ML)	A subfield of AI that enables systems to identify patterns in data and improve their predictive performance without explicit programming (Sarker, 2021).
Mobile Application (App)	A software program developed to run on smartphones, designed to provide specific functionalities such as rent prediction and visualization (Olaolu, 2022).

Random Forest	An ensemble machine learning technique that combines multiple decision trees to improve predictive accuracy and handle noisy data effectively (Adzanoukpe, 2025).
Rent Prediction	The use of machine learning techniques to estimate the fair rental value of housing units based on historical trends, features, and location (Goldwaser & Ge, 2023).
Simultaneous Localization and Mapping (SLAM)	A real-time computational method that enables AR systems to map physical spaces while tracking device position, enhancing spatial accuracy (Priyanka Das, 2020).
User Experience (UX)	The overall effectiveness, satisfaction, and ease with which users interact with a digital system, influenced by usability and design (Weichbroth, 2025).

Acknowledgements

First and foremost, I am deeply grateful to Almighty God for granting me strength, wisdom and perseverance throughout the preparation of this dissertation. Without His guidance and blessings, this work would not have been possible. I would also like to sincerely thank my supervisor, Prof. Bernard Shibwabo, for his invaluable guidance through this research.

Chapter 1: Introduction

1.1 Background to the Study

Urbanization in sub-Saharan Africa, especially in Nairobi, Kenya, has grown in recent years thus creating both economic opportunities and a rising challenge in access to affordable housing (UN-Habitat, 2022). In 2019 Nairobi's population was estimated at 4.4 million and is projected to exceed 5 million by 2030 according to the Kenya National Bureau of Statistics (Kenya National Bureau of Statistics, 2020) rapid growth has surpassed the city's housing supply leading to a significant affordability crisis, especially among low- and middle-income earners (Kenya Mortgage Refinance Company, 2023).

From the growing middle class and speculative investments, the real estate market in Nairobi continues to expand. However, the Cytonn Q3 2023 Real Estate report reveals that over 80% of new housing projects target higher-income earners therefore leaving a deficit of more than 2 million housing units highly affecting lower-income populations (Cytonn Investments, 2023). These dynamics have resulted in unregulated rental pricing, housing insecurity and growing informal settlements (Mwau & Sverdlik, 2020). Inspite of the scale of this crisis, digital rental platforms available in Nairobi operate mostly as static listing services, offering limited support for transparent decision-making (Baraka Mwau, 2020).

At the same time, Kenya has made considerable strides in digital connectivity. Mobile phone penetration is at 143%, whereas smartphone penetration is at 80.8% (Communications Authority of Kenya, 2025) . This increase in the digital space has provided fertile ground for intelligent, mobile-first housing solutions. However, most property listing sites fail to leverage this potential where they have basic listings without pricing transparency which lack spatial context and fraud detection mechanisms. As a result, renters, especially those unfamiliar with the market, often navigate the process blindly, leading to poor housing choices, financial exploitation and loss of trust in digital platforms.

Emerging technologies such as Machine Learning (ML) and Augmented Reality (AR) present a transformative opportunity to address these gaps. ML has been applied across various sectors to uncover data patterns and deliver predictive insights, including in pricing, risk analysis and fraud detection. For example, property market outcomes can be predicted with high accuracy by ML algorithms such as logistic regression and XGBoost (Żbikowski & Antosiuk, 2021) . Using historical rental as a training set, ML can deliver dynamic and fair rent predictions that guide renters toward cost-effective housing decisions.

In parallel, Augmented Reality (AR) has emerged as a tool for enhancing digital engagement by overlaying digital elements in physical environments. In real estate, AR enables users to visualize and interact with properties virtually before visiting them physically. Recent studies have shown that mobile AR solutions improve user engagement, spatial awareness and decision confidence in property searches (Pamungkas & Astrianty, 2024). Their research on an AR-based mobile app for boarding-house selection demonstrated that immersive, real-time visualization significantly enhances the decision-making experience, especially when paired with location-based data.

These technologies are already being used globally in sectors such as retail, healthcare, and logistics (McKinsey & Company, 2020). Looking at developed countries, real estate apps now offer virtual walkthroughs, predictive pricing insights and fraud alerts which are aimed at improving the quality and efficiency of decisions (Cacciamani et al., 2024). However, their integration into African housing markets remains largely untapped. Kenya's own real estate platforms rarely integrate advanced analytics or immersive media, resulting in a digital divide that affects the most vulnerable users (Winnie & Otieno, 2022).

Nairobi, with its growing urban population, mobile engaged youth and expanding informal housing sector presents a high-impact context for the deployment of a smart rental solution (Kenya National Bureau of Statistics, 2020; UN-Habitat, 2022). A mobile application that integrates ML and AR can serve as both a technological innovation and a social equalizer therefore providing fair pricing information and immersive property previews to users who would otherwise be disadvantaged by lack of access to quality housing information (Cacciamani et al., 2024; UN-Habitat, 2022) .

Therefore, Nairobi's housing affordability crisis combined with widespread mobile access and underutilized digital tools creates both a need and an opportunity for innovation. Integrating ML for price prediction and AR for property visualization into a unified mobile platform offers a data-driven, human-centered solution. This study proposes to design and evaluate such a solution.

1.2 Problem Statement

With a supply-demand gap exceeding 2 million units (Kenya Mortgage Refinance Company, 2023). Nairobi faces a shortage of affordable rental housing. Despite the rise of digital rental platforms, most function as static listing boards without intelligent tools to assess price fairness, identify fraud, or offer spatial insights. As a result, renters, especially youth, recent graduates and informal workers—are often misled by overpriced or misrepresented listings, with limited ability to verify value or neighborhood suitability (Onyango & Ondiek, 2021; UN-Habitat, 2022).

Globally, Machine Learning (ML) is increasingly used to predict property prices by analysing trends in location, property features and market data, often achieving over 85% accuracy (Żbikowski & Antosiuk, 2021; Zhang, 2023). Simultaneously, Augmented Reality (AR) enables interactive 3D previews and overlays that improve spatial understanding and decision confidence (Pamungkas & Astrianty, 2024). However, these innovations remain underutilised in Nairobi's rental ecosystem, leaving users with limited support during the housing search process (Abate et al., 2022).

Given the growing use of smartphones (Communications Authority of Kenya, 2025) and Nairobi's rapid urbanization (Cytonn Investments, 2023), there is a pressing need for a mobile solution that integrates ML for rent prediction and AR for property visualisation. This study proposes to fill that gap by designing a mobile application that enables fair pricing, reduces information asymmetry and enhances the rental decision process. By addressing current limitations in affordability, transparency and user experience, the solution will contribute to equitable housing access and align with Kenya's digital economy and smart city goals (World Bank, 2023).

1.3 Objectives of the Study

1.3.1 General Objective

The aim of this dissertation is to design and develop a mobile application that uses Machine Learning and Augmented Reality to support rent prediction and visualisation in Nairobi.

1.3.2 Specific Objectives:

a) To determine the key characteristics of rental housing prediction and visualisation systems relevant to the Nairobi housing market.

b) To review the existing methods and approaches in prediction and visualisation for rental housing.

c) To develop a mobile-based rent prediction and visualisation model using machine learning techniques for rental housing in Nairobi.

d) To validate the proposed mobile-based rent prediction and visualization model.

1.4 Research Questions

a) What are the key characteristics of rental housing prediction and visualisation systems relevant to the Nairobi housing market?

b) What methods and approaches are currently used in rental housing prediction and visualisation?

c) How can a mobile-based model be developed using machine learning techniques to accurately predict and visually represent rental housing prices in Nairobi?

d) How accurate and effective is the proposed mobile-based rent prediction and visualisation model in predicting rental prices and visualizing housing options?

1.5 Justification of the Study

Renters in Nairobi, particularly low- and middle-income earners, face increasing challenges in securing affordable housing due to inflated rental prices, lack of transparency and frequent exposure to fraudulent listings. While digital platforms for property listings exist, they largely function as static noticeboards, offering minimal support in assessing price fairness or visualizing properties. This poses a significant risk for urban dwellers who often make rental decisions without reliable data or contextual awareness. Despite Nairobi's growing digital ecosystem, the integration of advanced technologies such as Machine Learning (ML) and Augmented Reality (AR) into rental platforms remains underutilized. ML offers the potential to analyse vast datasets of property features, locations and historical prices to generate fair rent predictions, while AR can enhance user understanding by allowing immersive property previews. Globally, such technologies are transforming sectors like e-commerce and healthcare, but their combined application in Kenya's rental housing market—especially to support affordability—has not been adequately explored.

With over 54% smartphone penetration and access to public housing data, the technical feasibility for developing such a solution is strong. Frameworks like Google ARCore and Unity make AR deployment achievable within academic constraints, and open datasets from institutions such as the Kenya National Bureau of Statistics (KNBS) provide a solid foundation for training predictive models. Ethically, the system aims to empower renters by promoting fairness and transparency in rental decisions, ensuring data privacy, and minimising model bias across economic groups and neighborhoods. Practically, the mobile application prototype proposed in this study offers a timely, innovative, and socially responsive intervention to bridge the digital gap in Nairobi's housing sector and support equitable access to affordable living.

1.6 Scope

This study focuses on the development of a mobile-based rental housing application that integrates machine learning (ML) and augmented reality (AR) technologies to enhance rent prediction and property visualisation in Nairobi. The system will use existing rental data to train ML models for fair price estimation and employ AR to provide virtual walkthroughs. Additionally, the study also evaluates the model's performance in terms of prediction accuracy and user experience.

Chapter 2: Literature Review

2.1 Introduction

This chapter provides a comprehensive review of the existing literature relevant on a machine learning (ML) – based mobile application with augmented reality (AR) for rental housing prediction and visualisation in Nairobi. It explores theoretical and thematic research on housing affordability, technological tools for decision-making in real estate and recent innovations in mobile applications. By critically analysing relevant research work and identifying gaps in existing research through this chapter, a conceptual framework will inform the research problem and methodology.

2.2 Thematic Review

2.2.1 Housing Affordability and the Urban Rental Market in Nairobi

The shortage of affordable rental housing in Nairobi is well documented (UN-Habitat, 2022). Rapid urbanization has resulted in the expansion of informal settlements, with over 60% of residents living in poorly regulated and often exploitative rental conditions. Most digital rental platforms in Kenya, such as BuyRentKenya and Property24, lack predictive pricing or immersive spatial tools (Winnie & Otieno, 2022). Additionally, there is no integrated platform to guide tenants in making fair and transparent rental decisions thus they often struggle to navigate the housing market. In Table 2.1, the key issues in Nairobi rental sector are outlined.

Table 2. 1: Key Challenges in Nairobi Rental Sector (UN-Habitat, 2022 ; Winnie & Otieno, 2022)

Challenge	**Description**
Informal Settlements	Lack of regulation and basic amenities
Unregulated rent prices	Arbitrary rent hikes with no transparency
Limited Policy Implementation	Gap in enforcement of affordable housing strategies
Technology Gap	Static platforms, no predictive or visualization support

2.2.2 Machine Learning in Rental Prediction

Recent studies show a significant importance of machine learning (ML) in predicting rental housing prices using structured and unstructured data. ML algorithms make use of large datasets that contain housing attributes, location data and economic indicators to forecast rent levels. These models are trained on historical rental data and then evaluated using metrics like RMSE, MAE, and R^2 scores. As urban real estate markets continue to evolve, models such as neural networks, random forests and gradient boosting machines are being used to capture non-linear patterns and variable interactions that traditional statistical methods often miss (Adeoye & Owusu, 2022; Wang et al., 2023).

Linear Regression remains one of the most interpretable and commonly used baseline models for rental price prediction. It formulates rent as a weighted sum of features:

$$\hat{y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n$$

Here,X_i can include variables such as square footage, bedroom count, neighborhood and proximity to amenities. Though easy to train and interpret, linear regression often underperforms on complex, real-world housing datasets because it cannot model non-linear interactions or hierarchical effects.

Support Vector Regression (SVR) improves on this by handling nonlinearity through the kernel trick. SVR aims to fit a regression function $f(x) = w^T \Phi(x) + b$, that deviates from actual targets y_i by no more than an epsilon margin, while keeping minimal. The optimization problem balances margin minimization with penalty for outliers. In small-to-medium size datasets, SVR often yields better accuracy than linear models, especially when relationships between rent and predictors are nonlinear (Goldwaser & Ge, 2023). However, SVR's high computational cost and sensitivity to kernel selection make it less scalable for large datasets. Despite this promise, many studies find SVR requires careful hyperparameter tuning in areas like kernel bandwidth and regularization parameter. Thus, it is less accessible for non-expert deployment.

Gradient Boosting Models such as LightGBM and CatBoost have demonstrated high predictive accuracy for rental prices in structured data environments. These models build sequential decision trees where each new tree attempts to correct errors made by previous ones, thus minimizing loss over iterations. In (Goldwaser & Ge, 2023) there was an application of both LightGBM and CatBoost to housing price datasets, showing their superior ability to capture complex feature interactions while maintaining high accuracy. These boosting methods are well suited to handle variable importance ranking, missing data and categorical features efficiently. Their findings highlight how boosting-based models outperform traditional regressors in rental estimation tasks, particularly when transparency and precision are critical.

Machine learning (ML) techniques have also demonstrated effectiveness in modelling non-linear relationships among diverse housing features. A standard supervised learning formulation for rent prediction defines the problem as:

$$\boldsymbol{y} = \boldsymbol{f}(\boldsymbol{x}) + \varepsilon$$

Where:

a) y = predicted rent price

b) X = vector of input features (e.g., location, size, amenities)

c) f = learned prediction function (e.g., regression tree, neural network)

d) ε = error term or noise

2.2.3 Augmented Reality in Property Visualization

Augmented Reality (AR) enhances user decision-making through spatial immersion. By using Simultaneous Localization and Mapping (SLAM), AR applications overlay 3D digital content onto real environments, allowing users to interact with virtual assets that align with their physical surroundings. This creates an engaging way to explore rental properties remotely and in real time, even before visiting the actual location.

AR has come out as a practical solution for enabling users to explore spaces virtually. It overlays digital content on the physical environment and has been widely used in real estate applications to simulate property walkthroughs. Research shows that AR enhances spatial understanding, visual memory and overall satisfaction, especially among users who are unfamiliar with the physical layout of properties. This feature is particularly useful in urban housing scenarios where renters must evaluate multiple listings in a short time without physically visiting each site (Pamungkas & Astrianty, 2024).

The development of AR has been accelerated by toolkits like Google ARCore and Unity 3D, which support marker less tracking, environmental mapping and 3D rendering on mobile devices. These tools have made AR development more accessible, with ARCore proving particularly effective for enabling real-time tracking and surface detection on mid-range Android smartphones (Ekstrand & Lundqvist, 2020). Applications built with ARCore allow users to preview dimensions, interior layouts, and neighborhood overlays directly through their mobile cameras, reducing uncertainty in rental decisions.

In a study by Pamungkas & Astrianty (2024), AR-based real estate applications reduced decision-making time by 35% and increased user confidence by 40%. Users reported greater clarity regarding room sizes, layouts and spatial flow when using AR interfaces compared to traditional 2D photos or floor plans.

From a technical standpoint, AR in mobile real estate relies on SLAM-based pose transformations:

$$\begin{bmatrix} x^{'} \\ y^{'} \\ z^{'} \\ 1 \end{bmatrix} = P \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Here, (x, y, z) is transformed into a new coordinate $(x^{'}, y^{'}, z^{'})$ as seen through a camera using a matrix P.

Where, $P \in \mathbb{R}^{4\times4}$ is the pose matrix used by AR systems apply to apply spatial transformations estimated via ARCore/ARKit to anchor 3D assets in real space. Ensuring latency under 100 MS guarantees real-time responsiveness and mobile usability.

2.2.4 Key Characteristics of Rental Housing Prediction and Visualisation Systems

Rental housing prediction and visualisation systems are data-driven platforms designed to estimate rental prices and present property information through interactive or immersive interfaces. Based on an analysis of recent studies, these systems typically exhibit three interrelated core characteristics which are predictive modelling capability, visualisation functionality, and integrative decision-support mechanisms.

1. **Data-Driven Predictive Modelling**

The foundational characteristic of these systems is a data-driven prediction engine that leverages structured datasets—such as location, number of bedrooms, amenities, building type and historical rent values—to forecast property prices. Recent research emphasizes the use of machine learning algorithms including Random Forest, Artificial Neural Networks (ANN), and Light Gradient Boosting Machine (LightGBM), which outperform traditional regression models in handling non-linear and multivariate data relationships (Abolghasemi & Esmaeilbeigi, 2021; Kapoor & Narayanan, 2022). These algorithms learn patterns from historical data to estimate fair rental prices, detect anomalies and provide market insights. Predictive performance is often evaluated using metrics such as the Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) and Coefficient of Determination (R^2), as recommended in the scikit-learn (2020) documentation. The predictive component therefore serves as the analytical core of such systems.

2. **Visualisation and User Interaction Layer**

A second defining characteristic is the visualisation and interaction layer, which enhances user understanding and engagement. Many modern systems employ 3D modelling, interactive maps and Augmented Reality (AR) to enable users to explore rental spaces virtually. Studies such as Pamungkas and Astrianty (2024) demonstrate that visualisation not only improves user experience but also aids in decision-making by providing spatial and contextual cues that static listings cannot convey. This layer translates predictive insights into an accessible, visual format, thereby bridging the gap between analytical data and real-world perception.

3. **Integration and Decision-Support Functionality**

The third characteristic involves integration and decision support, where prediction results and visual components are embedded within a unified platform that delivers actionable insights to users. This integration may include comparative pricing dashboards, location-based analytics, or affordability indices (Goldwaser & Ge, 2023). Systems such as Zillow and Redfin integrate predictive outputs directly into user interfaces, helping users determine fair rent values or detect overpriced listings. Effective systems also employ localization features—such as currency, language and region-specific datasets—to enhance contextual relevance and trustworthiness.

4. Data Quality and Model Transparency

A cross-cutting characteristic across all effective systems is data quality management and model transparency. As Kapoor and Narayanan (2022) note, robust predictive systems must prevent data leakage, ensure unbiased sampling and maintain interpretability to foster reproducibility and user confidence. Techniques like stratified data partitioning and explainable AI modules are increasingly integrated to ensure that predictions are both reliable and ethically sound.

5. Relevance to the Nairobi Housing Market

A review of existing global and regional platforms reveals that while international systems such as Zillow and Redfin exemplify advanced integration of predictive analytics and user-friendly visualisation, comparable solutions remain limited in Kenya. Most local platforms such as BuyRentKenya and JumiaHouse primarily provide listing services without embedded price-prediction engines or visualisation components. Therefore, the key characteristics relevant to Nairobi's housing market include predictive models that account for local socioeconomic and spatial factors, context-aware visualisation that represents diverse urban housing types, and integration with local data sources such as KNBS rental indices. These features are essential for creating a reliable, transparent, and locally adaptive rental housing prediction and visualization system.

2.3 Theoretical Framework

In this section there is a walkthrough of three interrelated theoretical frameworks—Technology Acceptance Model (TAM), Smart City & Digital Equity and Human-Centered Design (HCD). These frameworks collectively inform the design, adoption and usability of the proposed ML + AR mobile application for rental housing in Nairobi.

2.3.1 Technology Acceptance Model (TAM)

The Technology Acceptance Model (TAM) remains widely used for evaluating user behavior toward emerging technologies. Recent research, such as (Olaolu, 2022), confirms that Perceived Usefulness (PU) and Perceived Ease of Use (PEOU) are crucial predictors of adoption for mobile apps in Africa. In this study on digital banking platforms in Nigeria, both PU and PEOU significantly influenced users' behavioral intention to adopt fintech apps.

Applying TAM to the Nairobi rental market context, the proposed mobile app must meet these two conditions where users must believe that it helps them make better rental decisions (PU) and that it is easy to use on their mobile devices (PEOU). ML-generated rent predictions and AR-based visual previews can improve decision-making and reduce cognitive burden, thereby enhancing both constructs.

Moreover, recent systematic evidence from Sub-Saharan Africa reaffirms that Perceived Usefulness (PU) and Perceived Ease of Use (PEOU) remain the strongest and most consistent predictors of mobile technology adoption, even in financially underserved or informal sectors. A 2024 meta-analysis of mobile fintech adoption in Africa concluded that these two TAM constructs significantly outweigh factors like cost or regulation when influencing user behavior. The study also highlights that incorporating trust-building mechanisms—such as transparent data, visual explanations, and low data consumption—improves adoption across mobile platforms (Yeboah et al., 2020).These findings directly support this study's inclusion of rent prediction transparency, AR-based visualizations and verified listing indicators in the app design.

2.3.2 Smart City & Digital Equity Framework

Smart City frameworks emphasize the use of data and technology for inclusive urban service delivery, anchored by the principle of digital equity. The World Bank (2023) stresses that urban interventions must ensure equal access to digital tools regardless of socio-economic status.

In Nairobi, increasing smartphone penetration and ICT access make mobile-powered solutions viable. However, platforms like BuyRentKenya and Property24 still provide static listings without price validation or spatial orientation tools (Winnie & Otieno, 2022). A smart city–aligned app should incorporate ML-driven rent estimates and AR previews to bridge informational gaps and promote fairness (Cacciamani et al., 2024).

Furthermore, smart city theory emphasizes contextual relevance and local data governance. By training ML models on Nairobi-specific rental datasets and designing AR visualizations suitable for informal housing structures, the application aligns with equitable service delivery and urban fairness principles.

2.3.3 Human-Centered Design (HCD)

Human-Centered Design (HCD) is formally codified in the ISO 9241-110:2020 standard, which defines seven interaction principles which include, suitability for user tasks, self-descriptiveness, expectation conformity, learnability, controllability, use-error robustness and user engagement. These principles underpin high usability in interactive systems (International Organization for Standardization, 2020). They emphasize that systems must be designed around real user workflows and contexts. For the proposed ML + AR rental app, applying ISO 9241-110:2020 ensures that features like rent prediction interfaces and AR walkthroughs map cleanly to user expectations and tasks, reducing cognitive load and improving usability.

A recent 2024 study by Vindigni demonstrates how aligning UX design with DIN EN ISO 9241 standards supports user inclusion, accessibility and satisfaction in socially inclusive digital applications (Vindigni, 2024). The paper shows that flow heuristics—adaptive interfaces that respond to user behavior—significantly enhance experience in AI-heavy systems. In our study, this suggests that adaptive interface elements — such as dynamic AR overlays that adjust based on user behavior — and progressive disclosure of predictive data will help users navigate the app intuitively, particularly those unfamiliar with rental jargon or AR navigation.

Recent empirical research highlights that human-centric usability issues—such as confusing information architecture, inconsistent interface design, performance bottlenecks and poor interaction flow—are significant barriers to successful mobile app adoption. A structured expert study published in 2025 identified these issues as the top five usability concerns across mobile apps, including AI-enhanced and AR-enabled platforms (Weichbroth, 2025). These findings underscore the necessity of applying rigorous usability assessment and iterative refinement in app development.

2.3.4 Predictive Model Theory

Predictive Modelling Theory provides the conceptual foundation for building systems that use historical data to estimate or forecast unknown values. It assumes that underlying patterns in past observations—such as property features, location, and amenities—can be captured by mathematical or algorithmic models and generalized to new cases. According to Abolghasemi and Esmaeilbeigi (2021), predictive modelling involves constructing and refining statistical or machine-learning functions that map input variables to target outcomes, enabling accurate estimation of continuous or categorical values.

This theory supports the use of machine learning algorithms such as ensemble decision-tree models and gradient-boosting frameworks. These algorithms learn from examples in a supervised manner to minimize prediction error while improving generalization to unseen data. In practice, the predictive-modelling process follows a structured pipeline that includes data preparation, feature engineering, model training, validation and performance evaluation. Each stage progressively improves the model's approximation of the true relationship between predictors and outcomes, ensuring that predictions remain both accurate and interpretable.

Modern implementations of predictive modelling increasingly rely on robust evaluation strategies such as cross-validation and stratified sampling to prevent overfitting and ensure representativeness across price categories. As highlighted by Kapoor and Narayanan (2022) failure to apply rigorous validation often leads to data leakage and inflated accuracy estimates—an issue this study mitigates through careful data splitting and benchmarking. In this research, predictive-modelling theory justifies the choice of Light Gradient Boosting Machine (LightGBM) as the primary algorithm, given its proven efficiency in structured datasets and its capacity to capture complex feature interactions.

2.4 Review of Current Approaches of ML in Rent Prediction

2.4.1 Machine Learning for Rent Prediction

Machine learning (ML) has transformed data-driven prediction in real estate by enabling systems to analyse large datasets, uncover patterns and make reliable forecasts. In rental price prediction, ML models make use of structured features such as number of bedrooms, location, square footage, proximity to amenities and historical pricing trends. The choice of algorithms significantly affects prediction accuracy, robustness and computational efficiency. Three of the most relevant models for rental prediction include Random Forest, XGBoost, and Artificial Neural Networks (ANNs).

2.4.1.1 Random Forest

Random Forest is an ensemble learning algorithm that builds multiple decision trees and aggregates their predictions. It is best for datasets that contain missing values, outliers and mixed data types either categorical or numerical. Each tree in the forest is trained on a random subset of the data using bootstrap aggregation, bagging, where the multiple subsets created for the training from random sampling might result in some data points appearing multiple times whereas others are left out. Also, the final prediction can be the majority vote, classification, where the class label thar receives the most vote is chosen or average, regression, of all trees.

Prediction formula (for regression):

$$\hat{y} = \frac{1}{K}\sum_{K=1}^{K} T_K(X)$$

Where:

a. K= number of trees

b. $T_K(X)$= prediction of the k-th tree for input features X

Random Forest is particularly useful in handling heterogeneous data typical of urban rental datasets in developing cities like Nairobi. For example, datasets from informal housing often contain missing or noisy records on features such as the presence of utilities, floor types or the number of occupants. In such a scenario, the Random Forest model remains robust, as it can still derive insights by averaging predictions across multiple decision trees trained on different data samples. The figure below illustrates how a Random Forest model uses multiple decision trees working together to predict rental prices based on housing features.

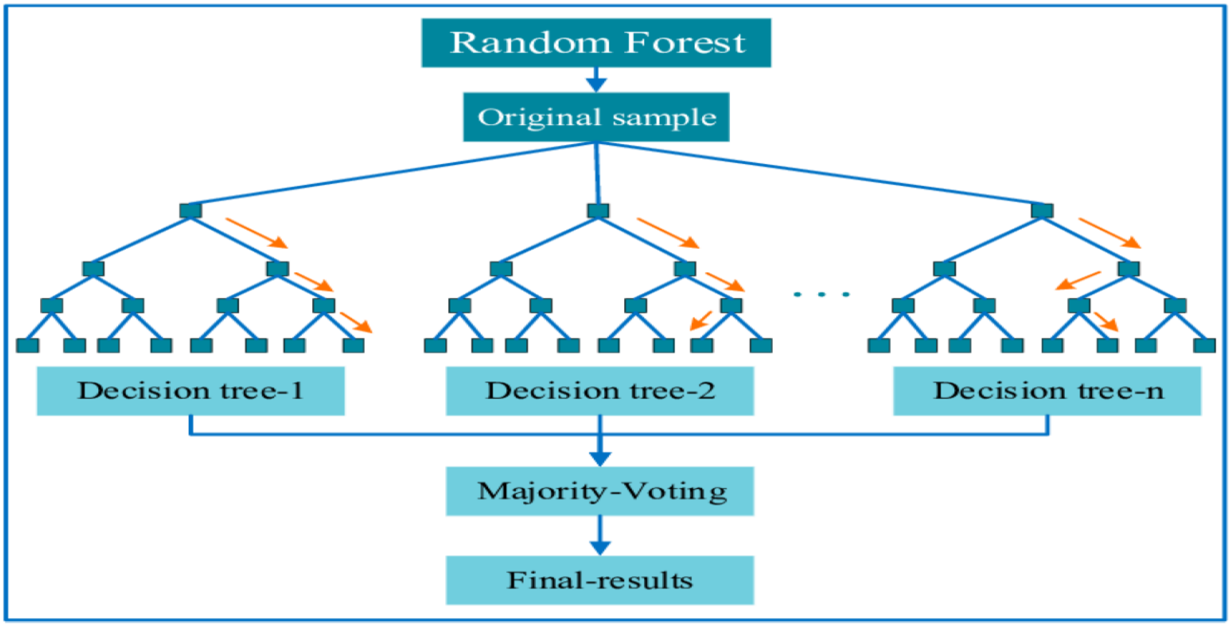


Figure 2. 1: Random Forest-based house price prediction model (Silpa Chaitanya et al., 2024)

In a study by Adzanoukpe (2025), Random Forest was applied to rental datasets from Ghana, which contained irregularities like those found in Nairobi's informal housing markets. The model was able to demonstrate resilience against missing values and noise, achieving robust prediction accuracy even where linear models underperformed. One notable outcome was the model's ability to rank feature importance, revealing that location, proximity to transport and sanitation facilities had the highest influence on rent variation. These findings underscore the suitability of Random Forest for environments like Nairobi, where rental data is often incomplete or inconsistently recorded.

2.4.1.2 LightGBM (Extreme Gradient Boosting)

LightGBM is a decision-tree-based ensemble algorithm that utilises the gradient boosting framework. Unlike Random Forest, which trains trees independently, LightGBM builds trees sequentially where each new tree corrects errors made by the previous ones. It uses a leaf-wise growth strategy instead of the level-wise method used in traditional gradient boosting, resulting in faster training and higher accuracy.

LightGBM is specifically optimized for efficiency and memory usage, making it highly suitable for real-time applications such as mobile rent prediction. It also supports native handling of categorical variables, which enhances its performance on structured/tabular data without requiring one-hot encoding.

Prediction formula:

$$\hat{y}_i = \sum_{m=1}^{M} f_m(x_i), f_m \in F$$

Where:

a. $\hat{y}_i$= predicted value for sample

b. f_m= individual regression tree

c. F = space of regression trees

Objective function:

$$L(\emptyset) = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{m=1}^{M} \Omega(f_m)$$

Where:

a. l= loss function (e.g., squared error)

b. Ω= regularization term to penalize complexity

LightGBM has demonstrated superior performance in structured real estate datasets, especially when capturing subtle patterns that linear models may overlook. In (Zhang, 2023), LightGBM was trained on a housing dataset from Nairobi's CBD and suburbs. It achieved an R^2 score of 0.89 and significantly reduced RMSE compared to Random Forest and SVR.

2.4.1.3 Artificial Neural Networks (ANNs)

Artificial Neural Networks (ANNs) are computing systems inspired by the biological structure of the brain. They consist of input, hidden and output layers. ANNs are well-suited for capturing complex nonlinear relationships in housing data, especially when unstructured data like images or text are involved.

Looking at the model structure of ANNs, it is made of three primary layers. The input layer receives structured or unstructured rental features such as location, size, amenities or even image data from listings. These inputs are passed through one or more hidden layers, where neurons apply learned weights, add biases and use non-linear activation functions such as ReLU or sigmoid to extract meaningful patterns. Each hidden layer refines the representations learned from the previous layer, enabling the network to model increasingly abstract relationships. The output layer produces the predicted rental price based on the cumulative transformation of input data. Throughout training, the model adjusts its internal parameters using backpropagation and gradient descent to minimize the difference between predicted and actual rents. As shown in Figure 2.2, a neural network is structured in layers that allow it to learn complex, non-linear relationships in rental pricing data.

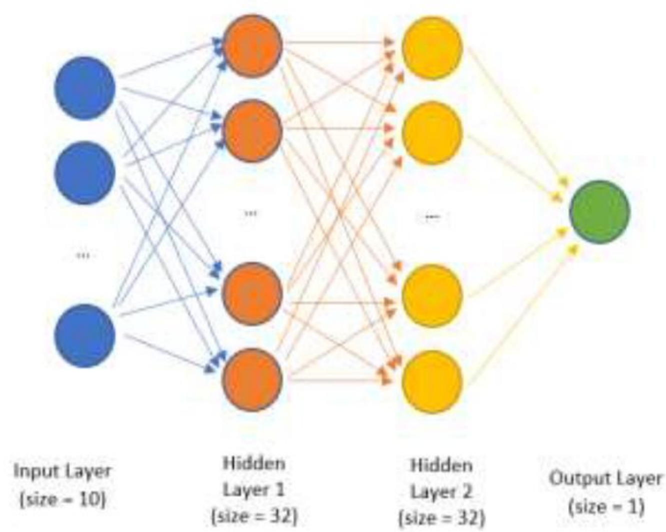


Figure 2. 2: Artificial Neural Network architecture used for house price prediction (Patil et al., 2022)

Forward Propagation Equation:

$$a^{(l)} = \sigma(W^{(l)}a^{(l-1)} + b^{(l)}$$

Where:

a. $a^{(l)}$= activation of layer

b. W= weights, = biases, = activation function (e.g., sigmoid)

In (Raju et al., 2023), Artificial Neural Networks (ANNs) were applied to a diverse set of urban rental datasets using a combination of regression techniques and deep learning architectures. The study highlighted the strength of ANNs in capturing nonlinear relationships between housing features such as location, access to amenities and visual cues derived from listing images. Notably, the integration of both structured data, rent history and number of bedrooms, plus unstructured data, street-view photos and reviews, led to a notable reduction in prediction error, with improvements in MAE of up to 10–15%. The ANN model's flexibility in adapting to complex, noisy environments suggests that it is particularly suitable for dynamic rental markets like Nairobi. However, the authors emphasize that ANN models require sufficient data preprocessing and normalization to avoid overfitting and maintain generalizability

2.4.1.4 Model Evaluation Metrics

Evaluating the performance of machine learning models is essential for understanding how well a model will perform on unseen data. In rent prediction, the most widely used metrics include Mean Absolute Error (MAE), Root Mean Square Error (RMSE) and Coefficient of Determination (R^2). Each of these metrics provides a different lens through which model accuracy and reliability can be assessed. When used together, these metrics offer a well-rounded assessment of a model's prediction quality, capturing both average performance and sensitivity to error magnitude. Looking at rental prediction systems, especially those deployed in dynamic urban settings like Nairobi, these metrics help developers ensure that the tool remains reliable and actionable across diverse property types. In the metrics, symbols (y_i) represents the actual value of the target variable for the i-th observation, ($\hat{y}_i$) is the predicted value generated by the machine learning model for the same i-th observation and n is the total number of data points or observations. Below is the description of the metrices.

i. MAE

Measures the average magnitude of errors in a set of predictions, without considering their direction. It is straightforward to interpret, as it informs on average, how much predicted rental values deviate from actual values in real currency units for instance in the Kenyan Shilling. MAE is especially valuable when you want a general sense of prediction accuracy and is less sensitive to outliers than RMSE. However, since it gives equal weight to all errors, it might not fully capture the cost of larger mistakes, which could be critical in high-value rental markets.

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|$$

ii. RMSE

Penalizes larger errors more than MAE due to the squaring of each error term before averaging. This makes RMSE an important metric when large errors are particularly undesirable such as predicting luxury apartment rents where even a small percentage error can translate into substantial financial misjudgments. However, its sensitivity to outliers can skew results if the dataset contains significant anomalies. In Nairobi's mixed-market data, using RMSE in combination with MAE helps provide a balanced view.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

iii. R^2

Also known as the coefficient of determination, indicates the proportion of the variance in the dependent variable that is predictable from the independent variables. A value of 1.0 means perfect prediction, while 0.0 means the model does no better than predicting the mean. It is useful for gauging the overall explanatory power of the model but does not provide detailed information about the size or direction of individual prediction errors. Furthermore, R^2 can be misleading in models that overfit the training data or in datasets with non-linear trends that the model fails to capture.

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_1 - \hat{y}_2)^2}$$

Figure 2.3 compares various machine learning models based on their accuracy and reliability in rent prediction scenarios.

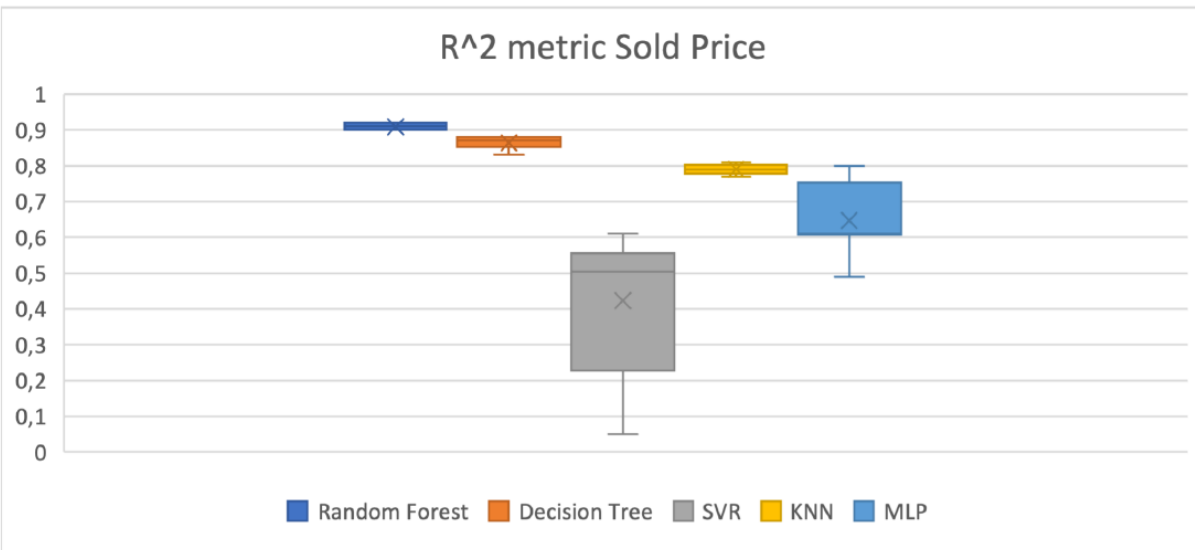


Figure 2. 3: Model performance comparison for housing price prediction

(Eren, 2023)

2.4.2 Augmented Reality in Real Estate

Modern AR applications like MagicPlan, IKEA Place and ARki enable users to walk through a virtual version of a property using only a mobile device. These tools rely on technologies like Simultaneous Localization and Mapping (SLAM), surface detection and motion tracking to accurately align digital content with real-world spaces. These AR platforms collectively illustrate how immersive spatial technology can elevate property evaluation beyond text and photo listings.

Integrating such applications into rent prediction tools could greatly enhance user trust and accessibility, particularly in Kenya's informal and semi-formal housing markets.

Augmented Reality development for real estate applications is increasingly supported by accessible toolkits and engines such as Google ARCore and Unity 3D. ARCore, developed by Google, enables Android devices to understand the environment and place virtual objects in space using Simultaneous Localization and Mapping (SLAM). Unity 3D, paired with the AR Foundation framework, allows developers to build cross-platform AR experiences for both Android (via ARCore) and iOS (via ARKit) (Syed et al., 2023).

The following sections explore real-world AR applications that exemplify how these platforms can be used to improve property visualization and rental decision-making.

2.4.2.1 MagicPlan

MagicPlan is mostly used for creating floor plans and property visualizations using mobile phone cameras. The app allows users to capture the dimensions of a space and generate a 2D or 3D model within minutes using SLAM technology and LiDAR if available. This is particularly useful in rental listings where no architectural blueprints exist. The key calculation performed by MagicPlan involves triangulating distances from the user's position using:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Where d is the distance between two scanned points. The app combines multiple measurements to construct an accurate room outline. As noted in (Syed et al., 2023), MagicPlan's ease of use, fast spatial capture and compatibility with mobile devices have contributed to its widespread adoption in affordable housing surveys and urban planning workflows across developing regions.

In Kenya, where many informal properties lack formal floor plans, tools like MagicPlan can be crucial for rapid spatial digitization, facilitating virtual walkthroughs and remote assessments. Its offline functionality and cloud synchronization will enable field officers to collect data even without internet connectivity, then seamlessly upload completed layouts once back online. This capability has been highlighted in developing urban contexts, where mobile spatial tools effectively streamline data collection and planning (Syed et al., 2023), providing a practical foundation for integrating such technologies into Kenyan housing surveys.

2.4.2.2 IKEA Place

IKEA Place is an AR application focused on interior design and furniture placement. It allows users to view true-scale models of furniture within their space using ARCore or ARKit. While originally created for home shopping, its precision rendering of spatial dimensions is valuable for tenants visualizing how rental spaces would function in daily use.

The visual rendering follows this transformation:

$$\vec{p}_{image} = K[R|t]\vec{P}_{world}$$

Where:

a. K =is the camera intrinsics matrix,

b. $[R|t]$=represents rotation and translation matrices,

c. $\vec{P}_{world}$=is the 3D point in world space,

d. $\vec{p}_{image}$=is the projected 2D point on the device screen.

As highlighted in (Alam et al., 2021), IKEA Place's integration with mobile AR platforms like ARKit and ARCore enables highly accurate scaling and object placement. This allows users to visualise furniture dimensions in real-time within their actual spaces, making it easier to assess whether items will fit before purchasing or renting. Such precision is especially valuable in urban housing markets where space optimization is essential for both comfort and functionality.

2.4.2.3 ARki

ARki is a more architecturally inclined AR application designed for visualizing interactive 3D models of buildings and interiors. Its real-time lighting adjustments and material simulation enable users to test spatial aesthetics before visiting a site. This makes it suitable for both property staging and technical visualization by landlords, renters, and inspectors.

ARki supports annotation overlays, allowing landlords or agents to tag specific areas such as plumbing lines, cracked walls directly in the 3D AR model. Its collaborative feature allows multiple stakeholders to view and interact with the model remotely, useful for joint rental or inspection decisions. As demonstrated in (Macchiaroli et al., 2025) architecturally focused AR tools, with real-time lighting, material simulation and collaborative annotation, have improved remote inspection workflows in West African property markets, allowing stakeholders to virtually assess structural and aesthetic elements before physical visits

Incorporating ARki into Nairobi's property sector could enable remote inspection and collaborative renting in areas where site access is difficult. Figure 2.4 outlines the architecture of an AR-based mobile system, showing how real-world scenes are overlaid with virtual rental data.

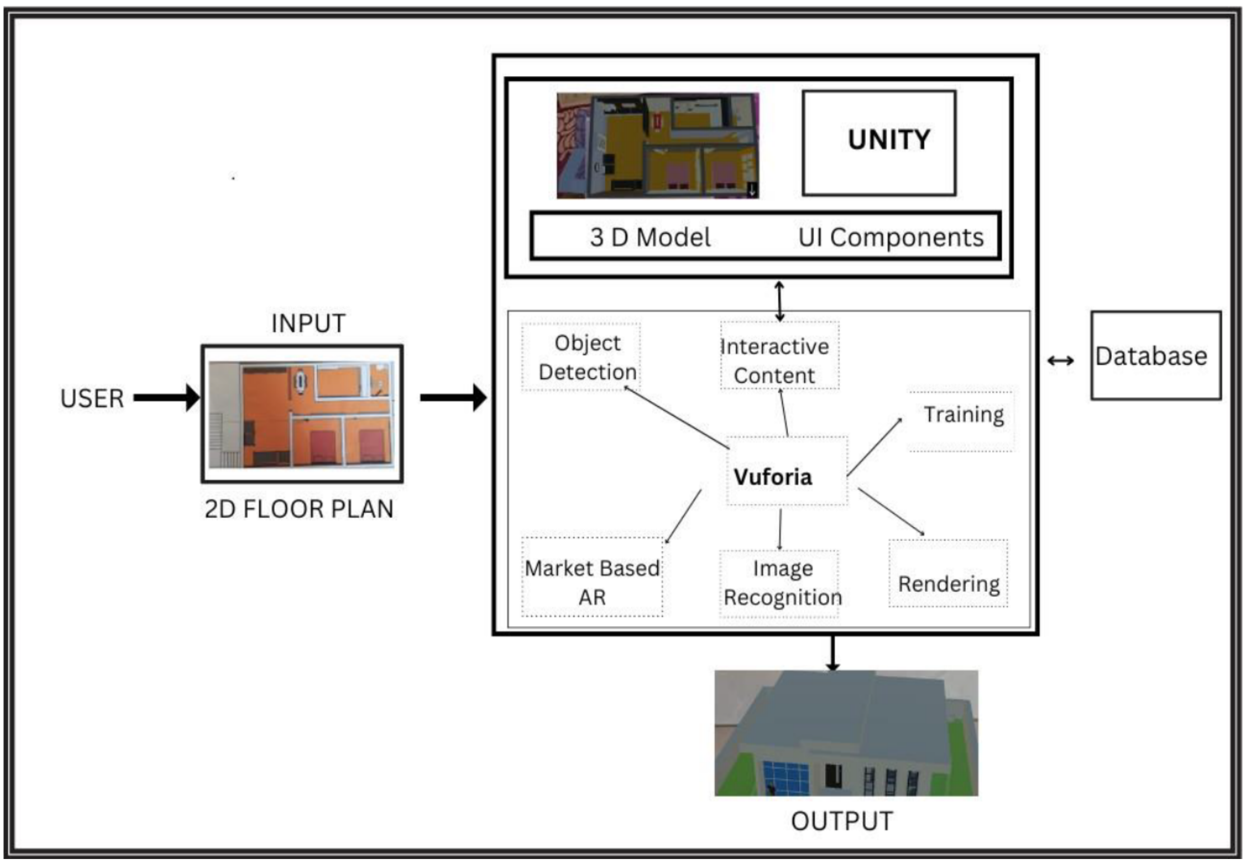


Figure 2. 4: Block diagram of an AR property visualization system (Aher et al., 2024)

2.5 Review of Existing Applications and Systems

Globally, several real estate platforms have adopted emerging technologies to enhance user experience and facilitate decision-making. In (Fu Ginger Zhe Jin Meng Liu et al., 2022) studies how machine learning algorithms are utilized to offer predictive pricing insights and housing value estimates. In the study looking at Zillow Application in the United States, regression-based models are used to forecast home values, drawing on historical prices, location, size and neighborhood features. However, these platforms fall short in offering immersive experiences, particularly lacking integration of Augmented Reality (AR), which could offer spatial context and deeper engagement. Figure 2.5 illustrates which housing features most influence rental price predictions in a Random Forest model.

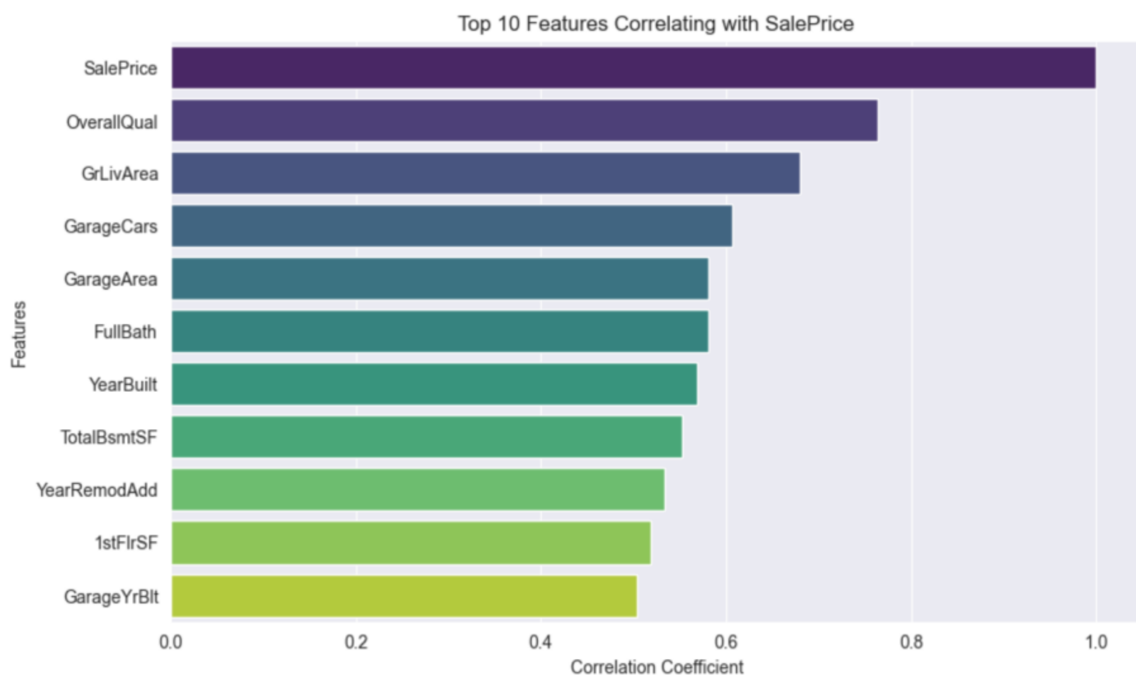


Figure 2. 5: Feature importance scores for rental prediction using Random Forest (Yang, 2025)

In Asia, MagicBricks and Housing.com, prominent platforms in India, have incorporated basic visualisation tools such as 3D tours and static floor plans to assist potential tenants or buyers (MagicBricks, 2021). While these enhance visualisation, the absence of dynamic AR experiences limits user interaction with space, especially in mobile-first contexts. MagicBricks, for instance, allows map-based filtering and basic VR previews but does not incorporate AR overlays for real-time property viewing or street-level navigation.

In the African context, especially in Kenya, existing real estate platforms such as BuyRentKenya, Property24 and Jumia House mostly serve as listing directories (Winnie & Otieno, 2022). They lack advanced decision-support tools like price verification, fraud detection and real-time market insights. Moreover, these platforms provide minimal support for price prediction or spatial visualisation, representing a missed opportunity for technological innovation. Table 2.2 illustrates a comparative analysis of existing real estate platforms including the proposed application.

Table 2. 2: Comparative Analysis of Real Estate platforms

Platform	**Country/Region**	**ML Integration**	**AR Features**	**Fraud Detection**	**Price Verification**	**Visualization Tools**
Zillow	USA	Advanced predictive models	None	Not available	Supported using Zestimate	Static images, basic maps
Redfin	USA	Predictive pricing engine	None	Not available	Supported	Photos, map integration
MagicBricks	India	Not integrated	Basic AR/VR in test phase	Not available	Not available	3D floor plans, virtual tours
BuyRentKenya	Kenya	Not integrated	None	Not available	Not available	Photos and Google Maps
Proposed App	Kenya	Custom ML models (local)	Full AR support planned	Integrated with ML alerts	Fair rent assessment module	AR walkthroughs, location overlays

2.6 Gaps in the Existing Literature and Practice

There is a significant gap in the availability of mobile platforms that integrate both machine learning (ML) and augmented reality (AR) technologies for rental housing support. Most systems are designed to perform isolated tasks—such as rental price prediction or basic 3D visualisation—but do not combine these capabilities into a single cohesive solution. An integrated ML-AR approach could enhance user experience by offering both data-driven insights and spatial immersion, helping users assess fair pricing while virtually exploring properties. Without this combination, current platforms offer only partial support for informed decision-making.

A second major gap is the limited localization of digital housing solutions for African urban contexts. Most existing platforms are modelled on systems developed in Western countries, with structured property markets and formal housing records. These systems often do not account for the informal housing structures, unregulated pricing, and fragmented infrastructure common in cities like Nairobi. Without localization, these tools perform poorly and are often not trusted by users. This disconnects calls for solutions designed around African socio-economic realities and urban dynamics.

Despite increasing smartphone access in Africa, AR remains underused in mobile rental apps. AR can greatly enhance interaction by enabling virtual property walkthroughs, furnishing simulations and real-time overlays of surrounding neighborhoods. However, most local platforms still rely on static content such as photos and text descriptions. Concerns about device compatibility and mobile data usage have contributed to this underutilization. Yet the technical capacity now exists to implement lightweight AR solutions suitable for mobile-first users in developing regions.

Fraud is another critical issue in online property listings, with many platforms lacking tools to detect or prevent misleading content. Scams involving fake listings, duplicate ads, or manipulated images are common, especially in urban rental markets. Without intelligent fraud detection, users are exposed to significant risk and must rely on manual verification methods. ML-based anomaly detection could identify suspicious patterns and improve platform trustworthiness, but such features are still largely absent from current systems.

Finally, most platforms do not provide tools to assess whether rental prices are fair in real time. In places like Nairobi, where pricing is often unregulated, renters have little guidance on market rates.

This lack of transparency can lead to exploitation. ML can address this gap by analyzing similar listings and neighborhood data to generate real-time fair price estimates. Integrating such functionality would empower users with actionable insights and encourage fairer pricing practices across the market.

2.7 Conceptual Framework

The proposed conceptual framework integrates two core technological modules into a unified mobile application tailored for Nairobi's urban housing market.

i. **ML-Based Rent Prediction Engine**

This module applies machine learning algorithms—such as Linear Regression, Random Forest and Gradient Boosting—trained on Nairobi-specific datasets including location, amenities, rental history and property type. The objective is to predict fair rental prices and detect anomalies such as overpricing or underpricing. Incorporating ensemble models ensures robustness and adaptability to non-linear trends in rental data.

ii. **AR-Based Visualization Module**

The augmented reality module provides users with immersive experiences through virtual property walkthroughs; furniture overlay simulation and spatial orientation. Using smartphone cameras and GPS data, this module enables users to "place themselves" in a property environment, enhancing trust and engagement.

iii. **Integration Layer and Decision Support**

Both modules are connected through a backend API and a mobile-friendly UI. The application supports interactive filters, real-time rent predictions, AR overlays, and fraud alerts, offering a comprehensive decision-support system. Figure 2.6 presents the proposed conceptual framework for the rental application, illustrating how the ML-based rent prediction engine, AR visualization module and integration layer interact through a mobile interface to provide users with immersive, data-driven insights.

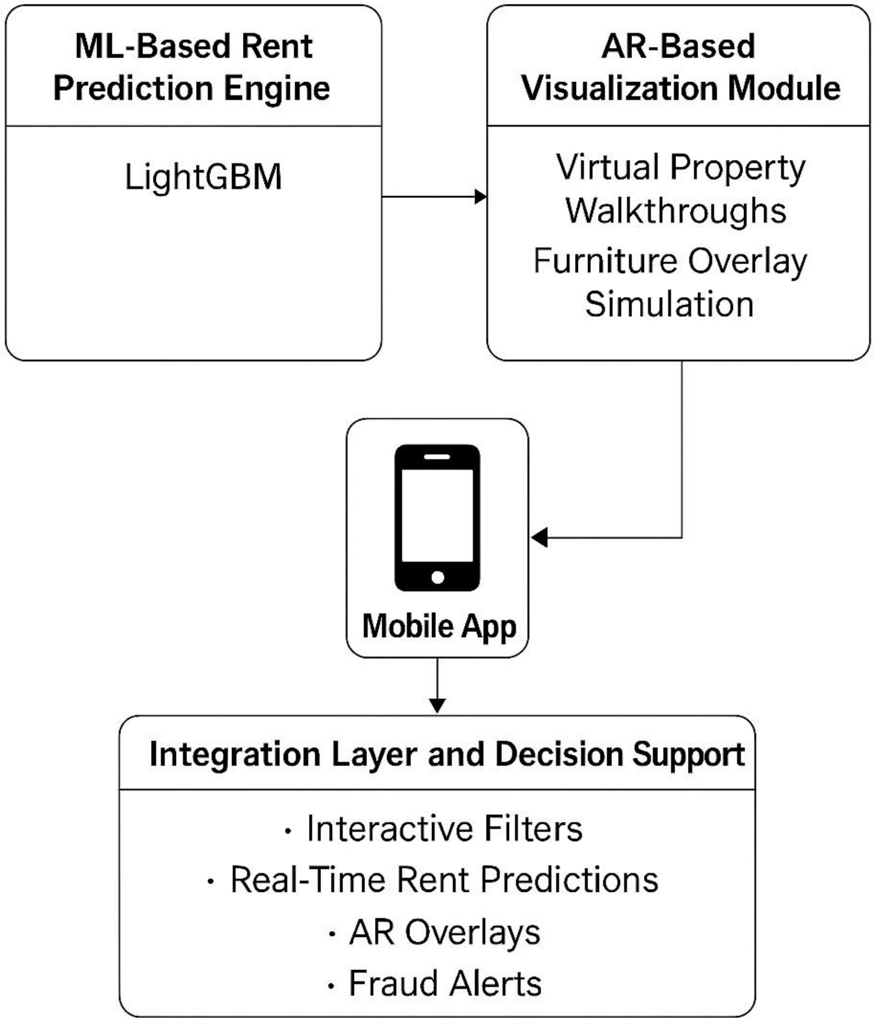


Figure 2. 6: Conceptual Framework for ML+ AR Rental Application

Chapter 3: Research Methodology

3.1 Introduction

This chapter describes the methodology that was used to design, develop and evaluate the proposed mobile application that integrates machine learning (ML) and augmented reality (AR) for rental housing prediction and visualisation in Nairobi. The chapter outlines the research design, case study context, data collection procedures, development of the proposed model, simulation software choices, data analysis methods and ethical considerations. Furthermore, it also addresses how research quality was ensured throughout the study. The approach draws on proven methodologies used in similar predictive modelling systems such as those by Żbikowski and Antosiuk (2021), Goldwaser and Ge (2023) and Pamungkas and Astrianty (2024).

To ensure that the study variables were clearly defined, measurable and linked to the study objectives, each variable was operationalized as shown in Table 3.1. Operationalization helped to translate abstract research concepts into specific, quantifiable indicators that can be measured and analyzed empirically. The key variables identified in this study included rental price, property features and user satisfaction. The operational definitions, measurement indicators, data sources, and corresponding analysis techniques are presented in Table 3.1.

Table 3. 1: Operationalization of Variables

Variable	Type	Measurement Indicator	Data Source	Analysis Technique
Rental Price (KES)	Dependent	Market rent per unit (monthly), price per square meter	Kenya National Bureau of Statistics (KNBS, 2024) datasets; verified online rental listings	LightGBM regression analysis using python to predict rent values
Property Features	Independent	Number of bedrooms, property type (apartment, bedsitter, maisonette), amenities (parking, Wi-Fi, water supply, security), and location coordinates	Verified online housing platforms such as BuyRentKenya and Property24	Feature importance and correlation analysis using Python libraries (Scikit-learn, Pandas)
User Satisfaction	Dependent	System Usability Scale (SUS) score ≥70,	Primary data collected from user testing	Descriptive statistical analysis using Python

		perceived ease of use, perceived usefulness	through SUS questionnaires	(Matplotlib, Pandas)

This operationalization framework ensures that each variable directly supports the study objectives and can be measured objectively using reliable indicators. The dependent variable is the predicted rental price, influenced by independent property features and validated through user satisfaction metrics obtained from prototype testing. These variables guided the data collection, model development, and evaluation processes described in later sections 3.5 and 3.7.

3.2 Research Design

This study adopted a Design Science Research (DSR) methodology structured around recent frameworks in information systems innovation. DSR enabled the creation and evaluation of artifacts—here, an ML + AR application prototype—that address real-world rental housing challenges while contributing theoretical knowledge (Kundisch et al., 2022). Additionally, the study included elements of applied and descriptive research, using observational data trends to define design requirements and problem context in Nairobi.

DSR supported iterative refinement of both the machine learning rent prediction engine and the augmented reality visualization module, enabling continuous improvement based on empirical feedback and stakeholder interaction.

3.3 Case Study Description

Nairobi, the capital city of Kenya, served as the geographical focus of this study. It presented a suitable context because of its rapid urbanisation, housing shortages and growing digital infrastructure. The housing affordability gap in Nairobi, as reported by the Kenya Mortgage Refinance Company (Kenya Mortgage Refinance Company, 2023), exceeds two million units. Informal settlements and price exploitation are rampant challenges affecting tenants, particularly in low-income areas. Nairobi also boasts a high mobile phone penetration rate (at 143% as of 2025), making it ideal for deploying a mobile-based solution that leverages machine learning and augmented reality (Communications Authority of Kenya, 2025; UN-Habitat, 2022).

3.4 Data Collection

Data required for this study was obtained from secondary datasets which include:

i. Rental listings from property platforms such as Jiji and KenyaPropertyCenter.

ii. Housing reports and open datasets from the Kenya National Bureau of Statistics (KNBS).

The data included attributes such as rental price, location coordinates, number of rooms, amenities, accessibility and property photos (Zhang, 2023) which were sampled using purposive sampling. Purposive sampling was adopted because the study seeks to capture housing units consistent with Nairobi's middle- and upper-income rental categories as defined by KNBS rent distribution data. Unlike simple random sampling, this approach ensured deliberate inclusion of property types for instance 1 to 3-bedroom apartments and maisonettes that align with formal housing standards reflected in the KNBS dataset.

This method allowed the researcher to maintain data consistency as random sampling may risk overrepresenting thus hindering the study's analytical focus. By referencing KNBS benchmarks, the purposive approach ensured that the dataset reflects authentic, verified and policy-aligned rent structures in Nairobi's formal housing market. The structure of the dataset used in this research is summarised in Table 3.1.

Table 3. 2: Sample Variables Available on Rental Datasets

Feature	**Type**	**Description**
Rental Price (KES)	Numeric	Monthly asking rent
Location Coordinates	Categorical	GPS coordinates (longitude, latitude)
Bedrooms	Numeric	Number of bedrooms
Amenities Score	Numeric	Derived score from amenities listed
Property Type	Categorical	Apartment, bedsitter, maisonette

3.4.1 Sampling Plan

To train and validate the ML model, at least 500 rental entries were to be collected to represent the distribution of formal rental housing across Nairobi. Nairobi City County accounts for about 66.7

percent of all residential properties in the country and has the largest share of rental units, highlighting its dominance as Kenya's primary housing market (Kenya National Bureau of Statistics, 2024) . In this study, the focus was on units suitable for middle- and upper-income households, such as apartments, maisonettes, and standalone houses. Listings were drawn from multiple neighborhoods across the city to capture a realistic distribution of formal rentals.

The target population for this study comprised of all formal rental housing units in Nairobi County, estimated at approximately 800,000 units according to the Kenya National Bureau of Statistics (Kenya National Bureau of Statistics, 2024). From this population, a purposive sample of 580 verified rental listings were selected from platforms such as Jiji and KenyaPropertyCenter.

The purposive approach is appropriate because it deliberately focused on listings that met the study's objectives which are those with complete information on rental price, property type, number of bedrooms and location coordinates. This method ensured inclusion of diverse housing categories that accurately represented the structure of Nairobi's formal rental market.

From Yamane's simplified sampling formula:

$$n = \frac{N}{1 + N(e)^2}$$

Where:

a. n = sample size

b. N = population size (800,000 units)

c. e = desired level of precision (0.05 or 5%)

Substituting the values gives:

$$n = \frac{800{,}000}{1 + 800{,}000(0.05)^2} = \frac{800{,}000}{1 + 2{,}000} \approx 400$$

A sample of approximately 400 records would therefore be statistically sufficient at a 95% confidence level. However, the study adopted a sample of 500 records to provide a safety margin that accounts for possible data loss during cleaning and model validation with the actual sample being 580 records.

The sample also meets the 10 × k rule recommended for machine-learning model development, where *k* represents the number of independent variables which are the predictors used. Given approximately 40–50 predictor features, a minimum of 400–500 observations are required to ensure robust model training and generalization. This sampling strategy guarantees that the collected data are scientifically representative, sufficient for predictive-model validation and aligned with established statistical and machine-learning adequacy standards.

Duplicate and incomplete housing entries were excluded to maintain data quality. Drawing on data in (Kenya National Bureau of Statistics, 2023), Kenya National Bureau of Statistics found that a significant proportion of urban tenants in Nairobi pay below KSh 20,000 per month, typically for smaller formal units such as bedsitters and studios located in estates like Embakasi, Zimmerman, and Umoja. Consequently, this study allocated about 100 rental entries to this category to ensure adequate representation of affordable formal housing, which remains underrepresented in digital listings but critical for affordability analysis. The middle range, covering rentals between KSh 20,000 and 50,000, included roughly 200 units and represents the largest portion of the formal market. KNBS data show that rents for one- to three-bedroom urban apartments generally range from KSh 15,000 to 45,000, reflecting the cost patterns for middle-income earners who dominate Nairobi's rental demand (Kenya National Bureau of Statistics, 2023). Finally, the upper range, comprising rentals above KSh 50,000, consisted of about 200 units to capture higher-end properties such as maisonettes and townhouses. This classification captures a realistic cross-section of formal rentals while excluding informal or unregulated housing. It ensures that the dataset covers affordable, yet desirable housing options commonly occupied by working- and middle-class residents, without skewing the sample toward either low-cost or luxury extremes.

The Kenya National Bureau of Statistics House Rent and Consumer Price Index dataset provides benchmark information on average rental values across Nairobi, categorized by number of bedrooms, dwelling type, and sub-county location. This data guided the sampling thresholds and validation of the 580 collected records from digital platforms. Kenya National Bureau of Statistics median rent values served as reference points during data cleaning and verification; any listings that deviate significantly (±30%) from Kenya National Bureau of Statistics benchmarks will be flagged as potential outliers or inconsistencies. This integration ensures that the final dataset accurately represents Nairobi's formal rental housing distribution and strengthens the credibility of the predictive model by grounding it in nationally verified statistics. The final dataset closely aligned with this intended distribution, comprising 118 low-income, 192 middle-income, and 269 high-income listings, with slight variations reflecting the natural distribution of available rental data and minor overlaps between categories.

3.4.2 Inclusion and Exclusion Criteria

To ensure that the data used in this study was accurate, reliable, and representative of Nairobi's formal rental market, clear inclusion and exclusion criteria were established as follows:

Inclusion Criteria

The study included:

i. Verified rental listings obtained from recognized and credible online platforms such as BuyRentKenya, Property24, and other licensed real estate portals.

ii. Properties located within Nairobi County, covering both high- and low-income residential areas to capture price variations across sub-counties.

iii. Rental units with complete and verifiable data, including rent amount, location coordinates, property type, number of bedrooms, and available amenities.

iv. Residential units with one to three bedrooms, representing the most common formal rental housing categories in Nairobi.

v. Listings updated within the last twelve months, ensuring that data reflect current market trends.

Exclusion Criteria

The study excluded:

i. Incomplete listings lacking critical information such as rent amount, property type, or geographic location.

ii. Properties in informal settlements or unregulated housing units that lack verifiable or standardized rental information.

iii. Duplicate or inconsistent entries appearing on multiple platforms.

iv. Outlier data points whose rental prices deviate by more than ±30% from the benchmark values reported in the Kenya National Bureau of Statistics (KNBS, 2024) Housing Survey.

v. Commercial or non-residential properties, as the study focuses exclusively on residential rental units.

These criteria ensured that the dataset used for model training and evaluation was reliable, representative and suitable for the development of a machine learning–based rental prediction system.

3.5 Model and Framework Development

The mobile application integrated two main modules: the machine learning rent prediction engine and the augmented reality property visualization component. These two components were integrated into a single mobile platform accessible via Android smartphones.

3.5.1 Machine Learning Module

The predictive model used LightGBM (Light Gradient Boosting Machine), a powerful and efficient gradient-boosted decision tree algorithm known for its high performance with structured data. LightGBM operates by building decision trees sequentially, where each tree corrects the residual errors of the previous ones, gradually improving the overall prediction accuracy. It supports categorical features natively and uses a histogram-based learning method to speed up training while reducing memory usage.

The model was trained using supervised learning techniques, where the rental price is the target variable and features such as location, number of bedrooms, amenities score, and property type act as inputs (as shown in Table 3.1). During training, LightGBM minimizes a loss function, typically Mean Squared Error for regression tasks, while applying regularization to prevent overfitting. After training, the model predicted rental prices for new inputs based on patterns learned from the dataset. The objective function to minimize loss and apply regularization, typically expressed as:

$$L(\Theta) = \Sigma\, l(y_i, \hat{y}_i) + \Sigma\, \Omega(f_k)$$

Where:

a. $l(y_i, \hat{y}_i)$ is the loss function, such as mean squared error between actual and predicted values
b. $\Omega(f_k)$ is the regularization term penalizing model complexity
c. f_k represents each decision tree in the ensemble

This formula ensures the model balances fitting the training data well and generalizing to unseen data by preventing overfitting. The predictions are processed within the application and presented separately to users, while the AR module is used independently for spatial visualization of property images. The goal is to forecast fair rental prices and detect anomalies such as overpricing, underpricing or outlier trends that deviate significantly from the market norm. Figure 3.1 presents a visual overview of the proposed model development and deployment pipeline, illustrating how data flows from preprocessing to LightGBM prediction and finally to augmented reality visualisation on the mobile interface.

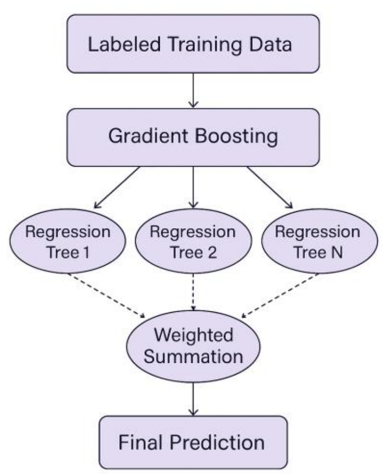


Figure 3. 1: Working process of LightGBM

The model development process includes:

i. Data cleaning and preprocessing

This foundational step involved handling inconsistencies and preparing the raw dataset for model training. Missing values which can be unspecified amenities were imputed using median values or removed where appropriate. Categorical variables such as property type and location were encoded using label encoding and one-hot encoding to make them suitable for input into the LightGBM algorithm, which handles categorical features efficiently. The data was also normalized or standardized where necessary to ensure uniform scaling, especially for continuous features such as rent amount and property age.

ii. Feature selection and engineering

Feature selection was conducted to identify the most informative variables contributing to rental price prediction. Techniques such as correlation analysis and feature importance metrics such as Gini importance from preliminary LightGBM runs were used. Additionally, new variables which are engineered features were created — for instance, the amenities score, a composite metric aggregating binary indicators of key amenities like Wi-Fi, parking, security and proximity to transport. This improved the model's ability to capture latent patterns in the rental data and reduced noise from irrelevant variables.

iii. Model training and testing split

The dataset was split into a training set (80%) and a test set (20%) using stratified sampling to maintain the distribution of price ranges across the subsets. The model was trained on the training set using supervised learning, where the LightGBM algorithm learned to map input features to the rental price. The 20% test was reserved for final evaluation to ensure the model generalized well to unseen data. Table 3.2 provides a practical illustration of how stratified sampling was applied to maintain a consistent distribution of rent price categories across both the training and test datasets.

Table 3. 3: Stratified Sampling Example of Rental Price Ranges

Price Range (KES)	**Frequency in Dataset**	**Proportion (%)**	**Sample in Train Set**	**Sample in Test Set**
< 20,000	100	20%	80	20
20,000 – 50,000	200	40%	160	40
> 50,000	200	40%	160	40
Total	**500**	**100%**	**400**	**100**

iv. Hyperparameter tuning through cross-validation

To optimize model performance, hyperparameters such as the number of leaves, learning rate and boosting rounds were fine-tuned using k-fold cross-validation (typically with k=5). In this method, the training set was split into k parts, and the model trained k times, each time using a different fold as the validation set and the rest for training. This process ensured robustness and reduced overfitting by validating the model across multiple partitions of the data. Grid search and random search techniques were used to explore the optimal hyperparameter combinations. Table 3.3 summarizes the key phases of the model development process and outlines the specific tasks undertaken at each stage to ensure efficient and accurate rent prediction.

Table 3. 4: Model Development Pipeline and Associated Tasks

Phase	**Description**
Data Preprocessing	Cleaning, imputing missing values, encoding categorical variables

Feature Engineering	Creating derived features like amenities score and interaction terms
Model Training (LightGBM)	Learning rental price predictions using decision tree-based boosting
Cross-Validation	Tuning hyperparameters and assessing performance using stratified k-fold
Model Evaluation	Measuring RMSE, MAE, and R^2 on holdout data
Output Integration	Feeding predictions into the AR visualisation component

3.5.2 Augmented Reality Module

The AR module provides immersive visualisation of rental units by overlaying digital elements onto real-world environments using mobile phone cameras. Features include:

i. 3D room previews using markerless tracking

ii. Location-based overlays of price estimates

iii. Interactive property tours with spatial dimension annotations

Development leveraged ARCore (for Android) and Unity 3D, which support real-time rendering and are compatible with most mid-range smartphones in Kenya. The AR module specifically uses markerless tracking, for instance visual-inertial techniques that leverage camera input and motion sensors (VIO) to place and align virtual content without requiring physical markers. This approach is supported by recent advancements in mobile AR, such as those described by Liu, Zhao and Braud (2023; 2024), who propose robust visual-inertial odometry frameworks like MobileARLoc to enhance localization and reduce drift in markerless AR applications.

While augmented reality (AR) applications in real estate can incorporate fully three-dimensional (3D) models and virtual walkthrough capabilities, the implementation in this study adopted a lightweight, image-based approach suitable for mobile deployment. The AR module focused on overlaying property images and descriptive information within the user's physical environment using surface detection and spatial anchoring.

This design choice ensured efficient performance, reduced computational requirements and maintained compatibility with a wide range of mobile devices, while still achieving the objective of enhancing spatial understanding of property information. Interaction is therefore achieved through device movement, allowing users to view the anchored content from different perspectives.

3.5.3 Integration Layer and Application Workflow

The integration layer is the middleware that ensures seamless communication and data flow between the ML rent prediction engine, AR visualisation module and the user-facing mobile interface. It plays a critical role in coordinating responses, synchronizing outputs and managing real-time interactions.

Firebase was employed as the backend service due to its lightweight integration with mobile applications, real-time database capabilities and built-in user authentication. When a user launches the mobile app and selects a location or inputs housing preferences, the data is captured via a form or interactive filter interface within the app and sent to Firebase's cloud functions, which act as the bridge to the ML model.

Once received, the user data is processed and sent to the LightGBM model, which runs server-side to ensure scalability and performance. The model computes a predicted rental price based on the provided inputs such as number of bedrooms, amenities score and location. The prediction is returned through Firebase to the frontend of the application.

The predicted rent value is then handed over to the Unity + ARCore engine, which integrates the value within the visual context. For example, the rent value may be overlaid on a virtual 3D walkthrough of a property, shown next to specific rooms or placed above a building when viewed through the AR lens. This gives the user a spatially contextualized understanding of rental pricing, improving transparency and decision-making.

The integrated application therefore comprises three synchronized workflows:

i. User Input → Firebase → LightGBM → Predicted Rent

ii. Predicted Rent → ARCore Visual Layer (Unity) → On-screen AR overlay

iii. User Feedback Loop → Selection/Adjustment → Re-query ML Model

This architecture ensures modularity, responsiveness and extensibility.

The structure and interaction of these modules is shown in Figure 3.1.

3.6 Choice of Simulation and Development Tools

The research tools selected, and technologies was guided by the methodological requirements of the study—particularly, the development, testing, and validation of a mobile application integrating machine learning (ML)–based rent prediction and augmented reality (AR)–based visualization. Each tool was selected to ensure compatibility, reproducibility and efficient workflow management across the machine learning and mobile development phases.

1. **Programming and Model Development Environment**

The study employed Python (version 3.11) as the primary programming language because of its rich ecosystem for data analysis and machine learning. The following Python libraries and frameworks were used:

a. **Scikit-learn:** Will be for implementing baseline regression models and evaluating performance metrics such as RMSE, MAE, and R^2.

b. **LightGBM:** For developing the primary gradient-boosted decision-tree model optimized for structured rental datasets.

c. **Pandas and NumPy:** For data cleaning, transformation, and numerical computation.

d. **Matplotlib and Seaborn:** For graphical representation of model performance and data visualization.

All Python coding, training, and experimentation were to be conducted in Google Colab, which provides a GPU-enabled, cloud-based environment suitable for high-performance computation and collaborative reproducibility. However, the development process was later transitioned to a local environment using a command-line interface and text editor. This shift provided greater control over file management, dependency configuration and execution workflows, while also reducing reliance on internet connectivity and session limitations associated with cloud platforms. The local setup enabled efficient data preprocessing, model training, and evaluation using libraries such as pandas, NumPy, and LightGBM.

2. **Data Management and Storage Tools**

Data extracted from secondary sources, which are KNBS datasets and verified rental platforms, were organized using SQLite and CSV repositories for lightweight local storage. Cleaning scripts were versioned using GitHub to maintain integrity and traceability throughout the development cycle.

3. **Mobile Application Development Tools**

The mobile prototype was developed using the Flutter SDK with the Dart programming language. Flutter provides a cross-platform environment that supports both Android and iOS deployment, ensuring accessibility to the target population. The Android Studio IDE was used for code compilation, debugging, and device emulation during app testing.

4. **Augmented Reality (AR) Development Tools**

The AR component was implemented using Unity 3D integrated with ARCore SDK (for Android) and ARKit SDK (for iOS). Unity's 3D rendering capabilities, physics engine and camera integration make it suitable for developing immersive property walkthroughs. The ARCore SDK enables markerless tracking, surface detection, and real-time environment mapping through Simultaneous Localization and Mapping (SLAM) techniques.

5. **Visualisation and Analytical Tools**

Interactive dashboards and summary charts were created using Tableau Public and Python Matplotlib to visualise rental distribution, price trends, and model predictions. These tools facilitate clear presentation of insights derived from the data and support decision-making analysis in Chapter 4.

6. **Version Control and Collaboration**

All scripts, datasets, and documentation were maintained under a GitHub repository, with version tracking for each stage of development—from data preprocessing and model training to app integration. This promotes transparency, reproducibility, and peer collaboration.

7. Testing and Development Tools

a. **Firebase** was used for backend data synchronization and basic authentication during the usability testing phase.
b. **Postman** was employed to test API endpoints that connect the machine-learning backend with the mobile front-end.
c. **System Usability Scale (SUS)** forms as seen in Appendix B were administered digitally using Google Forms or printed out to collect user feedback during validation.

In summary, these tools collectively ensured that the ML model can be efficiently trained and validated, the AR environment can be rendered in real time on mobile devices, and the overall system can be developed, tested, and maintained under academic research standards. A detailed supporting document was prepared alongside this study document. It outlines all the tools used in this study alongside their specific functions, software versions, configurations, and licensing information to ensure transparency, reproducibility, and ethical compliance in the research process. Table 3.4 summarizes the selected tools and their justifications.

Table 3. 5: Justification of Selected Tools

Tool	Purpose	Justification
Python	ML model development	High accuracy libraries and open datasets
Android Studio	App development	Kotlin is the recommended language for modern Android development due to its conciseness and full support by Google

Unity & ARCore	AR visualisation	Combines immersive rendering with real-time spatial mapping
Firebase	Cloud storage and sync	Real-time updates and cross-device access
SQLite	Local storage	Offline access and fast local querying

3.7 Validation and Data Analysis

This section outlines the procedures that were to be used to validate the developed prototype, focusing on whether the system meets its intended purpose of providing accurate rent predictions and delivering a usable AR-based property visualization experience.

3.7.1 Test Environment and Setup

Validation will be conducted using both technical and user-centred approaches:

a) **Hardware and Software:** Validation was carried out primarily on the researcher's mid-range Android smartphone that supports ARCore, reflecting the type of device commonly available to Nairobi users. This ensured consistency of results across all participants.

b) **Software stack:** The prototype was deployed via Android Studio, with ML predictions executed through LightGBM and AR rendering supported by Unity 3D + ARCore.

c) **Data Sources:** A hold-out dataset of Nairobi rental listings, separate from the training set, was used for validation to ensure independence from model development.

3.7.2 Validation of the Machine Learning

The rent prediction engine was validated for accuracy and generalization using the following strategies:

1. **Cross-Validation**: K-fold cross-validation (k=5) was used during model training to prevent overfitting and ensure robustness.

2. **Hold-out Validation**: A 20% subset of the dataset was reserved for final validation. Predictions from this set were compared against actual rental prices.

3. **Performance Metrics**: Validation was be quantified using:

    a. Root Mean Squared Error (RMSE) – penalizing large errors.

    b. Mean Absolute Error (MAE) – showing average deviation in Kenyan Shillings.

    c. Coefficient of Determination (R^2) – measuring explanatory power.

4. **Acceptance Thresholds**: The model will be considered valid if it achieves RMSE and MAE values within ±KSh 2,000 (or ≤1.5% of actual rents) and an R^2 value ≥0.8.

3.7.3 Validation of AR Visualisation Module

The AR module was validated quantitatively using the **System Usability Scale (SUS)**, a widely accepted and standardized instrument for measuring perceived usability.

a. **User Sessions**: A purposive sample of 15–20 Nairobi tenants interacted with the prototype on the researcher's ARCore-enabled smartphone.

b. **SUS Questionnaire**: After completing interaction tasks such as exploring property layouts, viewing rent overlays, each participant completed the 10-item SUS questionnaire, rating ease of use, clarity, and overall satisfaction on a Likert scale.

c. **Scoring**: Individual SUS responses were converted to a 0–100 scale where Odd-numbered items were scored as (Scale Value − 1) and even-numbered items as (5 − Scale Value), with the total multiplied by 2.5 and the mean SUS score across all participants will represent the usability rating of the AR module.

d. **Validation Criteria**: The AR module was considered valid if the mean SUS score is **≥70**, indicating good usability by industry standards.

3.7.4 Validation of the Integrated Prototype

The integrated ML + AR prototype was validated using a combination of technical benchmarking and user usability scores:

1. **Market Benchmarking**: Predicted rental prices were compared against actual Nairobi rental data to confirm accuracy of the ML predictions.

2. **SUS-Based Usability Validation**: The overall system (rent prediction + AR visualization) was validated through the same SUS instrument, measuring whether users find the combined features effective and usable in supporting rental decision-making.

3. **Validation Criteria**: The integrated prototype was considered valid if:

    a. ML predictions achieve RMSE and MAE within ±KSh 2,000 and $R^2 \geq 0.8$.

    b. The overall SUS score ≥70, showing that end users perceive the app as usable.

3.7.5 Data Analysis of Validation Results

Validation results were analyzed quantitatively:

a. **ML Module**: Predictions were validated using RMSE, MAE, and R^2 metrics on the hold-out dataset.

b. **AR Module & Integrated System**: Usability validated through SUS scores. Scores were benchmarked against SUS interpretive ranges (e.g., ≥70 = Good usability, ≥85 = Excellent usability).

3.7.6 Acceptance Criteria

The prototype was considered valid if:

a. **ML Module**: RMSE and MAE ≤ KSh 2,000 (or ≤1.5% of actual rents), and $R^2 \geq 0.8$.

b. **AR Module**: Mean SUS score ≥70.

c. **Integrated System**: Mean SUS score ≥70 and ML predictions align with Nairobi rental market benchmarks.

3.8 Ethical Considerations

The study fully adhered to ethical codes of conduct by ensuring that no sensitive personal information was collected and that all analysis was based solely on publicly accessible secondary datasets. The study respected local cultural norms regarding property ownership, privacy, and gender sensitivity in housing. Results are used solely for academic and technological improvement of rental transparency and not for commercial exploitation. User data was anonymized, respecting community perceptions and digital ethics. For participants who are unable to read or write, the System Usability Scale (SUS) questionnaire was to be read out loud to them in a language they understand best.

The dataset was gathered using open-source web crawling tools from publicly listed rental platforms and government housing data, ensuring no personally identifiable data is used. Additionally, no modifications were applied to the open-source algorithms used and all implementation efforts remained within the academic and ethical boundaries of transparency, reproducibility and responsible data use. All participants are to benefit equally through access to the validated application during pilot testing. Findings are to be shared with community housing groups, local innovation hubs, and county digital innovation offices to ensure equitable access. No participant was denied information or technical benefit arising from this study. The tools and instruments used, including the System Usability Scale (SUS) questionnaire and supporting software environments, are described in a support document alongside this study document for review and ethical verification. Ethical clearance was sought from the Strathmore University Institutional Ethics Review Committee (SU-IERC) before the research commenced.

3.9 Research Quality Assurance

To ensure the research remained valid, reliable and credible, the following quality assurance measures were implemented:

i. Trusted Data Sources: Training and evaluation of data was obtained from reputable sources such as KNBS and leading property platforms to ensure accuracy.
ii. Multi-Metric Model Evaluation: RMSE, MAE, and R^2 metrics were used to evaluate ML model performance, validated with cross-validation to avoid overfitting.
iii. Supervisor Reviews: Iterative feedback was obtained from my academic supervisor.

iv. Clear Documentation: All processes, datasets, tools and implementation details are recorded to ensure transparency, repeatability and replicability.

These strategies align with contemporary frameworks for quality assurance in AI-based systems as discussed by Felderer and Ramler (2021).

3.10 Research Budget

This part presents the estimated budget required to successfully conduct the research study. The budget covered key project components such as data collection, prototype testing, report preparation, and contingency expenses. The detailed breakdown is provided in Appendix C.

3.11 Research Implementation Timeline

This section outlines the planned schedule for conducting all research activities from proposal approval to final submission. The comprehensive timeline was presented in the form of a Gantt chart in Appendix D. The chart shows the sequence and duration of each research activity to ensure the project is completed within the stipulated university deadlines.

3.12 Summary

This chapter has outlined the research methodology for developing a mobile application that integrates machine learning (ML) and augmented reality (AR) to support rental decision-making in Nairobi. The design science research approach was adopted, supported by structured data collection procedures, appropriate system development tools, a quantitative validation framework, and ethical safeguards. A research budget and timeline were also discussed about in this chapter. The ML rent prediction model was validated using performance metrics such as RMSE, MAE, and R^2, while the AR visualization module and integrated prototype was validated through the System Usability Scale (SUS). By combining technical accuracy with user-centered usability assessment, the methodology ensured that the resulting prototype is both reliable and practical. Ultimately, the study sought to deliver a validated solution that enhances affordability, transparency, and access to quality rental housing information in urban Kenya. The next chapter presents the system analysis and design specifications for the proposed solution.

3.13 Dissemination of Study Results

The results and outcomes of this research will be disseminated through multiple academic and community channels to ensure transparency, accessibility, and equitable sharing of benefits. A final copy of the project report and system prototype will be deposited in the Strathmore University Digital Repository for academic reference and future research use.

Findings will also be presented at university research seminars, academic conferences and shared as open-access technical reports to allow other scholars and practitioners to benefit from the study.

Participants who take part in the usability testing will be informed of the results through email or WhatsApp summaries once the evaluation is complete. In addition, policy briefs summarizing the key insights will be shared with the Nairobi County Department of Housing, relevant digital innovation offices, and community housing groups to promote the practical use of the findings in improving rental housing services and supporting digital transformation in the housing sector.

This approach ensures that the benefits of the research are distributed and accessible to both academic and non-academic stakeholders.

Chapter 4: System Analysis and Design

4.1 Introduction

This chapter provides detailed system analysis and design for the proposed study. From this chapter, there is a translation of the conceptual framework and methodological decisions presented in Chapters 2 and 3 into a technically coherent and implementable system blueprint. In line with Design Science Research (DSR) principles, the chapter focuses on defining system requirements, architectural design, component interactions, and data structures that collectively support the development of a functional prototype. The chapter is organized into system requirements analysis, system architecture, system design models (UML diagrams), data design, user interface design, and the integration of machine learning and augmented reality components.

4.2 System Requirement Analysis

System requirement analysis establishes a clear understanding of what the proposed system must accomplish and the conditions under which it must operate. In ML-driven systems, requirement specification is particularly important because predictive accuracy alone does not guarantee usability or real-world adoption (Huang et al., 2021). System requirement analysis establishes a clear understanding of what the proposed system must accomplish and the conditions under which it must operate. Requirements have been derived from the study objectives, the problem statement, gaps identified in existing rental platforms, and user needs within the Nairobi rental housing context. Both functional and non-functional requirements are defined to ensure that the system meets operational, technical, and usability expectations.

4.2.1 Functional Requirements

Functional requirements describe the specific services and capabilities that the system must provide to its users.

i. The system should allow users to enter property attributes, including location, house type, number of bedrooms, floor area, and available amenities.
ii. The system should preprocess user-input property data to ensure compatibility with the trained machine learning model.
iii. The system should predict a fair rental price using a trained Light Gradient Boosting Machine (LightGBM) model.

iv. The system should display the predicted rental price together with relevant property details and pricing feedback.

v. The system should allow administrators and landlords to enter the quoted rental price for each listed property.

vi. The system should compare the quoted rental price with the machine learning–predicted rental price.

vii. The system should classify rental prices as underpriced, fairly priced, or overpriced based on predefined comparison thresholds.

viii. The system should support property visualisation through augmented reality using approved property images and digital visual assets.

ix. The system should render property visual content in an augmented reality (AR) view within the mobile application.

x. The system should allow administrators and landlords to upload property images or capture visuals using the device camera for use in property visualisation.

xi. The system should allow administrators to review, manage, and approve uploaded property images and visual assets.

xii. The system should ensure that only administrator-approved property media are used in the visualisation module.

xiii. The system should display the property visualisation together with the quoted rental price and predicted rental price.

xiv. The system should allow users to interact with the AR visualisation by moving the mobile device to view anchored property visual content from different angles and perspectives.

xv. The system should store prediction results, quoted prices, price classifications, and user interaction logs for evaluation and analysis purposes.

xvi. The system should allow administrators to update rental datasets.

xvii. The system should support model updating and redeployment when new rental data becomes available.

These functional requirements ensure that the system directly addresses the challenges of rental price uncertainty and limited property visualization faced by tenants in Nairobi.

4.2.2 Non-Functional Requirements

Non-functional requirements define the quality attributes and constraints of the system. Recent studies emphasize that non-functional requirements such as usability, performance, and reliability are central to the acceptance of intelligent systems, especially mobile applications integrating machine learning (Fred & Ru, 2024; International Organization for Standardization & International Electrotechnical Commission, 2023). Non-functional requirements define the quality attributes and constraints of the system.

i. **Performance:** The system should generate rental predictions within a response time of less than three seconds under normal operating conditions.

ii. **Usability:** The mobile application should be easy to learn and use, targeting a System Usability Scale (SUS) score of at least 70.

iii. **Reliability:** The system should provide consistent predictions and stable AR visualization without frequent crashes or failures.

iv. **Scalability:** The system architecture should support future expansion to additional urban areas beyond Nairobi.

v. **Security:** Access to backend services should be controlled, and sensitive system components should be protected from unauthorized access.

vi. **Compatibility:** The mobile application should be compatible with Android devices that support ARCore.

vii. **Maintainability:** The system should be modular to facilitate updates to the machine learning model and AR components.

4.3 System Architecture

The system architecture defines the high-level structure of the proposed solution and the interaction between its major components. Layered and service-oriented architectures are widely recommended for machine learning systems because they allow independent evolution of data pipelines, models, and user interfaces (Jarrahi et al., 2023). The system architecture defines the high-level structure of the proposed solution and the interaction between its major components. A layered architectural approach has been adopted to promote modularity, scalability, and maintainability, consistent with architectures used in similar ML-based predictive systems.

4.3.1 Architectural Overview

The system comprises four primary layers:

i. **Presentation Layer:** This layer consists of the mobile application interface through which users interact with the system. It handles data input, result visualisation, and AR rendering.

ii. **Application Layer:** This layer manages business logic, request handling, and communication between the presentation layer and backend services.

iii. **Machine Learning Layer:** This layer hosts the trained LightGBM model responsible for rental price prediction.

iv. **Data Layer:** The data layer stores structured rental datasets, property attributes, quoted rental prices, predicted rental prices, price classification results, approved visual assets, and system interaction logs.

The separation of concerns across these layers enhances system robustness and allows independent evolution of individual components.

4.3.2 Component Interaction

When a user submits property details or performs a property search, the presentation layer forwards the request to the application layer via a RESTful API. The application layer validates the property attributes and quoted rental price provided by the administrator or landlord, then invokes the machine learning layer to generate a predicted rental price using the trained Light Gradient Boosting Machine (LightGBM) model. The predicted value is compared with the quoted price to determine whether the property is underpriced, fairly priced, or overpriced. Approved visual assets are then retrieved, and a three-dimensional house model is generated and rendered in the presentation layer as an integrated augmented reality (AR) visualization, displaying the house model together with the quoted price, predicted price, and price classification. Figure 4.1 outlines the system architecture design.

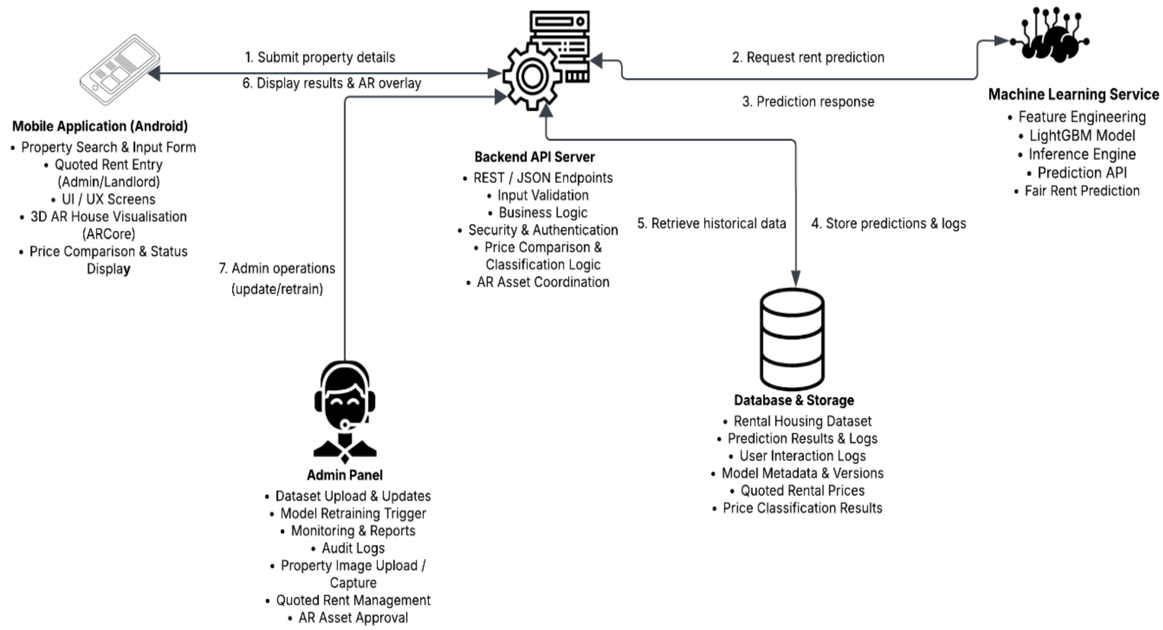


Figure 4. 1 : System Architecture

4.4 System Design Models

System design models provide visual representations of system behaviour and structure. Unified Modelling Language (UML) remains a widely accepted approach for documenting complex software systems, including ML-enabled applications, because it improves stakeholder communication and reduces design ambiguity (Kiratiratanapruk et al., 2022). System design models provide visual representations of system behavior and structure. Unified Modeling Language (UML) diagrams have been used in this study to model system interactions, workflows, and static relationships.

4.4.1 Use Case Diagram

The use case diagram identifies system actors and the functionalities they can perform.

Actors:

a. Tenant/User

b. Landlord

c. System Administrator

Use Cases:

Tenant/User

a. Search for property
b. View quoted rental price
c. View price comparison and classification (underpriced, fairly priced, or overpriced)
d. View and interact with the AR visualization of property images and details.

Landlord

a. Create and manage property listings
b. Upload or capture property visuals for AR visualization
c. Enter quoted rental price

System Administrator

a. Manage property visuals
b. Approve property images
c. Update rental datasets
d. Retrain and redeploy the prediction model
e. Monitor system performance and pricing trends

The use case diagram as shown in Figure 4.2 illustrates how different users interact with the system to achieve specific goals.

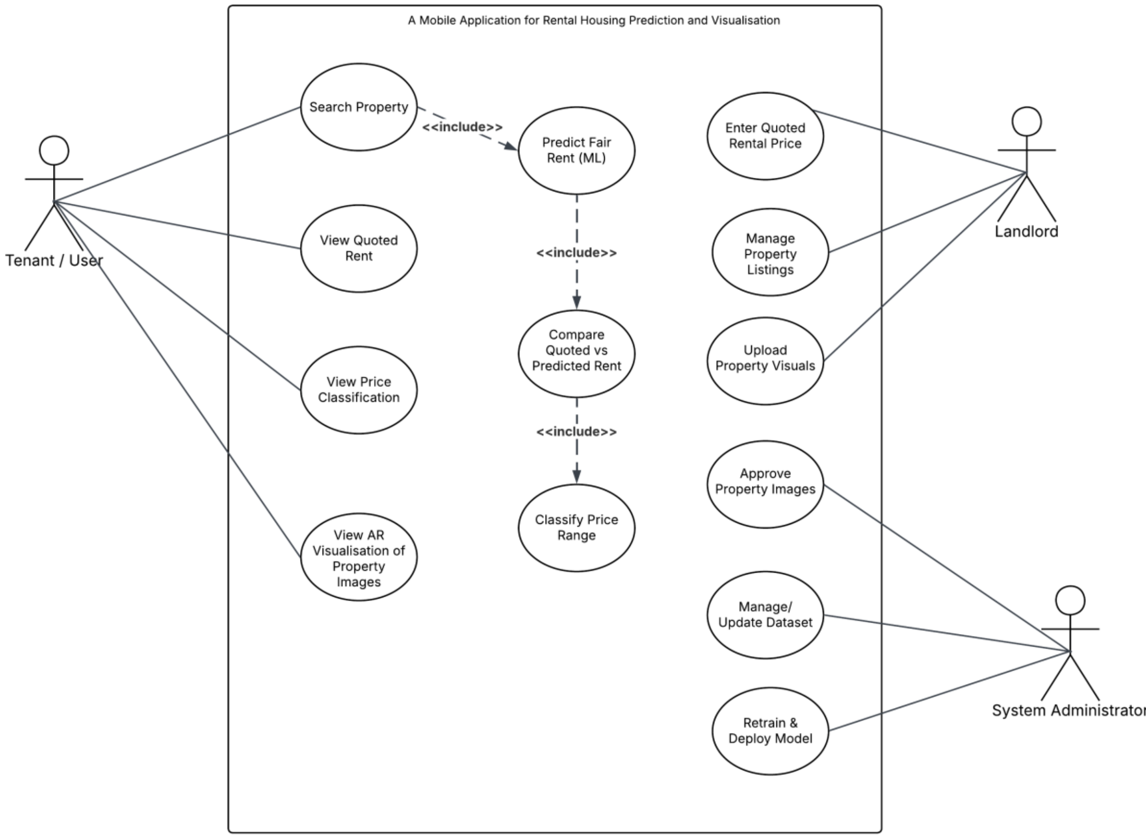


Figure 4.2: Use Case Diagram

4.4.2 Sequence Diagram

The sequence diagram depicts the dynamic interaction between system components during the rental prediction process.

Sequence Flow:

i. The tenant/user searches for a property and selects a listing within the mobile application.
ii. The landlord or administrator uploads/captures property visuals and enters the quoted rental price for the listing.
iii. The mobile application sends the property attributes to the backend API for processing.
iv. The backend validates the input data and forwards relevant property attributes to the machine learning model.
v. The machine learning model generates a predicted (fair) rental price and returns it to the backend.

vi. The backend compares the quoted rental price with the predicted rental price and determines a price classification (underpriced, fairly priced, or overpriced).

vii. The backend returns the price band, price classification and approved property visual assets to the mobile application.

viii. The mobile application renders the AR property visualisation and displays the quoted rental price, price band, and price classification to the tenant/user.

Figure 4.3 which is the sequence diagram, illustrates the chronological interaction between the system components involved in the rental price prediction and augmented reality visualization process.

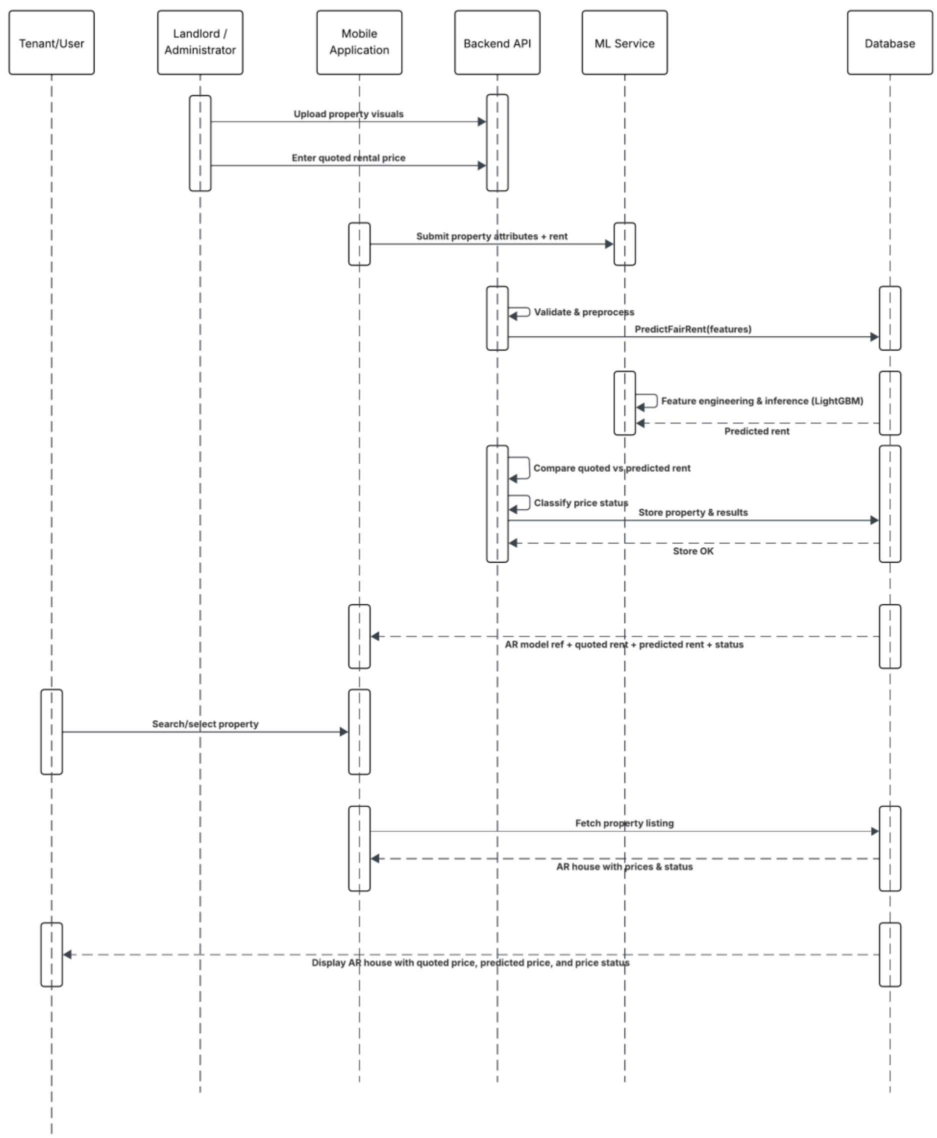


Figure 4. 3 : Sequence Diagram

4.4.3 Class Diagram

The class diagram represents the static structure of the system and relationships among system entities.

Key Classes and Responsibilities:

a. **User:** Stores user-related attributes and supports interaction with the rental prediction and AR visualization functions. Users are categorized by roles (tenant, landlord, administrator) to enable role-based access and functionality.
b. **Tenant**: Represents end-users who search for properties and view the AR house visualization together with quoted and predicted rental prices and price classification results.
c. **Landlord**: Manages property listings, uploads or captures property visuals, and enters the quoted rental price for each property listing.
d. **Property:** Represents property characteristics such as location, size, and amenities used as inputs for rental price prediction.
e. **PropertyMedia (ARAsset)**: Stores property visual assets such as uploaded images or captured visuals used to enhance AR house visualization. It includes metadata such as media type, storage reference, uploader role, and approval status to ensure that only verified assets are used in AR rendering.
f. **PredictionRequest:** Captures prediction requests associated with a selected property and tracks request status, timestamps, and relevant request metadata.
g. **PredictionModel:** Encapsulates the machine learning logic (LightGBM inference) used to process property features and generate predicted (fair) rental prices.
h. **PricingVerification (PriceComparator)**: Compares the quoted rental price with the machine learning–predicted price and determines the price classification whether underpriced, fairly priced, or overpriced based on defined thresholds.

i. **PredictionResult:** Stores prediction outputs, including predicted rental price, confidence score where applicable, and contextual metadata such as average market rent indicators.

j. **ARVisualisation:** Manages the rendering of the three-dimensional house model within the mobile application environment and supports user interactions such as rotating, zooming, and repositioning the AR house view. It retrieves approved media assets and pricing outputs to present an integrated visualization.

k. **Dataset:** Stores and manages historical rental data used for training and retraining the prediction model.

l. **Admin:** Performs system administration tasks including dataset updates, media approval, monitoring, and machine learning model retraining and redeployment. Administrators also manage and approve AR visual assets and may enter or validate quoted rental prices for listings.

Figure 4.4 presents the UML class diagram, illustrating the static structure of the system by showing the key classes, their attributes, methods, and relationships involved in rental price prediction and augmented reality visualization.

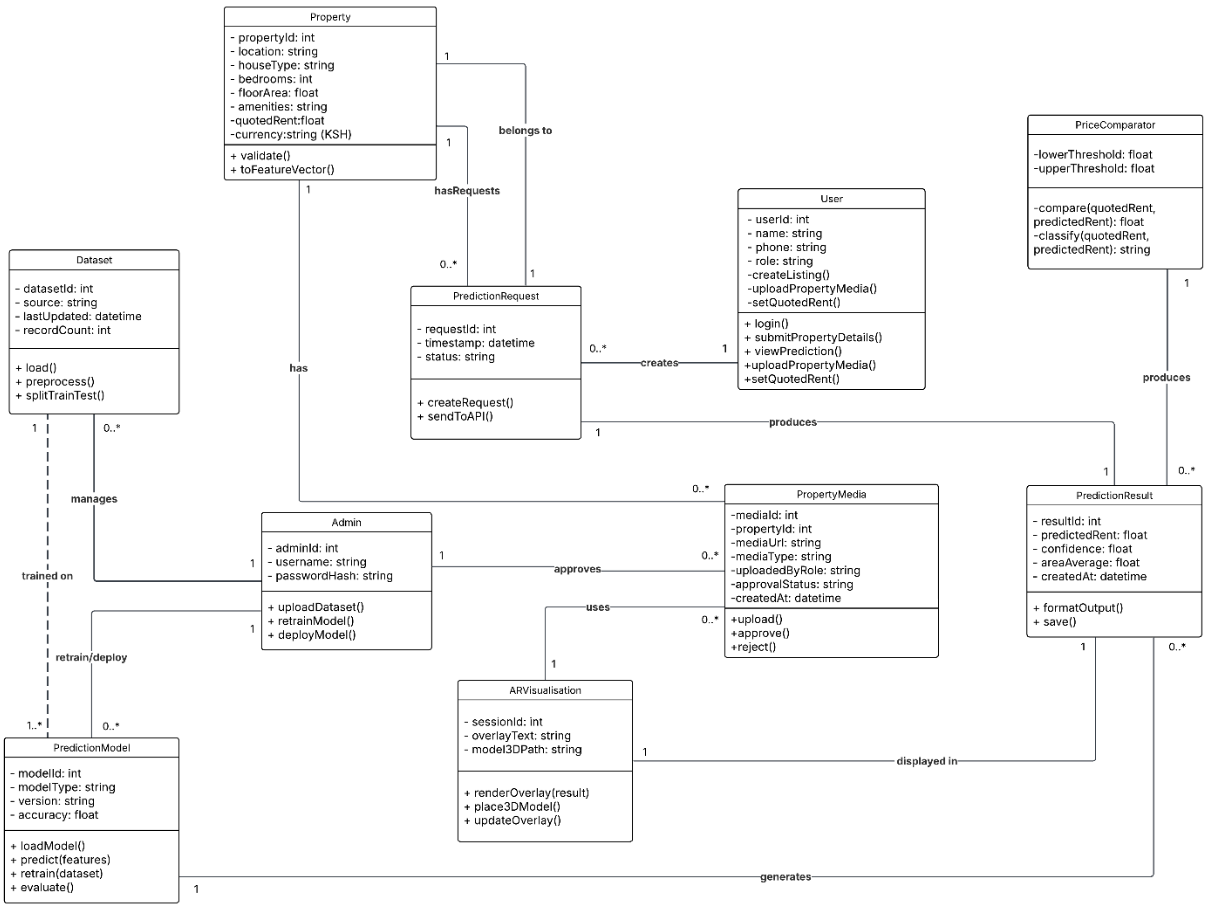


Figure 4. 4 : Class Diagram

4.5 Data Design

Data design defines how data is structured, stored, and accessed within the system. Recent research highlights that well-designed data schemas are essential for maintaining data quality, traceability, and reproducibility in machine learning systems (International Organization for Standardization & International Electrotechnical Commission, 2023). Data design defines how data is structured, stored, and accessed within the system.

4.5.1 Dataset Structure

The rental dataset used in this study is a structured tabular dataset comprising property-level attributes and corresponding rental price information. Key attributes include property location, house type, number of bedrooms, floor area, available amenities, and historical rental prices. These features are selected based on their relevance to rental price determination in housing market studies and their suitability for machine learning–based prediction.

The dataset serves as the foundation for training the rental price prediction model and for providing contextual benchmarks during system operation. During preprocessing, categorical attributes are encoded and numerical attributes are normalized to ensure compatibility with the machine learning model. The dataset is periodically updated by system administrators to reflect current market conditions, enabling accurate rental price prediction and supporting price verification by comparing quoted rental prices with predicted fair market values.

4.5.2 Database Schema

The database schema is designed to support the storage and management of key system data, including user accounts, property listings, property media, prediction results and user interaction records. The Property entity serves as the central table, storing core information about each rental listing and linking it to the user who uploaded it. Associated property images are stored in the PropertyMedia table.

In addition, the PredictionResult table stores rental price predictions generated by the machine learning model, while the InteractionLog table records user activities such as property views and prediction requests. Overall, the schema ensures data integrity, efficient data retrieval, and proper organisation of information required for the operation of the system. The database schema as shown in Figure 4.5 illustrates the structure of the system database and the relationships between the main entities used in the rental housing prediction and visualization system.

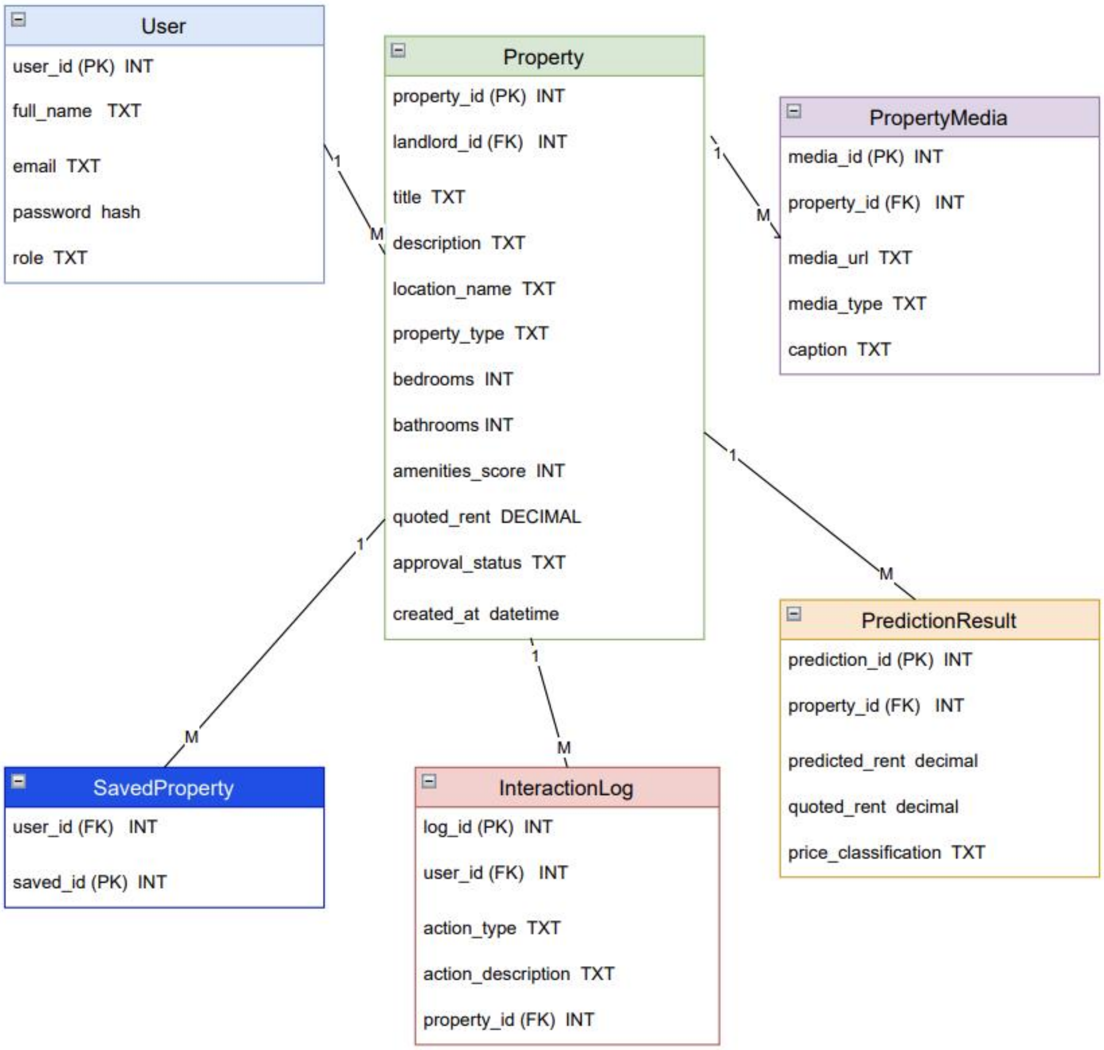


Figure 4. 5 : Database Schema of the Proposed Rental Housing Prediction and Visualisation System

4.5.3 Entity Relationship Diagram

The Entity Relationship Diagram (ERD) illustrates the logical structure of the database used in the proposed rental housing prediction and augmented reality visualization system. It shows the main entities in the system, their attributes and the relationships between them.

The User entity stores information about system users, including tenants, landlords and administrators. The Property entity contains details of rental listings such as location, house type, number of bedrooms, amenities and quoted rent. Each property is uploaded by a user and can have multiple associated media files stored in the PropertyMedia entity.

The PredictionRequest entity records requests made by users to obtain rental price predictions for specific properties. The prediction outcomes are stored in the PredictionResult entity, which contains the predicted rent and related prediction information. In addition, the ARSession entity records augmented reality interactions when users visualize properties within the application.

Overall, the ERD as seen in Figure 4.6 illustrates how these entities are connected to support the core functions of the system, including property management, rental price prediction, media handling, and AR-based property visualization.

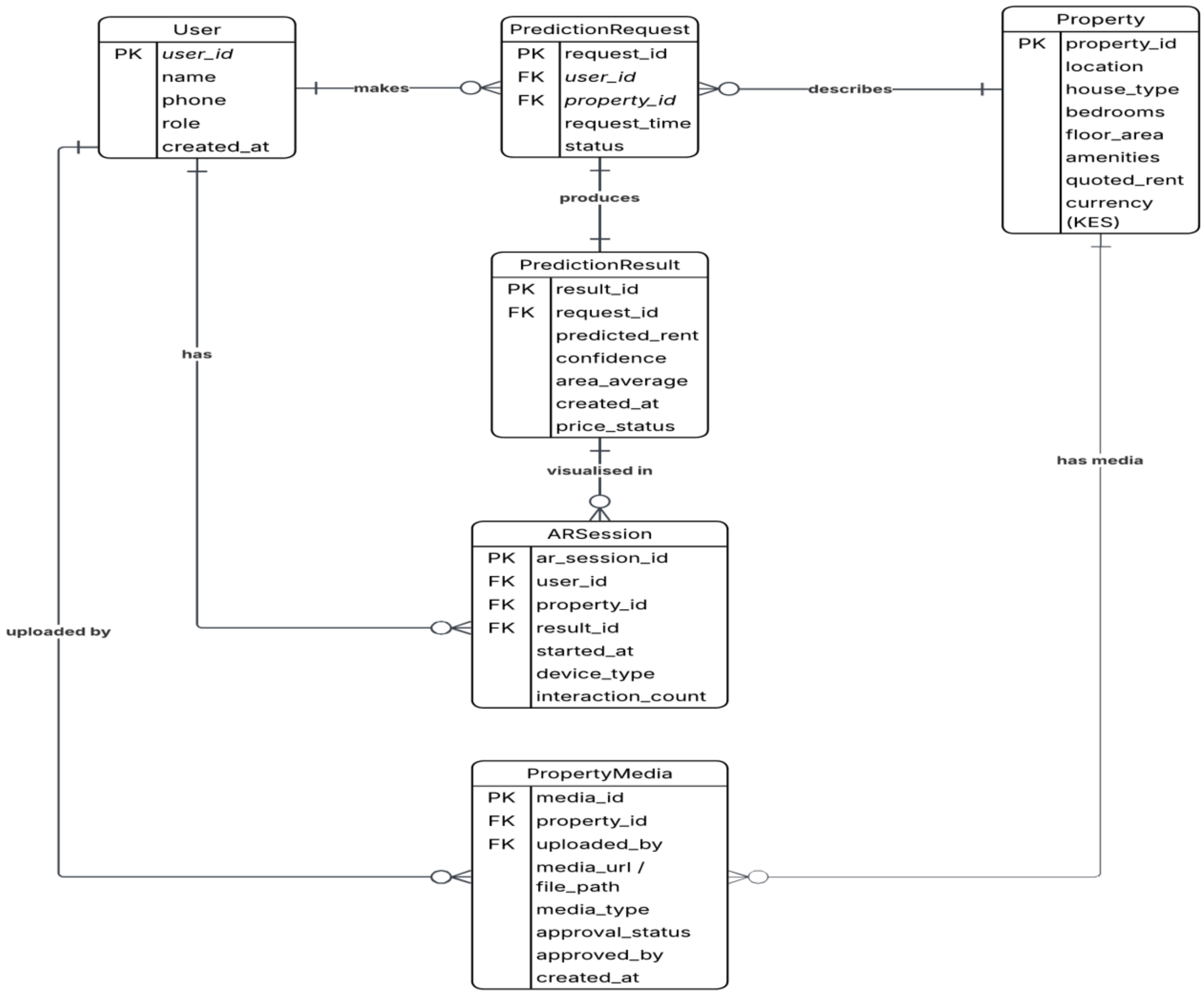


Figure 4. 6 : Entity Relationship Diagram

4.6 User Interface Design

The user interface was designed following human-centered design principles to enhance usability and accessibility. Studies on intelligent mobile systems indicate that user trust and perceived usefulness are strongly influenced by interface clarity and interaction design, particularly when predictive algorithms are involved (Asaad & Avksentieva, 2023; Brasoveanu et al., 2020). The user interface will be designed following human-centered design principles to enhance usability and accessibility.

4.6.1 Mobile Application Screens

Key screens include:

a. **Sign-In Screen** for user authentication and role identification, directing users to tenant- or landlord-specific interfaces upon successful login**.**

b. **Tenant Home** – Property Search and Listing Screen for browsing approved property listings and filtering results by location, house type, and number of bedrooms.

c. **Landlord Home –** Manage Properties Screen for viewing uploaded listings, monitoring approval status, and accessing the property upload function.

d. **Property Upload Screen (Landlord)** for entering property details, specifying the quoted rental price, and uploading or capturing property images for administrative approval.

e. **Property Details and Price Comparison Screen** for displaying comprehensive property information, uploaded visuals, the quoted rental price and the price classification outcome (underpriced, fairly priced, or overpriced).

f. **AR House Visualization Screen** where tenants view AR-enhanced property visuals together with property information and interaction features such as movement-based viewing, scaling, and repositioning.

Wireframes will be designed to ensure logical navigation, clarity of information, and minimal cognitive load on users.

a. Sign-In Screen Wireframe

Figure 4.7 presents the Sign-In Screen wireframe, which serves as the initial access point to the mobile application. The screen enables users to authenticate using their registered credentials and ensures secure access to system functionalities. Upon successful authentication, the system identifies the user's role and routes the user to the appropriate interface, either the tenant property browsing interface or the landlord property management interface.

The sign-in screen is designed to be minimal and intuitive, reducing user friction during authentication while supporting role-based navigation and access control within the application.

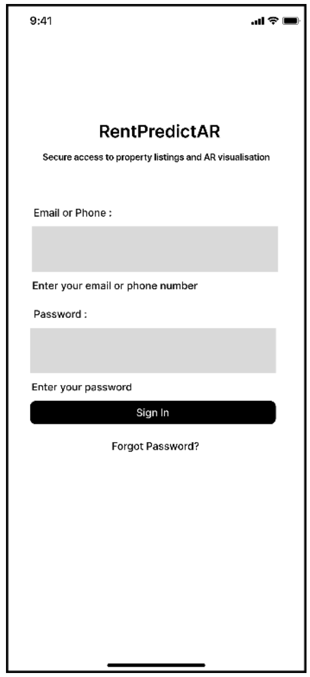


Figure 4. 7 : Sign-In Screen Wireframe

b. Home Screen Wireframe

Figure 4.8 illustrates the Home Screen wireframe, which serves as the primary landing interface after successful user authentication. The home screen provides role-based navigation, directing tenants and landlords to their respective core functionalities within the mobile application.

For tenants, the home screen offers clear access to property browsing and search functionality, enabling users to view available rental listings and initiate property selection. For landlords, the home screen provides access to property management functions, including viewing previously uploaded listings and navigating to the property upload interface. Common navigation elements such as user account access and saved items are also available from this screen.

The home screen is designed to be intuitive and uncluttered, ensuring logical navigation and quick access to key actions while guiding users toward informed decision-making through property exploration and rental price verification workflows.

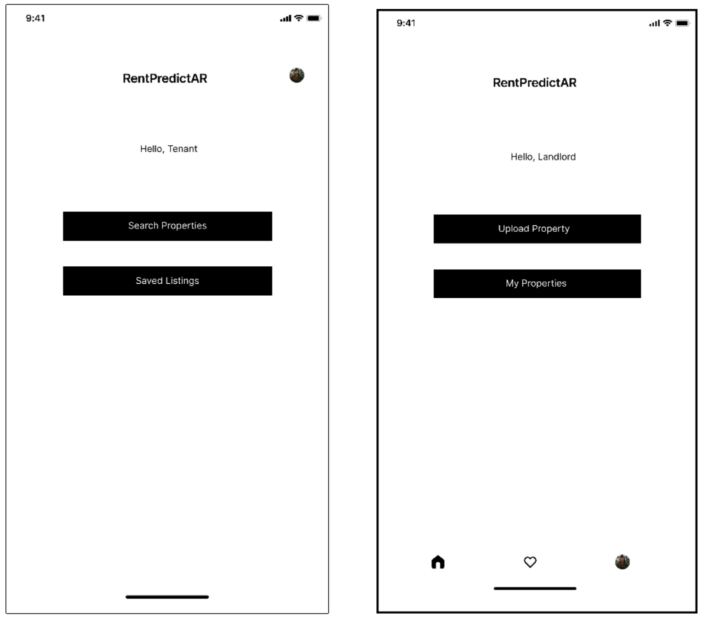


Figure 4. 8 : Home Screen Wireframe for Tenant and Landlord

c. PropertyUpload Screen Wireframe

Figure 4.9 shows the Property Upload Screen wireframe, which enables landlords to create and submit new property listings through the mobile application. The screen provides structured form fields for entering essential property attributes, including location, house type, number of bedrooms, floor area and available amenities, together with a mandatory field for specifying the quoted rental price. These attributes are captured and stored for use in rental price prediction and comparison processes within the system.

In addition to textual property details, the interface allows landlords to upload or capture property images using the mobile device camera. These images are stored as property media and are later used as visual assets within the augmented reality (AR) visualisation module. Uploaded images are submitted for administrative approval to ensure quality and authenticity before being made available for AR-based visualisation.

The property upload screen is designed to guide landlords through a clear and sequential listing creation process, ensuring that all information required for rental price verification and AR-based property visualisation is accurately captured.

.

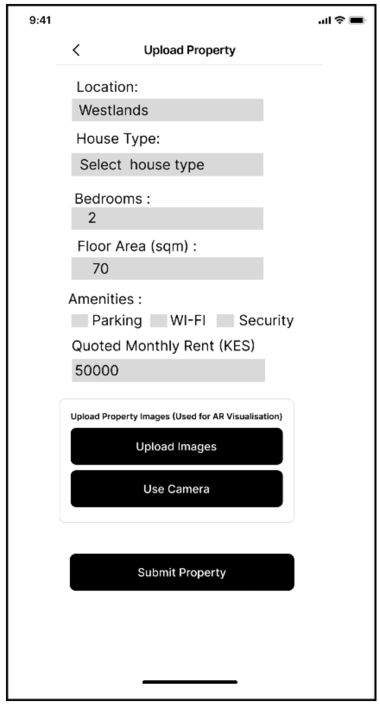


Figure 4. 9 : Property Upload Screen Wireframe

d. Property Details and Price Comparison Screen Wireframe

Figure 4.10 presents the Property Details and Price Comparison Screen wireframe, which is displayed when a tenant selects a property from the search results. This screen provides a comprehensive view of the selected property, combining descriptive information, uploaded visuals, and rental price analysis to support informed decision-making.

The screen displays key property attributes such as location, house type, number of bedrooms, floor area and available amenities, together with approved property images submitted by the landlord. The quoted rental price entered during property listing is shown alongside the predicted fair rental price generated by the machine learning model. To enhance clarity, the system highlights the price classification outcome—underpriced shown by orange colour, fairly priced shown by green colour, or overpriced shown by red colour—based on the comparison between the quoted and predicted values.

An action button is provided to allow the tenant to proceed to the augmented reality (AR) house visualization, where the property can be explored in three dimensions. The property details and price comparison screen is designed to clearly separate descriptive information from analytical results, ensuring clarity of information while enabling tenants to evaluate pricing fairness before engaging with AR-based visualization.

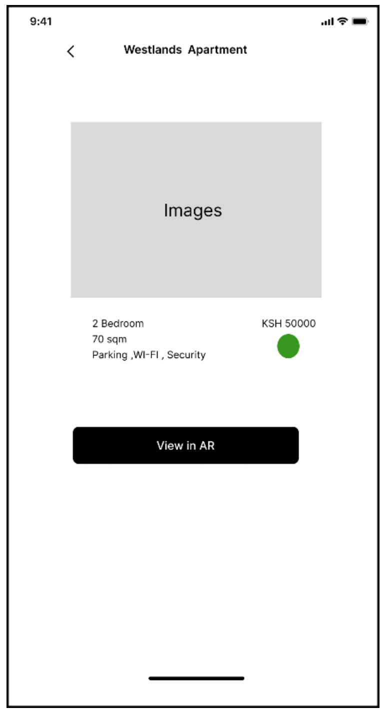


Figure 4. 10 : Property Details and Price Comparison Screen Wireframe

e. AR House Visualization Screen Wireframe

Figure 4.11 presents the AR visualization screen wireframe, which enables tenants to explore a selected property using augmented reality (AR). This screen is accessed after a property has been selected and its details reviewed, ensuring that the AR visualization is contextual and supports user understanding.

The AR interface overlays property images within the user's physical environment using the mobile device camera. Approved property images uploaded by the landlord are used directly as visual elements and are anchored onto detected flat surfaces within the environment. This allows the virtual content to remain fixed in position while responding dynamically to camera movement, enabling users to view the property from different perspectives.

The screen supports interaction through device movement rather than direct manipulation, allowing users to explore the spatial placement of the property image by moving around it. Property-related information such as descriptive details is displayed alongside the AR view to enhance user understanding.

The AR visualization screen is designed to complement property browsing by providing spatial context and improving user engagement, enabling tenants to better interpret property information before making rental decisions.

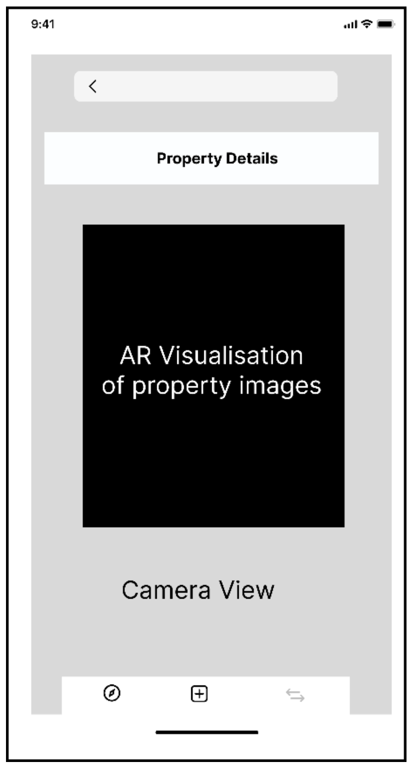


Figure 4. 11 : AR Visualisation of Property Images

4.7 Integration of Machine Learning and Augmented Reality

The integration layer connects the machine learning prediction module with the AR visualization module. Recent research demonstrates that combining predictive analytics with immersive visualization improves decision-making by enhancing contextual awareness and user engagement

(Billinghurst, 2021; Fombona-Pascual et al., 2022). In the proposed system, rental price predictions generated by the LightGBM model are used within the application to support rental analysis, while property metadata such as location, number of rooms and associated images are used for visualization purposes.

The system adopts an image-based AR approach, where property images uploaded by landlords are directly used as visual elements within the AR environment.

During runtime, the AR module uses ARCore to capture real-time camera input and apply Simultaneous Localization and Mapping (SLAM) techniques to detect flat surfaces within the user's physical environment. Once a surface is detected, users can place the property image onto the surface, where it is anchored within the real-world coordinate system. This ensures that the virtual content remains spatially fixed and responds dynamically to camera movement, thereby achieving spatial visualization.

Within the AR interface, only property-related information such as images and descriptive details are displayed. The rental price prediction results are presented separately within the application and are not directly rendered in the AR view. This design choice simplifies the AR experience and maintains clarity in user interaction.

The integration of machine learning and AR therefore allows the system to provide both predictive insights and spatial visualization within a single platform, while maintaining a clear separation between analytical outputs and visual interaction. Figure 4.12 presents the ARCore-based spatial visualization process.

Figure 4. 12 : AR Spatial Tracking and Visualisation Process

4.8 Chapter Summary

This chapter has comprehensively presented the system analysis and design of the proposed machine learning and augmented reality-based rental housing application. It detailed system requirements, architectural design, UML models, data structures, user interface considerations and component integration. The design choices adopted in this chapter are consistent with best practices in information systems design and machine learning system development, as documented in prior scholarly work. The outputs of this chapter provide a solid technical foundation for system implementation and testing, which are discussed in the next chapter.

Chapter 5: System Implementation and Testing

5.1 Introduction

This chapter presents the implementation and testing of the proposed mobile application that integrates Machine Learning (ML) and Augmented Reality (AR) for rental housing prediction and visualization in Nairobi. Guided by the methodology in Chapter Three and the system design in Chapter Four, this chapter describes how the proposed solution is translated into a functional prototype.

It outlines the implementation of the mobile application, the ML-based rent prediction engine, the AR visualization module and the backend integration layer. The chapter also reports on system testing and validation, where the predictive accuracy of the ML model and the usability of the AR-enabled application is evaluated. The findings presented in this chapter provide the basis for discussion in the subsequent chapter.

5.2 Development Environment and Tools

This section describes the hardware and software environment used in developing and testing of the proposed ML–AR mobile application. The selected tools and platforms are aligned with those outlined in Chapter Three and are chosen to support efficient development, integration, and validation of the system components.

5.2.1 Hardware Environment

System development and testing is conducted using a mid-range development laptop and an ARCore-supported Android smartphone. The development laptop provides sufficient computational resources to support mobile application development, data preprocessing, and machine learning experimentation.

The Android smartphone is used to deploy and evaluate the augmented reality visualization module under realistic usage conditions. Testing the AR module on a physical device enables validation of camera performance, surface detection, overlay stability and user interaction within real-world environments. This approach ensures that the system reflects the actual operating conditions of target users in Nairobi rather than relying solely on emulated environments.

Overall, the hardware environment is sufficient for prototype development, AR testing and preparation for machine learning integration.

5.2.2 Software Environment

The mobile application is developed using Android Studio, with Kotlin as the primary programming language. Android SDK components are used to implement user interface elements, activity navigation and access to device features such as the camera and sensors.

The machine learning and data preparation components are implemented using Python. Key libraries used include Pandas and NumPy for data cleaning and preprocessing, scikit-learn for preprocessing utilities and evaluation support and LightGBM for gradient boosting–based rent prediction modeling.

During the prototype phase, processed datasets are stored in CSV format to support rapid inspection, iterative refinement and seamless interaction between data preparation scripts and subsequent machine learning or backend components.

5.2.3 Development and Integration Tools

The development of the proposed rental housing prediction and visualization system required the integration of several development frameworks, programming libraries and third-party APIs. These tools enable the implementation of the mobile application, the machine learning prediction engine, the augmented reality visualization module and the backend data storage system. The selected tools are chosen based on their compatibility with mobile platforms, support for machine learning deployment, scalability and ease of integration with cloud-based data services.

Table 5.1 summarizes the main development tools, frameworks and APIs used in the implementation of the system together with their roles within the system architecture.

Table 5.1: Development and Integration Tools Used in System Implementation

Tool / Framework	**Purpose in the System**
Android Studio	Primary Integrated Development Environment (IDE) used for developing the Android mobile

	application, designing user interfaces, and managing application activities.
Kotlin Programming Language	Used to implement the mobile application logic which includes authentication, property browsing, AR module control, and interaction with backend services.
Python	Used for machine learning model development in areas of data preprocessing, feature engineering, and training of the rental price prediction model.
LightGBM Library	Machine learning framework used to train the rental price prediction model due to its efficiency, high accuracy and ability to handle structured tabular data.
Scikit-learn	Used for model evaluation, preprocessing utilities and performance metric calculation such as RMSE and MAE.
NumPy and Pandas	Used for data manipulation, cleaning, feature preparation and dataset transformation during machine learning model training.
Firebase Firestore	Cloud-based NoSQL database used for storing property listings, property images, landlord uploads and prediction results.
Firebase Authentication	Used for managing user authentication including login and registration functionality within the mobile application.
Firebase Storage	Used for storing property images uploaded by landlords and retrieving them for display in the application.
Google ARCore SDK	Will provide augmented reality capabilities including camera tracking, surface detection

	and placement of virtual rental information overlays in the real environment.
Google Play Services for AR	Enables ARCore functionality on supported Android devices and manages AR session compatibility.
Gradle Build System	Used for dependency management, application packaging, and automated building of the Android application.

The integration of these tools enables seamless communication between the machine learning model, backend database, mobile application interface and augmented reality module. The combined environment provides the necessary infrastructure to support data processing, model prediction, real-time visualization and user interaction within a unified system.

5.3 System Implementation

This section presents the implementation of the proposed rental housing prediction and augmented reality visualization system. The implementation translates the system design described in Chapter Four into a functional software system composed of a mobile application, a machine learning prediction model, an augmented reality visualization module and a backend data storage system.

The implementation process involves the development of the Android mobile application using Android Studio and Kotlin, the training of the rental price prediction model using Python and LightGBM and the integration of augmented reality functionality using Google ARCore. Firebase services are used to manage authentication, data storage and communication between system components.

The following subsections describe the implementation of the mobile application interface, the machine learning prediction engine, the augmented reality visualization module and the backend database system.

5.3.1 Mobile Application Implementation

The mobile application served as the primary interface through which users interact with the system. The application enabled tenants to browse available rental properties, landlords to upload property listings, and users to obtain rental price predictions and visualize rental information using augmented reality. The application is structured as a multi-activity Android application where each activity handles a specific system functionality.

The core application structure included activities for user authentication, property upload and browsing, rental price prediction and augmented reality visualization. Navigation between these activities is implemented using Android intents to ensure smooth user interaction across the system.

a. Authentication Interfaces

User authentication has been implemented to control access to system services. Firebase Authentication is used to manage user login and registration. During registration, users provide an email address and password which are validated and securely stored in the Firebase authentication system. After successful authentication, users are redirected to the application dashboard where they can access system features.

The authentication interfaces implemented in the application are shown in Figure 5.1.

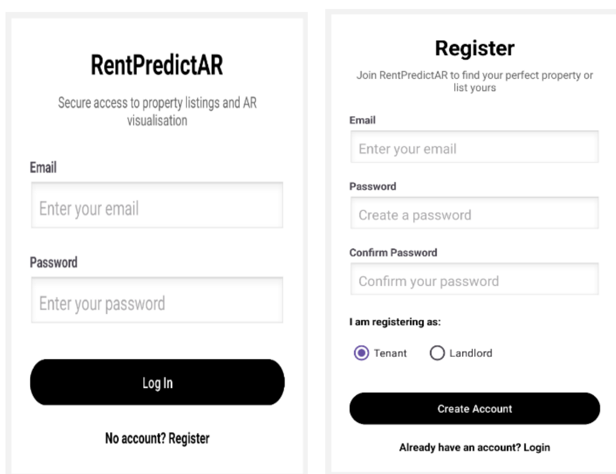


Figure 5. 1: Authentication Interfaces of the Mobile Application (Login and Registration Screens)

b. Property Management Interfaces

The property management component enables landlords to upload rental property listings and allows tenants to browse available properties. The property upload interface provided input fields for property attributes such as location, number of bedrooms, rental price, property description and property images. Uploaded property information is stored in Firebase Firestore while property images are stored in Firebase Storage.

Tenants can browse available properties through a list interface implemented using a RecyclerView component. When a property is selected, the system displays detailed information including property description, price, and images.

The implemented property management interfaces are illustrated in Figure 5.2.

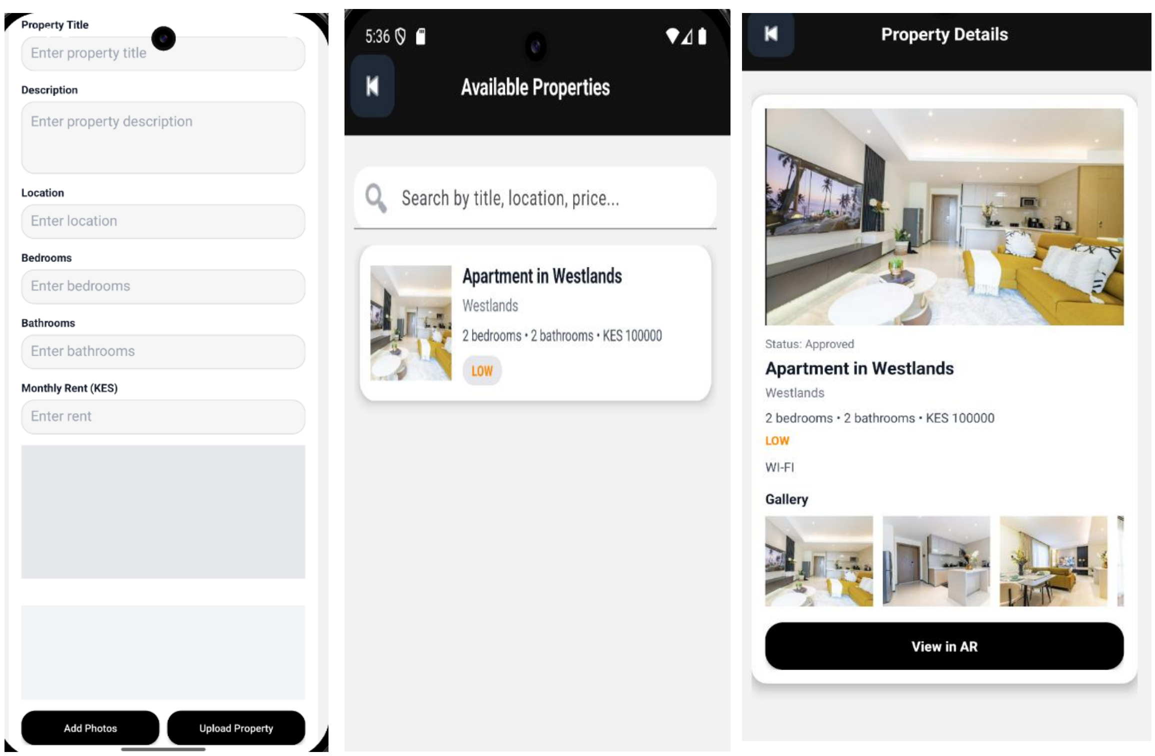


Figure 5.2 : Property Management Interfaces (Property Upload, Property List, and Property Details Screens)

c. Rental Price Prediction Interface

From the rental prediction module, estimation of the rental prices based on selected property features is conducted. This prediction module is in the property upload section where the interface collects key property attributes including location, number of bedrooms and number of bathrooms. These inputs are processed by the machine learning prediction module which generates an estimated rental value based on the trained LightGBM model.

After the prediction is generated, the system compares the predicted rental value with the rental price specified by the landlord. Based on this comparison, the system calculates a rent ratio and classifies the property into a rent band category such as underpriced, fair or overpriced.

The prediction interface implemented in the mobile application is presented in Figure 5.3.

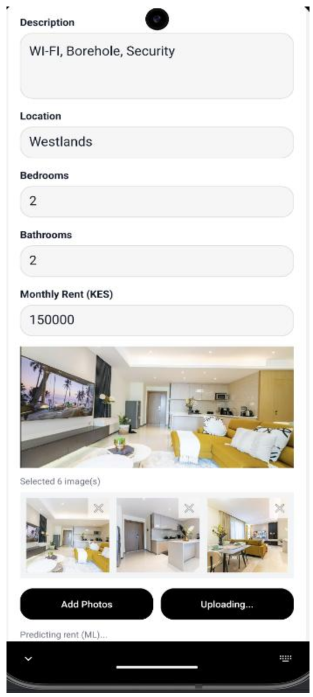


Figure 5. 3 : Rental Price Prediction Interface within the Mobile Application

d. Augmented Reality Visualization Interface

The augmented reality module enables users to visualize property information directly within the physical environment using the smartphone camera. The AR functionality is implemented using the Google ARCore SDK, which provides capabilities such as motion tracking, environmental understanding and virtual object placement.

When AR mode is activated, the device camera scans the surrounding environment to detect horizontal surfaces. Once a suitable surface is identified, the system places a virtual overlay displaying property information and property images within the user’s real-world environment, enabling spatial visualization of the rental property. This functionality enhances the user’s ability to better understand property characteristics and supports more informed rental decision-making.

The implemented augmented reality visualization interface is shown in Figure 5.4.

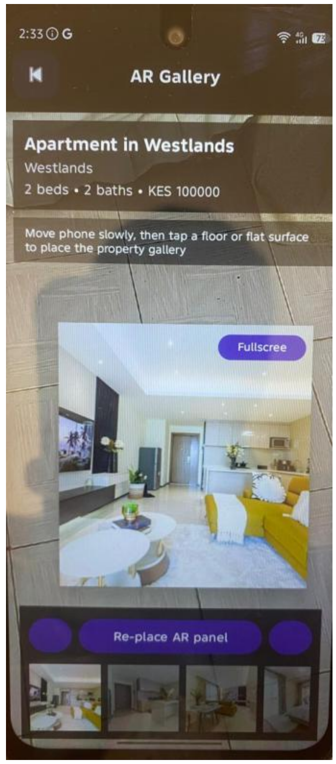


Figure 5. 4 : Augmented Reality Visualisation of Property Information

e. Resulting Mobile Application Functionality

Through the integration of these interfaces, the mobile application provided a complete workflow that enables users to authenticate into the system, browse available properties, upload property listings, generate rental price predictions and visualize property information using augmented reality. The mobile application therefore serves as the central interaction platform connecting the machine learning prediction engine, the backend database and the augmented reality visualization module.

5.3.2 Machine Learning Model Implementation

This section describes the implementation of the machine learning model used to predict rental housing prices within the proposed system. The objective of the model is to generate fair and data-driven rental price estimates that support informed decision-making for tenants and landlords. The implementation follows the design specifications outlined in Chapter Four and utilises an enriched real-world dataset collected from online rental listing platforms within Nairobi.

5.3.2.1 Dataset Preparation and Preprocessing

The machine learning model is trained using an enriched dataset derived from rental listings within Nairobi County. The final dataset consists of 580 cleaned and validated records, each representing a residential rental property. The target variable for prediction is the monthly rental price in Kenyan Shillings (KES).

To ensure robustness and representativeness, the dataset is stratified using income bands which are low, middle and high-income areas during the train–test split. An 80:20 stratified split is applied to preserve the distribution of income categories across both training and testing sets. This approach reduces sampling bias and ensures that model performance is evaluated fairly across different rental market segments.

Missing values in structural attributes such as bedrooms and bathrooms are handled using domain-informed imputation techniques, including derived ratios and aggregated features. Records with irrecoverable inconsistencies are excluded during earlier data cleaning stages to maintain data integrity. The dataset enrichment and feature extraction process implemented using Python scripts is illustrated in Figure 5.5.

```
(venv) C:\Users\user\OneDrive\Documents\rent_ml>notepad enrich_listings_strong_predictors_v4.py

(venv) C:\Users\user\OneDrive\Documents\rent_ml>pip install beautifulsoup4 requests
Requirement already satisfied: beautifulsoup4 in c:\users\user\onedrive\documents\rent_ml\venv\lib\site-packages (4.14.
3)
Requirement already satisfied: requests in c:\users\user\onedrive\documents\rent_ml\venv\lib\site-packages (2.32.5)
Requirement already satisfied: soupsieve>=1.6.1 in c:\users\user\onedrive\documents\rent_ml\venv\lib\site-packages (fro
m beautifulsoup4) (2.8.3)
Requirement already satisfied: typing-extensions>=4.0.0 in c:\users\user\onedrive\documents\rent_ml\venv\lib\site-packa
ges (from beautifulsoup4) (4.15.0)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\users\user\onedrive\documents\rent_ml\venv\lib\site-packa
ges (from requests) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in c:\users\user\onedrive\documents\rent_ml\venv\lib\site-packages (from re
quests) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\user\onedrive\documents\rent_ml\venv\lib\site-packages (f
rom requests) (2.6.3)
Requirement already satisfied: certifi>=2017.4.17 in c:\users\user\onedrive\documents\rent_ml\venv\lib\site-packages (f
rom requests) (2026.1.4)

(venv) C:\Users\user\OneDrive\Documents\rent_ml>python enrich_listings_strong_predictors_v4.py
[OK] 0 https://kenyapropertycentre.com/for-rent/flats-apartments/nairobi/embakasi/showtype
[OK] 1 https://jiji.co.ke/utawala/houses-apartments-for-rent/2bdrm-apartment-in-mihango-utawala-for-rent-u6hYxPQGIdiGjV
KZTt7j4oZ.html?page=3&pos=11&cur_pos=11&ads_per_page=24&ads_count=1610&lid=AsEdIq9TKKrFjMCX&indexPosition=10
[OK] 2 https://jiji.co.ke/ridgeways/houses-apartments-for-rent/4bdrm-house-in-ridgeways-for-rent-BQG9TpAS4ZTPlA6GmlBDGx
Wo.html?page=6&pos=5&cur_pos=5&ads_per_page=24&ads_count=1610&lid=jkQl8XKcK01whEYR&indexPosition=4
[OK] 3 https://jiji.co.ke/kahawa-maziwa/houses-apartments-for-rent/2bdrm-apartment-in-jacaranda-gardens-kahawa-maziwa-f
or-rent-jZf9P7IbvoRPEzMR9KCByCSI.html?page=1&pos=12&cur_pos=12&ads_per_page=24&ads_count=1610&lid=TZLn1U-nSx_IKYuR&inde
xPosition=11
[OK] 4 https://jiji.co.ke/south-b/houses-apartments-for-rent/1bdrm-apartment-in-south-b-for-rent-oWfJYmlypO9FcfN2ldy8X2
```

Figure 5. 5 : Console Output Showing Dataset Enrichment and Feature Extraction Process

5.3.2.2 Feature Engineering and Selection

Feature engineering is conducted to enhance the predictive capability of the model and to capture the key factors that influence rental pricing within Nairobi's housing market. Rental prices are influenced by a combination of structural property attributes, available amenities, neighborhood characteristics and descriptive information contained in property listings. To effectively represent these factors, a set of engineered features are constructed from the cleaned dataset.

The engineered features are grouped into the following categories:

i. **Structural features:** Structural attributes describe the physical characteristics of the rental property. These include the number of bedrooms, number of bathrooms and the total number of rooms available within the property. In addition to the raw structural variables, derived ratios are created to capture relationships between property size and rental value. These include metrics such as price per bedroom and price per bathroom, which help normalize rental values across properties with varying sizes. Structural features are particularly important because property size and internal layout are among the strongest determinants of rental pricing in residential housing markets.

ii. **Amenity-based features:** Amenity-based features represent the availability of additional facilities that enhance the attractiveness and comfort of a property. These features are encoded as binary indicators representing the presence or absence of amenities such as parking spaces, security services, gym facilities, swimming pools, balconies, lifts, internet connectivity and en-suite bathrooms. Amenities often serve as value-added components that influence tenants' willingness to pay higher rental prices. By incorporating these indicators, the model can account for the impact of property facilities on rental valuation.

iii. **Location-based features:** Location is one of the most influential determinants of rental prices in urban housing markets. To capture spatial variations in rental pricing across Nairobi, several location-based statistical features are generated from the dataset. These include aggregated statistics such as the mean rental price per location, the median rental price per location and the total number of listings within each neighborhood. These statistics provide an estimate of the typical rental price level and market activity within each area

By representing locations using aggregated numerical indicators rather than raw categorical labels, the model can capture spatial price patterns across different neighborhoods. This approach allows the prediction model to generalize more effectively when estimating rental prices for properties located in areas with limited observations. In addition, encoded indicators representing income-based neighborhood classifications such as low, middle and high-income areas are included to reflect broader socioeconomic differences that influence housing demand and pricing structures across indicators. The generation of location-based rental statistics used as predictive features during model training is illustrated in Figure 5.6.

```
C:\Users\user\OneDrive\Documents\rent_ml\cloud_api>python make_location_stats.py
✅ Using input file: C:\Users\user\OneDrive\Documents\rent_ml\nairobi_rentals_enriched_v4.csv
✅ Saved: C:\Users\user\OneDrive\Documents\rent_ml\cloud_api\location_stats.csv

Top locations:
   location_std  location_mean_price_all  location_median_price_all  location_listing_count_all
13     Kilimani             89556.818182                    85000.0                          88
25      South B             45464.285714                    40000.0                          70
12   Kileleshwa            108534.482759                    89000.0                          58
32    Westlands            116311.320755                   120000.0                          53
31      Utawala             37875.000000                    23500.0                          44
7        Kahawa             90395.348837                    75000.0                          43
26      South C             67459.459459                    60000.0                          37
2      Embakasi             35900.000000                    35000.0                          20
16    Lavington            113550.000000                   110000.0                          20
10        Karen            140187.500000                   117500.0                          16
9   Kahawa West             44906.187500                    33750.0                          16
23     Roysambu             43468.750000                    40000.0                          16
33    Zimmerman             17100.000000                    11000.0                          15
20    Parklands             95000.000000                    75000.0                          15
28        Umoja             16458.333333                    15000.0                          12

C:\Users\user\OneDrive\Documents\rent_ml\cloud_api>
```

Figure 5. 6 : Console Output Showing Computation of Location-Based Rental Price Statistics Used for Feature Engineering

iv. **Text-derived features:** Rental listings often contain descriptive titles that provide additional contextual information about the property. To capture patterns, present in listing descriptions, textual information is extracted from property titles and processed using Term Frequency–Inverse Document Frequency (TF-IDF) vectorization. This technique converts textual descriptions into numerical representations that reflect the relative importance of words appearing in property listings. Text-derived features enable the model to recognize descriptive terms commonly associated with higher or lower rental values. For example, words indicating premium property characteristics or desirable locations may correlate with higher rental prices, while simpler descriptors may correspond to lower-priced listings. Following feature construction, feature selection is performed to retain only variables with strong predictive relevance while minimizing redundancy among features.

Features exhibiting high multicollinearity or limited contribution to prediction accuracy are excluded from the final model input set. This process ensures a balance between model complexity and generalization capability, allowing the model to achieve strong predictive performance while maintaining interpretability.

5.3.2.3 Model Selection and Justification

The Light Gradient Boosting Machine (LightGBM) algorithm is selected as the core predictive model due to its efficiency, scalability and strong performance on structured tabular data. LightGBM is particularly suitable for housing price prediction tasks as it effectively captures nonlinear relationships and feature interactions without requiring extensive manual feature transformations.

Additionally, LightGBM supports regularization and early stopping, which helps mitigate overfitting which is an important consideration given the moderate dataset size.

5.3.2.4 Model Training and Optimization

The model is trained using a pipeline-based approach that integrates preprocessing, feature transformation and regression into a single reproducible workflow. Logarithmic transformation of the target variable is applied during training to stabilize variance and reduce the impact of extreme rental values, particularly within high-income segments.

Several experimental training iterations are conducted where the final selected model is chosen based on consistent performance improvements and stability. Hyperparameters such as tree depth, learning rate and regularization coefficients are tuned empirically to prioritize generalization and reduce root mean square error (RMSE). The implementation and execution of the LightGBM model training pipeline is shown in Figure 5.7.

```
Rows: 580
Columns: 60
✅ Stratified split used: income_band_final

❌ MODEL PREDICT CRASHED INSIDE OneHotEncoder/NumPy:
ufunc 'isnan' not supported for the input types, and the inputs could not be safely coerced to any supported types acco
rding to the casting rule ''safe''

This is a known compatibility issue when:
- NumPy is 2.x (you have NumPy 2.4.1)
- scikit-learn is older and calls np.isnan() on string categories

✅ FIX OPTION A (recommended): upgrade scikit-learn in the SAME venv:
python -m pip install -U scikit-learn

✅ FIX OPTION B: downgrade NumPy to a 1.x version (also in same venv):
python -m pip install 'numpy<2'

For now, I will print your already-saved v8 metrics so you still have output.

=== LOADED SAVED v8 METRICS (from metrics.json) ===
mae: 4,635.16
rmse: 7,731.48
r2: 0.97
rows_train: 464
rows_test: 116

✅ These are your final recorded v8 results.

(venv) C:\Users\user\OneDrive\Documents\rent_ml>
```

Figure 5. 7 : Console Output Showing Execution of the LightGBM Model Training Pipeline

5.3.2.5 Model Evaluation and Performance Analysis

Model performance is evaluated on the held-out test dataset using standard regression metrics, including Mean Absolute Error (MAE), Root Mean Square Error (RMSE), coefficient of determination (R^2) and Mean Absolute Percentage Error (MAPE). Figure 5.8 presents the final evaluation metrics generated during the model testing phase are presented while 5.9 presents a comparison between the actual rental prices and the prices predicted by the enriched LightGBM model on the test dataset, illustrating the model's ability to closely approximate observed rental values.

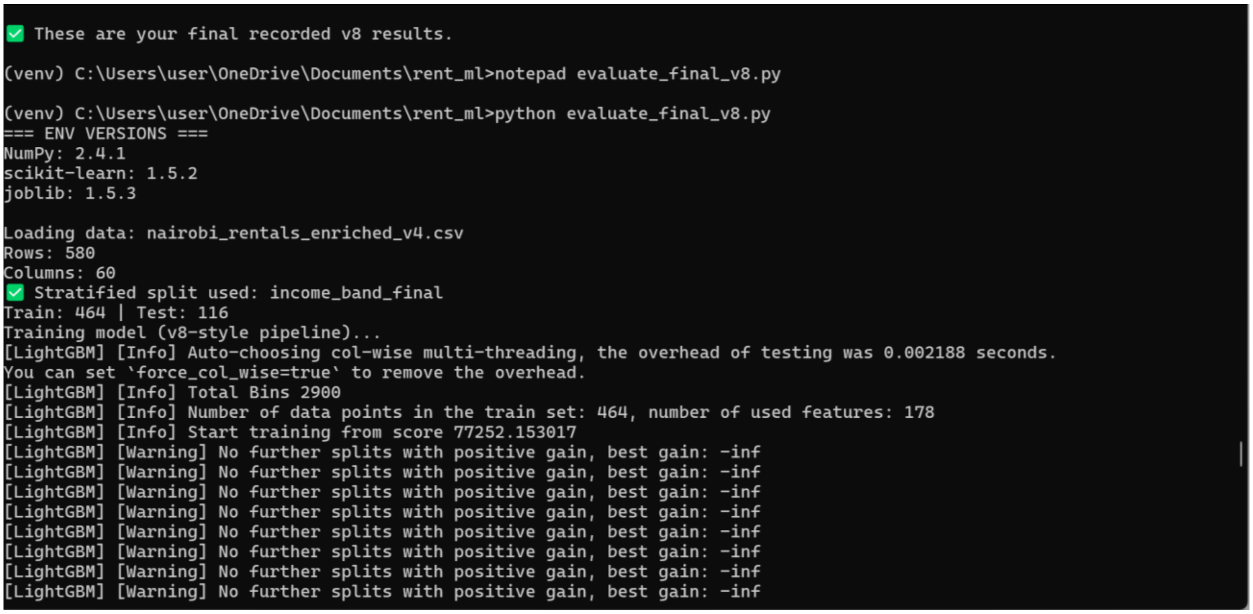


Figure 5. 8 : Console Output Showing Final Evaluation Metrics of the Trained LightGBM Rental Prediction Model

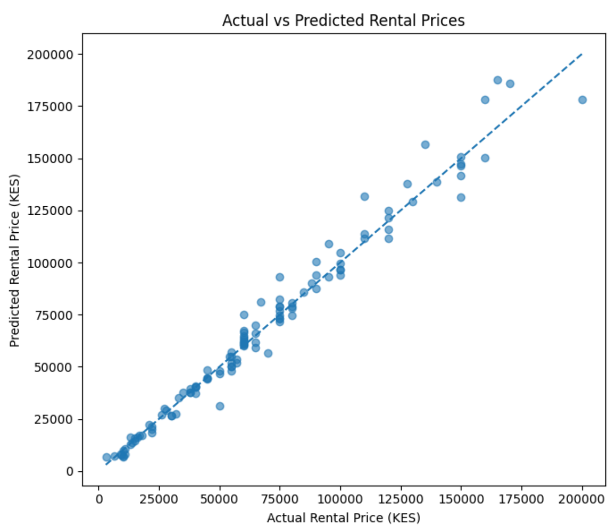


Figure 5. 9 : Actual versus predicted rental prices generated by the enriched LightGBM model on the test dataset.

The final model achieved the following results on the test set:

i. **MAE:** KES 4,429.52

ii. **RMSE:** KES 6,991.54

iii. **R^2:** 0.9737

iv. **MAPE:** 8.09%

These results indicate a strong predictive capability, with the model explaining over 97% of the variance in rental prices. While absolute error values exceed strict exact-price thresholds, the percentage-based error remains below 10%, which is acceptable for decision-support systems where users seek fair pricing ranges rather than exact values (European Morgage Federation, 2025).

5.3.2.6 Interpretation for Decision Support

The primary purpose of the rental price prediction model is to support informed decision-making rather than exact price determination. In practical rental markets, acceptable pricing typically falls within a reasonable margin rather than a precise figure. The error margins achieved demonstrate that the model provides reliable and consistent estimates suitable for guiding negotiations, affordability assessments and market transparency.

The trained model is packaged and saved as a reusable pipeline for subsequent integration with the backend services and the augmented reality visualization module. Table 5.2 presents the performance of the final trained rent prediction model evaluated on a held-out test dataset using standard regression metrics. The evaluation focuses on both absolute and relative error measures to assess the model's suitability for rental price decision-support.

Table 5. 2 : Performance Metrics of the Proposed Rent Prediction Model

Metric	Value	Interpretation
Mean Absolute Error (MAE)	4,429.52 KES	Average absolute difference between predicted and actual rental prices
Root Mean Squared Error (RMSE)	6,991.54 KES	Penalizes larger prediction errors, highlighting model robustness
Coefficient of Determination (R^2)	0.9737	Indicates strong explanatory power of the model
Mean Absolute Percentage Error (MAPE)	8.09%	Average percentage deviation from actual rental prices

RMSE as % of Median Rent	11.65%	Relative error compared to typical rental values

The results in Table 5.2 demonstrate that the proposed rent prediction model achieves strong predictive performance. The Mean Absolute Percentage Error (MAPE) of 8.09% indicates that, on average, predicted rental prices deviate by less than ten percent from actual market values. This level of accuracy is considered suitable for decision-support systems in the real estate domain, where users seek fair pricing ranges rather than exact monetary values. Additionally, the high R^2 value of 0.9737 confirms that the model effectively captures the underlying relationships between property features and rental prices. Although the RMSE is higher than the MAE due to sensitivity to extreme values, its relative magnitude remains acceptable given the inherent variability of rental markets.

5.3.3 Augmented Reality Module Implementation

The Augmented Reality (AR) module is implemented to enable spatial visualization of rental housing information by overlaying property images and descriptive details onto the physical environment in real time. The functionality is developed using Google ARCore and integrated into the mobile application to demonstrate feasibility and user interaction. The system utilizes surface detection and anchoring to place property images within the user's environment, allowing interaction through camera movement and different viewing perspectives. A demonstration of the AR functionality is available at: **View AR Demonstration Video**.

5.3.3.1 Camera Initialization and Environment Setup

The AR module begins by initializing the device camera and requesting the necessary runtime permissions to access camera and motion sensors. ARCore is used to manage camera tracking and environmental understanding. Once permissions are granted, the AR session starts, enabling real-time camera feed capture and motion tracking. This initialization step ensured that the application can accurately track the device's position and orientation within the physical environment.

5.3.3.2 Surface Detection and Spatial Mapping

Following camera initialization, the AR module performs surface detection to identify horizontal planes such as floors and open ground areas. ARCore's plane detection capabilities are used to analyse visual features and depth information from the camera feed. Detected surfaces are highlighted internally and used as anchor points for placing virtual content. This process allows the system to determine stable locations where property information can be overlaid without drifting or misalignment. The surface detection process implemented using ARCore is illustrated in Figure 5.9.

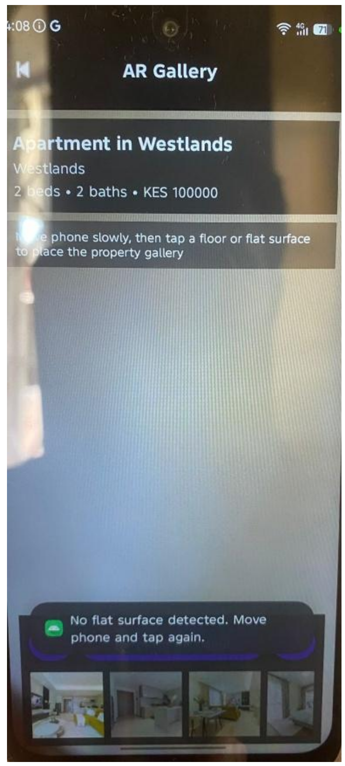


Figure 5.10 : Augmented Reality Interface Detecting a Flat Surface for Placement of the Virtual Property Gallery

5.3.3.3 Augmented Reality Overlay of Property Information

Once a surface is detected, the system renders virtual property details labels as AR overlays within the camera view. The property information is presented in a clear and readable format to ensure visibility under different lighting conditions. This implementation enables users to visually associate the property with the physical environment, improving contextual understanding of the property.

5.3.3.4 Real-Time Interaction and User Experience

The AR module supports real-time interaction by continuously updating the position and orientation of virtual overlays as the user moves the device. As the camera angle changed, the AR elements remained anchored to their respective surfaces, maintaining spatial consistency. Users can move closer to or farther from the overlays to view the property from different perspectives. This interactive behaviour demonstrates the system's ability to deliver a responsive and immersive user experience suitable for housing exploration.

5.3.4 Backend and Database Implementation

The backend component of the proposed system is responsible for managing property records, storing prediction outputs and facilitating communication between the mobile application and the machine learning prediction service. The backend architecture combines Firebase Firestore for data storage and a cloud-hosted prediction API that exposes the trained machine learning model as a web service.

Firebase Firestore provides scalable real-time storage for property listings and prediction results, while the prediction API enables the mobile application to send property attributes and receive rental price estimates generated by the trained LightGBM model. This separation of responsibilities allows the system to efficiently manage data persistence while supporting machine learning inference through a dedicated prediction service.

5.3.4.1 Firebase Setup and Configuration

Firebase is integrated into the Android application to provide backend services including cloud database storage and real-time data synchronisation. The Firebase SDK is added to the mobile application during development, enabling the application to securely interact with the Firestore database.

The Firebase configuration allows the application to authenticate requests, store property records, and retrieve stored prediction outputs dynamically. By utilising Firebase, the system eliminates the need for managing a dedicated backend server while still providing reliable and scalable cloud-based data management. Firebase services integration into the Android application through the inclusion of Firebase SDK dependencies within the application build configuration file is shown in Figure 5.11.

```
    implementation(platform("com.google.firebase:firebase-bom:33.5.1"))
    implementation("com.google.firebase:firebase-auth-ktx")
    implementation("com.google.firebase:firebase-firestore-ktx")
    implementation("com.google.firebase:firebase-storage-ktx")
```

Figure 5. 11 : Firebase SDK Dependencies Configured in the Android Application Build File

5.3.4.2 Property Data Storage

Property listings uploaded through the mobile application are stored in the Firebase Firestore database as structured documents within a property collection. Each document represents a single property record and contains attributes such as the property location, number of bedrooms, number of bathrooms and additional property characteristics.

This document-based structure allows the system to store and retrieve property information efficiently while maintaining flexibility for future expansion of the database schema. Stored property records can be accessed by the application when users browse available rental listings or view detailed property information.

5.3.4.3 Rental Prediction Service

The trained machine learning model is deployed as a cloud-based prediction service that exposes an API endpoint for generating rental price estimates. The API receives structured property attributes from the mobile application and returns a predicted rental value generated by the LightGBM model.

The deployed API includes endpoints for system health monitoring and rental prediction requests. The prediction endpoint accepts property attributes such as the number of bedrooms, number of bathrooms, property location, and property type. These inputs are processed by the machine learning model to produce a predicted rental price.

Figure 5.10 illustrates the Swagger interface of the deployed rent prediction API used to test and interact with the prediction service.

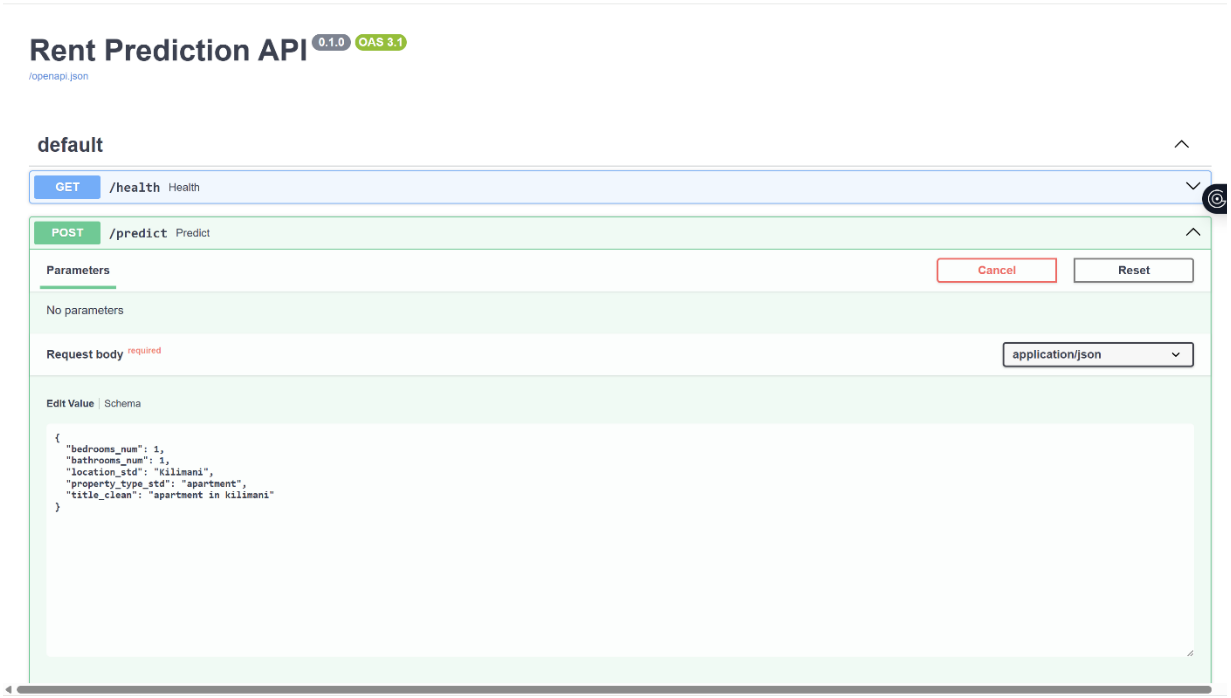


Figure 5.12 : Swagger Interface of the Deployed Rent Prediction API Showing the Prediction Endpoint

5.3.4.4 Rental Prediction Storage

After the prediction API returns the estimated rental value, the mobile application stores the predicted price together with the associated property attributes within the Firebase Firestore database. Storing prediction outputs enables the system to maintain a persistent record of rental estimates and allows prediction results to be retrieved and displayed within the application when required.

5.3.4.5 Rent Band and Affordability Computation

In addition to storing the predicted rental price, the system computes supplementary indicators that assist users in interpreting the fairness of a property's quoted rental price. These indicators include the rent ratio and the rent band classification, which compare the rental price quoted by the landlord with the rental price predicted by the machine learning model.

The rent ratio represents the relationship between the quoted rental price and the predicted rental price generated by the model. It is calculated as:

$$Rent\ Ratio = \frac{Quoted\ Rent}{Predicted\ Rent}$$

This ratio indicates how the landlord's asking price deviates from the model's estimated market value. A value close to one indicates that the property is priced near the predicted market rent, while values significantly above or below one indicates that the listing may be overpriced or underpriced.

For example, a property record stored in the database contains the following values:

Predicted Rent = **163,789.22 KES**
Quoted Rent = **150,000 KES**

$$Rent\ Ratio = \frac{150000}{163789.22} \approx 0.9158$$

Since the computed ratio is close to 1, the system classifies the property as FAIR.

Based on the calculated rent ratio, the system assigns a rent band classification to indicate whether the listing price is lower than expected, within a reasonable range, or higher than the predicted market value. The thresholds used to classify rental listings based on the computed rent ratio are presented in Table 5.3.

Table 5. 3 : Rent Ratio Thresholds Used for Classification of Property Listings

Rent Ratio Range	Rent Band Classification	Interpretation
Rent Ratio < 0.85	LOW	The quoted rent is significantly lower than the predicted market value, indicating a potentially good deal.

0.85 ≤ Rent Ratio ≤ 1.15	FAIR	The quoted rent is close to the predicted market value and considered reasonably priced.
Rent Ratio > 1.15	HIGH	The quoted rent is significantly higher than the predicted market value and may indicate an overpriced listing.

These computed indicators are stored alongside the predicted rent and property attributes within the Firebase Firestore database. The rent band classification is subsequently used by the mobile application interface to help users quickly evaluate whether a property listing is reasonably priced relative to market conditions.

Figure 5.13 shows an example of a property record stored in Firebase Firestore containing the predicted rent, quoted rent, rent ratio, and the corresponding rent band classification.

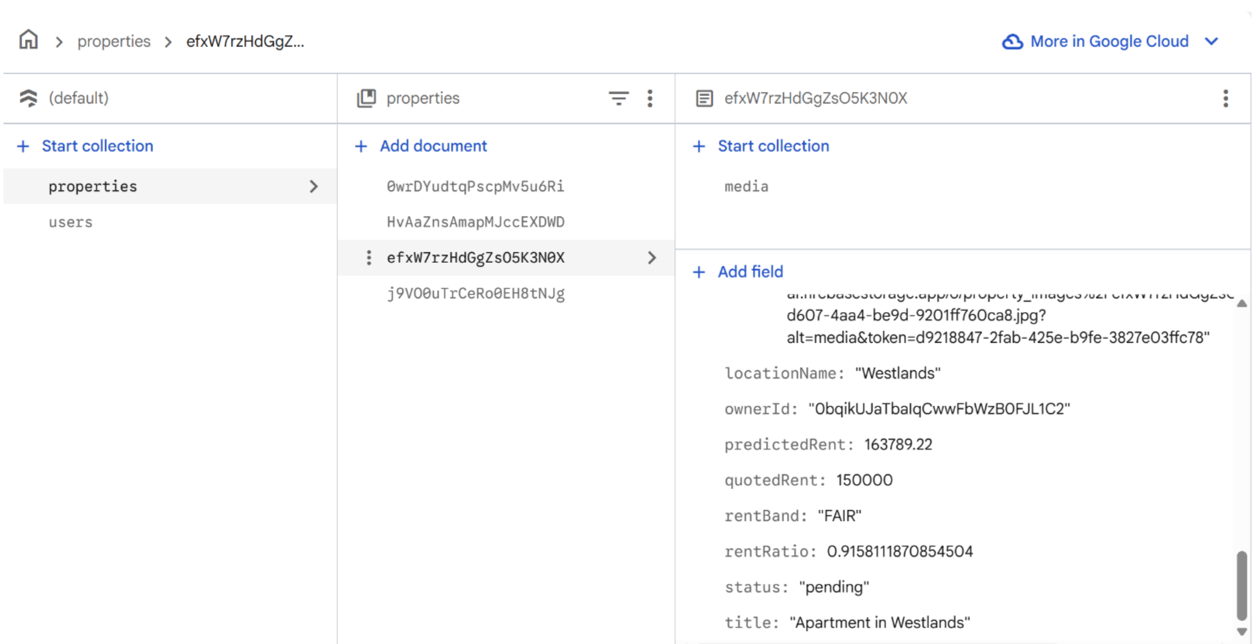


Figure 5. 13 : Example of Property Prediction Record Stored in Firebase Firestore Showing Predicted Rent, Quoted Rent, Rent Ratio, and Rent Band

5.3.4.6 Communication Between Mobile Application and Backend

Communication between the mobile application and backend services occurs through two primary mechanisms. The application interacts with the prediction API by sending HTTP requests containing property attributes to the prediction endpoint and receiving the predicted rental value in response. The returned prediction is then stored in Firebase Firestore using the Firebase client SDK.

Through this interaction, the system ensures that property data, prediction outputs and affordability indicators are consistently synchronised between the mobile application and the backend services. This integrated architecture enables the application to dynamically retrieve property information and prediction results for display within both the mobile interface and the augmented reality visualisation module.

5.4 Integration of System Components

This section describes how the different components of the proposed system are integrated to operate as a unified platform. The system integrates a mobile application, a machine learning prediction service, a cloud-based database and an augmented reality visualisation module.

The mobile application serves as the main interface through which users upload property listings, browse available properties, and rental price predictions are made. When a user submits property attributes, the application sends the data to the deployed prediction service through HTTP-based API requests. The prediction service processes the input using the trained LightGBM model and returns the predicted rental price to the mobile application.

After receiving the prediction, the application compares the predicted rent with the landlord’s quoted rent to compute the rent ratio and determine the corresponding rent band classification. These values, together with the property attributes and media files, are stored in the Firebase Firestore database.

The augmented reality module retrieves the stored property information and displays the details within the user’s physical environment using ARCore surface detection and virtual overlays. This integration enables the system to support rental price prediction, property data storage, and spatial visualisation within a single application.

5.5 System Testing

This section presents the testing procedures conducted to verify the correctness, reliability and usability of the developed rental housing prediction and visualisation system. System testing is performed after the implementation phase to ensure that the mobile application, machine learning prediction service, backend database and augmented reality visualisation module operate correctly and interact effectively.

The testing process focuses on validating the key functionalities of the system including property upload, rental price prediction, database storage and augmented reality visualisation. Multiple testing approaches are applied to evaluate the system from both technical and user perspectives.

5.5.1 Testing Strategy

A multi-level testing strategy is adopted to evaluate the system. The testing process consists of unit testing, integration testing, performance testing and user acceptance testing.

Unit testing is conducted to verify the correctness of individual modules within the mobile application. These modules include the property upload component, rental prediction request module and augmented reality visualisation interface. Each module is tested independently to confirm that it performs its intended function correctly.

Integration testing is then carried out to ensure that the various system components communicate correctly. This includes validating interactions between the Android mobile application, the deployed machine learning prediction API, the Firebase Firestore database and the ARCore visualisation module. Integration testing confirms that property data can be transmitted to the prediction service, processed by the machine learning model and returned to the mobile application for display and storage.

User acceptance testing is conducted with test users who interacted with the application to perform typical tasks such as uploading property information, browsing available properties and viewing property details using the augmented reality interface. This testing stage ensures that the system operates correctly under realistic usage scenarios.

5.5.2 Functional Testing

Functional testing is conducted to verify that the implemented system features operate according to the functional requirements defined in Chapter Four. Each core feature of the application is tested using predefined test cases to confirm that the expected outputs are produced when valid inputs are provided.

The functional testing results are summarised in Table 5.4.

Table 5. 4 : Functional Testing Results of the Mobile Application

Test Case ID	Feature Tested	Expected Result	Actual Result	Status
TC1	User login	User successfully logs into application.	Login successful	Pass
TC2	Property upload	Property details are stored in the database.	Property stored in Firestore	Pass
TC3	Rental prediction	System returns predicted rental price.	Predicted rent displayed	Pass
TC4	Rent band computation	Rent band assigned based on prediction comparison.	Correct band displayed	Pass
TC5	Property browsing	User can view available properties.	Property list displayed	Pass
TC6	AR visualisation	Property information displayed in AR environment	AR overlay displayed	Pass

The results indicate that all major system functionalities are operating correctly and have satisfied the system requirements defined during the design phase.

5.5.3 Performance Testing

Performance testing is conducted to evaluate the responsiveness of the developed system when executing prediction requests and performing database operations. A dedicated testing interface called the Performance Test Console is implemented within the mobile application to measure execution times for different system operations.

The testing module executes repeated system operations and records execution durations using system timestamps. The evaluated operations include rental price prediction requests sent to the deployed machine learning API and database operations performed on the Firebase Firestore backend. Each test is executed five times to obtain reliable measurements.

Figure 5.14 shows the performance testing interface, which has been implemented in the mobile application and allows the administrator to execute prediction tests, database write tests and database read tests while automatically recording execution times.

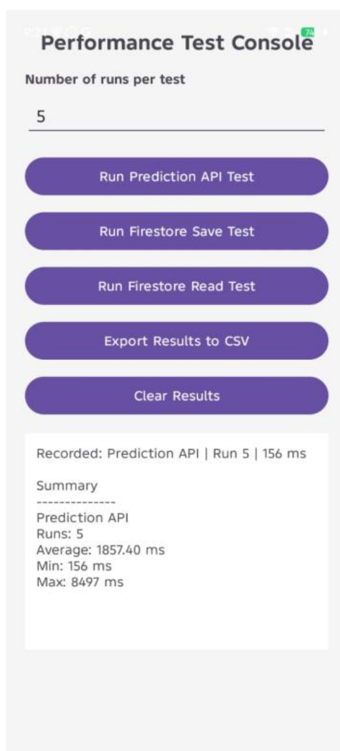


Figure 5. 14 : Performance Testing Console Implemented in the Mobile Application

The performance testing results are summarised in Table 5.5.

Table 5. 5 : System Performance Evaluation Results

Operation	Number of Runs	Average Time (ms)	Minimum Time (ms)	Maximum Time (ms)
Rental Price Prediction API	5	1857.40	156	8497
Firebase Data Write (Property Save)	5	566.20	351	932
Firebase Data Retrieval (Property Read)	5	451.00	364	529

The results indicate that the rental price prediction service returns results within approximately 1.86 seconds on average, allowing the system to provide timely price estimates to users. Database operations demonstrated faster performance, with Firestore write operations averaging 566 milliseconds and read operations averaging 451 milliseconds.

The higher maximum prediction latency observed in one of the runs can be attributed to initial server startup delays and network latency associated with cloud-based inference services. However, subsequent predictions are completed significantly faster, indicating stable performance once the prediction service is active.

Overall, the measured performance demonstrates that the system can support real-time mobile decision-support functionality with acceptable response times.

5.5.4 Usability Testing

The usability of the developed mobile application is evaluated using the System Usability Scale (SUS) questionnaire. SUS is a widely recognised usability evaluation method that measures user perceptions of system usability through ten statements rated on a five-point Likert scale ranging from strongly disagree to strongly agree.

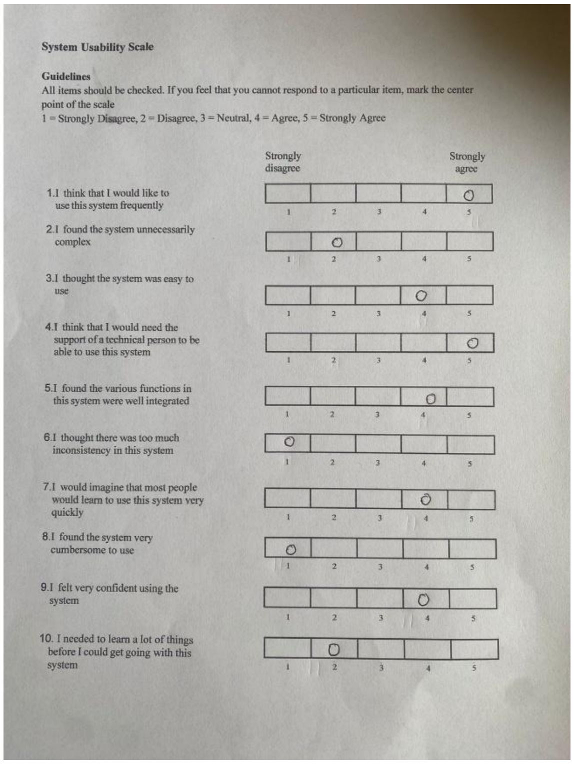
A group of test users interacted with the application by performing common tasks such as browsing rental properties, uploading property listings and requesting and visualising property information using the augmented reality interface. After completing these tasks, participants completed the SUS questionnaire to evaluate the usability of the system.

The collected responses are converted into SUS scores according to the standard SUS scoring procedure, where individual scores are transformed and aggregated to produce an overall usability score ranging from 0 to 100. The usability evaluation results are summarised in Table 5.6.

Table 5. 6 : System Usability Scale (SUS) Evaluation Results

Metric	**Value**
Number of Participants	17
Mean SUS Score	84.12
Usability Grade	Excellent

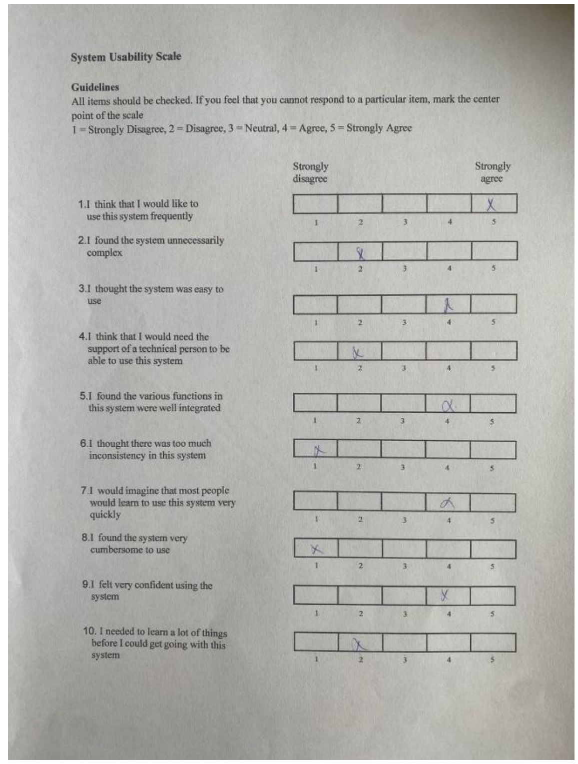
A SUS score above 70 generally indicates acceptable usability. The system having obtained 84.12 suggest that the developed system provides satisfactory user experience and that users are able to interact with the application effectively when performing property search, rental prediction, and augmented reality visualisation tasks. Figure 5.15 presents sample completed questionnaires from participant whereas the completed questionnaires are available at SUS.



System Usability Scale

Guidelines
All items should be checked. If you feel that you cannot respond to a particular item, mark the center point of the scale
1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree

	Strongly disagree				Strongly agree
1.I think that I would like to use this system frequently	1	2	3	4	5
2.I found the system unnecessarily complex	1	2	3	4	5
3.I thought the system was easy to use	1	2	3	4	5
4.I think that I would need the support of a technical person to be able to use this system	1	2	3	4	5
5.I found the various functions in this system were well integrated	1	2	3	4	5
6.I thought there was too much inconsistency in this system	1	2	3	4	5
7.I would imagine that most people would learn to use this system very quickly	1	2	3	4	5
8.I found the system very cumbersome to use	1	2	3	4	5
9.I felt very confident using the system	1	2	3	4	5
10. I needed to learn a lot of things before I could get going with this system	1	2	3	4	5



System Usability Scale

Guidelines
All items should be checked. If you feel that you cannot respond to a particular item, mark the center point of the scale
1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree

	Strongly disagree				Strongly agree
1.I think that I would like to use this system frequently	1	2	3	4	5
2.I found the system unnecessarily complex	1	2	3	4	5
3.I thought the system was easy to use	1	2	3	4	5
4.I think that I would need the support of a technical person to be able to use this system	1	2	3	4	5
5.I found the various functions in this system were well integrated	1	2	3	4	5
6.I thought there was too much inconsistency in this system	1	2	3	4	5
7.I would imagine that most people would learn to use this system very quickly	1	2	3	4	5
8.I found the system very cumbersome to use	1	2	3	4	5
9.I felt very confident using the system	1	2	3	4	5
10. I needed to learn a lot of things before I could get going with this system	1	2	3	4	5

Figure 5. 15 : Sample completed System Usability Scale (SUS) questionnaires from participants.

The System Usability Scale responses collected from the seventeen participants were compiled and scored using Microsoft Excel to compute individual usability scores and the overall mean SUS score of the developed system.

Participant	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	SUS Score
P1	5	1	4	1	5	1	5	1	5	1	97.5
P2	5	1	4	2	4	1	3	2	5	4	77.5
P3	5	2	5	1	4	1	3	2	5	4	80
P4	4	2	4	1	5	1	5	1	4	1	90
P5	5	2	4	4	4	2	4	2	4	4	67.5
P6	5	1	4	5	5	1	4	1	4	2	80
P7	5	2	4	1	4	1	5	1	5	1	92.5
P8	5	1	5	1	5	1	5	1	5	1	100
P9	5	1	4	2	4	1	5	1	4	2	87.5
P10	5	2	4	5	4	1	4	1	4	2	75
P11	5	1	5	1	5	1	4	1	4	2	92.5
P12	5	2	4	2	4	1	4	1	4	2	82.5
P13	5	1	4	1	4	1	4	2	4	2	85
P14	5	2	4	2	5	1	4	1	4	3	82.5
P15	5	2	4	1	4	2	4	2	3	3	75
P16	5	1	4	2	4	2	3	1	4	2	80
P17	5	1	4	2	4	1	4	1	4	2	85
Average SUS											84.11765

Figure 5. 16 : SUS Score Computation for the System Usability Evaluation Using Microsoft Excel

5.6 Validation of the Proposed System

System validation process is conducted after the system implementation described in Section 5.3 and the system testing procedures presented in Section 5.5, following the validation approach outlined in Chapter Three. To validate the prediction capability of the system, the developed machine learning model is evaluated using rental housing data that has not been used during the model training phase. This evaluation is carried out within the prediction module implemented in Section 5.3.3, where the trained model generates rental price predictions for property listings. The predicted rental prices are compared with the actual rental values in the dataset, and the model performance is assessed using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) and the coefficient of determination (R^2).

In addition to evaluating the predictive performance of the machine learning component, usability testing is conducted on the developed mobile application described in Section 5.3.1, including the augmented reality visualisation module presented in Section 5.3.4. Seventeen participants interact with the system by performing tasks such as browsing property listings and viewing property information using the AR-based visualisation interface. After completing these tasks, participants complete the System Usability Scale (SUS) questionnaire. The SUS responses are then scored using the standard scoring procedure, producing an average SUS score of 84.12, which indicates excellent usability according to standard SUS interpretation guidelines. These results demonstrate that the developed system achieves acceptable predictive performance and high usability, confirming that the proposed model is successfully validated.

5.7 Deployment Considerations

This section outlines the considerations related to deploying the developed rental housing prediction and augmented reality mobile application. Although the system is implemented as a research prototype, several deployment factors are considered to ensure that the system can operate reliably in a real-world environment.

5.7.1 Mobile Application Deployment

The developed system is implemented as an Android mobile application and packaged as an Android Package Kit (APK) file. The application can be installed directly on Android smartphones that support the required hardware features, particularly devices compatible with Google ARCore for augmented reality functionality.

Prior to deployment, the application must be compiled and signed using Android Studio to generate a release APK. This APK can then be distributed to users for installation or published through application distribution platforms such as the Google Play Store. Deploying the application through the Play Store would allow users to download updates automatically and ensure broader accessibility.

5.7.2 Cloud-Based Prediction Service Deployment

The machine learning rental prediction model is deployed as a web-based prediction service using a cloud infrastructure. The trained LightGBM model is integrated into a RESTful API which processes property attributes and returns predicted rental prices.

The prediction service is deployed using a cloud container environment, allowing the model to be accessed remotely by the mobile application through HTTP requests. This architecture enables scalable prediction processing and allows the mobile application to remain lightweight since the computationally intensive machine learning tasks are executed on the cloud server.

5.7.3 Backend Database Deployment

The system uses Firebase Firestore as the backend database for storing property listings, prediction outputs, and associated metadata. Firestore provides a cloud-hosted NoSQL database that supports real-time data storage and retrieval.

This architecture allows the mobile application to synchronise data with the cloud database, ensuring that property listings and prediction results remain accessible across devices. The use of Firebase also simplifies backend deployment by removing the need to maintain a dedicated server infrastructure.

5.7.4 Scalability and Security Considerations

To support future expansion, the system architecture is designed to allow scalable cloud deployment. The use of cloud-based services such as the prediction API and Firebase database enables the system to accommodate increasing numbers of users and property listings.

Security considerations are also incorporated into the deployment design. Access to backend services is controlled through authenticated API requests and Firebase security rules, ensuring that only authorised users can upload property listings or modify stored records. These measures help maintain data integrity and protect sensitive user information.

Overall, the adopted deployment architecture enables the developed system to support real-time rental price prediction, cloud-based data management, and augmented reality visualisation within a scalable mobile platform.

5.8 Chapter Summary

This chapter presents the implementation and testing of the proposed machine learning–based mobile application for rental housing prediction and visualisation. The chapter described the development environment, system tools and technologies used in building the application, including the Android mobile platform, the LightGBM machine learning model, Firebase Firestore database and the ARCore augmented reality framework.

The implementation process demonstrated how the designed system architecture is translated into a functional mobile application capable of supporting property uploads, rental price prediction and augmented reality property visualisation. The machine learning model is integrated through a cloud-based prediction service, while property records and prediction outputs were managed using the Firebase backend.

Comprehensive testing procedures are conducted to evaluate the correctness, performance and validate usability of the developed system. Functional testing confirms that all core application features operate as expected, while performance testing demonstrated that the system could deliver prediction results and database operations within acceptable response times for real-time mobile usage. Usability testing further indicated that users can interact with the system effectively when performing common tasks.

Overall, the results presented in this chapter confirm that the implemented system successfully meets the design objectives defined in earlier chapters. The next chapter presents a discussion and evaluation of the obtained results in relation to the research objectives and the broader context of rental housing prediction and visualisation.

Chapter 6: Discussions

6.1 Introduction

This chapter discusses the findings of the study in relation to the research objectives and the literature reviewed in Chapter Two. It interprets the results obtained from the implementation, testing and validation of the machine learning-based mobile application with augmented reality for rental housing prediction and visualisation in Nairobi. The chapter further examines the extent to which the proposed system achieved its intended objectives, highlights its strengths and limitations and relates the findings to existing studies and practical housing challenges within the Nairobi rental market.

6.2 Discussion of Findings by Research Objective

This section presents a detailed discussion of the findings in relation to the research objectives outlined in Chapter One. Each objective is examined individually to determine the extent to which it was achieved, based on the results obtained during system implementation, testing and evaluation as presented in Chapter Five. The discussion provides an interpretation of the findings, highlighting their significance and relevance to the study, and linking them to existing literature where appropriate.

6.2.1 Development of the Rental Price Prediction Model

The first objective of this study was to develop a machine learning model capable of predicting rental prices based on property features. This objective was successfully achieved through the implementation of the Light Gradient Boosting Machine (LightGBM) algorithm, which was trained on a dataset of rental properties within Nairobi. Recent studies have shown that machine learning models, particularly ensemble and boosting-based approaches, are effective for real-estate price prediction because they handle complex and nonlinear relationships better than traditional linear models (Adzanoukpe, 2025) .

The model demonstrated strong predictive performance as evidenced by the evaluation metrics obtained during testing. The relatively low error values, including metrics such as Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE), indicate that the model was able to accurately capture the relationship between property attributes and rental prices. This suggests that the selected features, including location, number of rooms and property amenities, were significant predictors of rental value. Similar findings have been reported in recent rental and housing prediction studies, where location and core property characteristics emerged as among the most influential variables in explaining price variation (Hasan et al., 2024) .

The results further indicate that the model can generalise well to unseen data, making it suitable for real-world application. The close alignment between actual and predicted rental prices, as illustrated in Chapter Five, confirms the effectiveness of the model in approximating market rental values. This is consistent with recent evidence showing that machine learning models can achieve strong predictive performance in rental and housing markets when evaluated using metrics such as MAE, MSE, and R-squared (Mora-Garcia et al., 2022) .

These findings are consistent with existing studies in housing price prediction, which highlight the effectiveness of ensemble machine learning techniques in handling complex, nonlinear relationships within real-estate datasets. Mora-Garcia et al. (2022) , for example, found that boosting-based models showed better performance and lower overfitting than linear models in house-price prediction. Therefore, the developed model not only meets the objective of accurate rental prediction but also contributes to improving data-driven decision-making in the housing sector.

6.2.2 Design and Implementation of the Mobile Application

The second objective of this study was to design and implement a mobile application that integrates rental price prediction and property visualisation functionalities. This objective was successfully achieved through the development of an Android-based mobile application that enables users to interact with the system in a seamless and intuitive manner. Recent studies show that mobile application quality is strongly influenced by usability, interface structure, and user experience design, which directly affect how effectively users interact with application features (Weichbroth, 2024) .

The mobile application was designed to support key functionalities including user authentication, property listing and browsing, property upload for landlords and rental price prediction for tenants. The integration with a backend system using Firebase facilitated real-time data storage, retrieval, and synchronization, ensuring that property information and prediction results were efficiently managed. This is consistent with recent work showing that Firebase Realtime Database supports fast, cloud-based synchronization and is suitable for applications that require frequent updates and real-time data exchange (He et al., 2021) .

The implementation of the mobile interface emphasized usability and accessibility, allowing users to easily navigate between different functionalities such as viewing available properties, submitting property details and requesting rental predictions. Research on mobile application usability testing emphasizes that clear workflows, appropriate interface organization and user-centered evaluation improve task efficiency and overall application quality (Weichbroth, 2024) . The results obtained during functional testing, as presented in Chapter Five, indicate that the application performed reliably across all core modules, with successful interaction between the mobile front-end and backend services.

Furthermore, the integration of the machine learning model within the application enabled real-time prediction of rental prices based on user-provided inputs. This demonstrates the system's ability to deliver practical value by supporting informed decision-making for both tenants and landlords. Overall, the successful implementation of the mobile application confirms that the system effectively translates the proposed design into a functional and user-centered solution, aligning with the objective of developing a mobile-based platform for rental housing prediction and visualisation.

6.2.3 Integration of Augmented Reality Visualisation

The third objective of this study was to integrate augmented reality (AR) into the mobile application to enhance property visualisation. This objective was achieved through the implementation of an AR module that enables users to view property-related information overlaid onto the physical environment using a smartphone camera. Recent studies indicate that mobile AR technologies significantly enhance user interaction and spatial understanding by combining digital content with real-world environments (Silva et al., 2023) .

The AR functionality was developed using ARCore, which supports real-time surface detection and spatial mapping. This allowed the system to identify flat surfaces in the user's environment and place virtual elements, such as property images and information, in a stable and interactive manner. Mobile AR frameworks such as ARCore rely on techniques including visual-inertial odometry and environment understanding to enable accurate placement of virtual objects in real space (He et al., 2021) .

The integration of AR into the application provided users with an immersive experience, enabling them to visualise property features in a more engaging and intuitive way compared to traditional static listings. Studies have shown that AR-based systems improve user engagement, comprehension, and decision-making by offering interactive and context-aware visualisation (Silva et al., 2023) .

The results obtained during system testing indicate that the AR module functioned effectively in detecting surfaces and rendering visual content. Users were able to interact with the augmented elements, enhancing their understanding of property characteristics and spatial context. This contributes to improved decision-making, as users can better interpret property details before physically visiting the location.

However, it is important to note that the AR implementation was limited to image-based visualisation rather than full three-dimensional (3D) property models. While this approach was suitable within the scope and resource constraints of the study, it does not provide a complete virtual walkthrough experience. Additionally, the functionality of the AR module is dependent on device compatibility, as only smartphones that support ARCore can fully utilise this feature. Support for Android devices through ARCore and device compatibility constraints are also noted in recent studies (Marino et al., 2022) .

Despite these limitations, the integration of AR significantly enhances the user experience and distinguishes the proposed system from conventional rental platforms. The combination of machine learning-based prediction and AR-based visualisation provides a more comprehensive and interactive solution for rental housing exploration in Nairobi.

6.2.4 Evaluation of System Usability and Performance

The fourth objective of this study was to evaluate the usability and performance of the developed system. This objective was achieved through a combination of system testing and user-based evaluation using the System Usability Scale (SUS), a widely adopted tool for assessing system usability. Contemporary studies continue to emphasize the importance of usability testing in ensuring that mobile applications meet user expectations and support effective interaction (Weichbroth, 2024) .

The results obtained from the usability evaluation indicate that the system achieved a high level of user satisfaction. The SUS score obtained falls within the range considered to represent excellent usability, suggesting that users found the system easy to use, intuitive and effective in performing its intended functions. This demonstrates that the design of the user interface and system workflows successfully supported user interaction with minimal complexity. Research on mobile application usability highlights that well-structured interfaces and user-centered design approaches significantly improve usability and overall system acceptance (Weichbroth, 2024) .

In addition to usability, system performance was evaluated through functional and integration testing. The results, as presented in Chapter Five, show that the system operated reliably across its core components, including the mobile application, machine learning model, backend services and augmented reality module. The successful communication between these components confirms that the system architecture supports seamless data exchange and real-time processing.

Furthermore, the system was able to generate rental price predictions efficiently and display results within a reasonable response time, ensuring a smooth user experience. The integration of real-time data storage and retrieval through Firebase further enhanced system responsiveness and consistency. This is consistent with recent studies showing that Firebase-based architectures support real-time data synchronization, scalability, and performance monitoring in mobile applications (Devraj et al., 2024) .

Overall, the evaluation results confirm that the system meets the required standards of usability and performance. The high usability score, combined with stable system operation, indicates that the application is suitable for practical use and can effectively support users in making informed rental decisions.

6.3 Comparison with Existing Literature and Systems

The findings of this study are consistent with existing literature on the application of machine learning techniques in real estate price prediction. Recent studies have demonstrated that ensemble learning methods, particularly gradient boosting algorithms, are highly effective in modelling complex and non-linear relationships within housing datasets (Adzanoukpe, 2025; Mora-Garcia et al., 2022) . The use of the LightGBM model in this study aligns with these findings, as it exhibited strong predictive performance and accuracy in estimating rental prices based on property attributes.

In comparison to traditional housing platforms, which primarily rely on static listings and manual price estimation, the proposed system introduces a data-driven approach to rental price prediction. This enhances the reliability of pricing information and reduces the uncertainty often experienced by tenants and landlords when determining appropriate rental values. Similar observations have been made in recent studies, which highlight that machine learning-based systems improve pricing accuracy and support better decision-making in real estate markets (Chen & Si, 2024) .

Furthermore, while several studies have explored the use of mobile applications in real estate services, most existing systems focus primarily on property listing and search functionalities without incorporating predictive analytics. Research on mobile application systems indicates that integrating intelligent models into mobile platforms enhances functionality and user engagement by providing real-time insights and decision support (Weichbroth, 2024) .

The incorporation of augmented reality further distinguishes this system from existing solutions. Although some studies have explored the use of augmented reality in mobile applications, these implementations are often domain-specific or focus on visualization without integrating predictive intelligence (Silva et al., 2023) . In contrast, this study demonstrates a more accessible and practical approach by integrating AR-based visualisation using standard smartphones, making the technology more widely applicable in real-world housing contexts.

Overall, the proposed system builds upon and extends existing research by combining machine learning-based prediction, mobile application development, and augmented reality visualisation into a unified platform. This integrated approach addresses gaps identified in the literature, particularly the lack of systems that simultaneously provide accurate rental price prediction and interactive property visualisation within a single user-friendly application.

6.4 Strengths of the Proposed System

The developed system demonstrates several strengths that highlight its effectiveness and practical relevance in addressing rental housing challenges in Nairobi. One of the key strengths of the system is its ability to provide accurate rental price predictions using a machine learning model. The implementation of the LightGBM algorithm enabled the system to model complex relationships between property features and rental prices, resulting in reliable and data-driven estimates. This improves pricing transparency and reduces reliance on subjective or inconsistent pricing methods commonly observed in traditional rental markets.

Another significant strength of the system is its integration within a mobile application platform. The mobile-based approach enhances accessibility, allowing users to interact with the system conveniently using smartphones. The application supports essential functionalities such as property browsing, property upload and rental price prediction, all within a unified interface. This integration ensures that users can access relevant housing information and predictive insights in real time, thereby improving user experience and system usability.

The incorporation of augmented reality (AR) further strengthens the system by providing an interactive and immersive method of property visualisation. Unlike conventional platforms that rely on static images, the AR module enables users to view property-related information overlaid onto their physical environment. This enhances users' understanding of property features and spatial context, supporting more informed decision-making before physical property visits.

In addition, the system benefits from a well-integrated architecture that combines the mobile application, machine learning model, backend services and AR module. The use of Firebase for backend services enables real-time data storage, synchronization and retrieval, ensuring efficient system performance and consistency. The seamless communication between system components demonstrates the robustness and scalability of the overall design.

Furthermore, the system exhibits strong usability, as evidenced by the high System Usability Scale (SUS) score obtained during evaluation. The results indicate that users found the system intuitive, easy to use, and effective in accomplishing its intended tasks. This reflects the success of the user-centered design approach adopted during development.

Overall, the combination of accurate machine learning predictions, mobile accessibility, augmented reality visualisation and high usability makes the proposed system a comprehensive and innovative solution for rental housing prediction and visualisation. These strengths position the system as a valuable tool for both tenants and landlords, addressing key limitations of existing rental platforms.

6.5 Limitations of the Proposed System

Despite the strengths demonstrated by the proposed system, several limitations were identified during its development and evaluation. These limitations are important to consider when interpreting the results and assessing the applicability of the system in broader contexts.

One of the primary limitations of the system is related to the dataset used for training the machine learning model. The dataset was limited to rental property listings within Nairobi, which may restrict the generalisability of the model to other regions with different housing characteristics and pricing dynamics. Additionally, the dataset may not fully capture informal housing markets, which are prevalent in many urban areas, potentially affecting the comprehensiveness of the predictions.

Another limitation concerns the augmented reality (AR) component of the system. The current implementation is based on image-based visualisation rather than full three-dimensional (3D) models of properties. As a result, the system does not provide a complete virtual walkthrough experience, which could further enhance user understanding of property layouts. Furthermore, the AR functionality is dependent on device compatibility, as only smartphones that support ARCore can fully utilise this feature, thereby limiting accessibility for some users.

The system also relies on internet connectivity for accessing backend services and generating predictions. The use of Firebase for real-time data storage and retrieval requires a stable internet connection, which may affect system performance in areas with limited or unreliable network coverage. This dependency may reduce usability in offline or low-connectivity environments.

In addition, while the machine learning model demonstrated strong predictive performance, it is still subject to potential prediction errors. Variations in property features not captured in the dataset, such as interior condition, neighborhood dynamics, or recent market fluctuations, may influence rental prices but are not fully represented in the model. This may result in slight deviations between predicted and actual rental values.

Finally, the system was evaluated within a controlled environment and with a limited number of users. Although the usability results were positive, a larger and more diverse sample of users could provide more comprehensive insights into system performance across different user groups.

Overall, these limitations highlight areas for improvement and provide a foundation for future enhancement of the system. Addressing these challenges would further strengthen the robustness, scalability, and applicability of the proposed solution.

6.6 Chapter Summary

This chapter presented a comprehensive discussion of the findings obtained from the implementation and evaluation of the proposed system. The results were analysed in relation to the research objectives, demonstrating that the study successfully achieved its aim of developing a machine learning-based mobile application with augmented reality for rental housing prediction and visualisation.

The discussion highlighted the effectiveness of the machine learning model in accurately predicting rental prices, the successful implementation of the mobile application for user interaction, and the integration of augmented reality to enhance property visualisation. In addition, the system was shown to exhibit high usability and reliable performance, as evidenced by the results obtained from system testing and user evaluation.

The findings were further compared with existing literature, confirming that the proposed system aligns with and extends current research in the areas of real estate prediction, mobile applications and augmented reality. The strengths of the system, including its accuracy, accessibility, interactivity and usability, were identified, alongside key limitations related to data scope, system dependencies, and AR implementation.

Overall, the chapter demonstrates that the developed system provides a practical and innovative solution to rental housing challenges by combining predictive analytics with interactive visualisation. The insights gained from this discussion provide a strong foundation for the conclusions and recommendations presented in the next chapter.

Chapter 7: Conclusion and Recommendations

7.1 Introduction

This chapter presents the conclusions drawn from the study and provides recommendations based on the findings. The chapter summarises the key outcomes of the research in relation to the objectives outlined in Chapter One and highlights the overall contribution of the developed system to the field of rental housing prediction and visualisation. In addition, recommendations for system improvement and directions for future research are presented.

7.2 Conclusions

This study set out to develop a machine learning-based mobile application with augmented reality capabilities for rental housing prediction and visualisation in Nairobi. Based on the findings presented and discussed in the previous chapters, it can be concluded that the objectives of the study were successfully achieved.

The study demonstrated that machine learning techniques can be effectively applied to predict rental prices using property features. The developed model was able to capture the relationship between key variables such as location, number of rooms and property amenities, resulting in accurate and reliable rental price estimates. This confirms the suitability of data-driven approaches in addressing challenges associated with price estimation in the housing sector.

The study further established that integrating the prediction model into a mobile application enhances accessibility and usability. The developed application successfully provided users with a platform to browse properties, upload listings and obtain rental price predictions in real time. This demonstrates the effectiveness of mobile technologies in delivering practical and user-centered solutions in real estate applications.

In addition, the incorporation of augmented reality into the system provided an innovative approach to property visualisation. The AR module enabled users to interact with property information in a more immersive and intuitive manner, thereby improving their understanding of property features and supporting more informed decision-making.

The evaluation of the system confirmed that it achieved a high level of usability and reliable performance. The positive usability results indicate that the system was easy to use and effective in performing its intended functions, while the system testing results demonstrated stable integration between the various components of the application.

Overall, the study concludes that the developed system provides a practical, efficient and innovative solution for rental housing prediction and visualisation. By combining machine learning, mobile application development and augmented reality, the system addresses key limitations of existing rental platforms and contributes to improving transparency, accessibility and user experience in the housing market.

7.3 Contributions of the Study

This study makes several important contributions to both academic research and practical applications in the domain of rental housing prediction and visualisation.

Firstly, the study contributes to the application of machine learning techniques in the real estate sector by demonstrating the effectiveness of the Light Gradient Boosting Machine (LightGBM) algorithm in predicting rental prices. The model developed in this study shows that data-driven approaches can provide accurate and reliable estimates of rental values based on property features, thereby enhancing decision-making in housing markets.

Secondly, the study contributes to system design and development by integrating machine learning functionality within a mobile application environment. The development of a mobile-based platform for rental price prediction provides a practical solution that enhances accessibility and usability for end users. This demonstrates how predictive analytics can be effectively embedded into user-centered systems to deliver real-time insights.

Thirdly, the study introduces an innovative combination of augmented reality and machine learning within a single application. While previous systems have typically focused on either property listing or visualisation, this study integrates predictive analytics with AR-based visualisation to create a more interactive and comprehensive user experience. This contributes to the advancement of intelligent real estate systems by bridging the gap between data-driven prediction and immersive visual interaction.

In addition, the study contributes to addressing real-world housing challenges within the Nairobi context. By focusing on rental housing in an urban environment, the system provides a locally relevant solution that supports both tenants and landlords in making informed decisions. This enhances transparency in rental pricing and reduces uncertainty in the housing market.

Finally, the study contributes methodologically by demonstrating the feasibility of developing an end-to-end system that integrates data collection, machine learning model development, mobile application implementation and augmented reality within a single framework. This provides a reference model for future research and system development in related domains.

Overall, the contributions of this study highlight its significance in advancing both theoretical understanding and practical solutions in the field of rental housing prediction and visualisation.

7.4 Recommendations

Based on the findings of this study, several recommendations are proposed to enhance the system. Firstly, future work should incorporate a larger and more diverse dataset, including data from different regions, to improve the accuracy and generalisability of the machine learning model.

Secondly, the augmented reality component can be enhanced by integrating full three-dimensional (3D) models to enable virtual walkthroughs of properties, thereby improving user experience.

Thirdly, the system should be optimised to support a wider range of devices and reduce dependence on continuous internet connectivity to improve accessibility.

Additionally, the existing search functionality can be enhanced by incorporating advanced filtering options such as location-based filtering and price range selection, allowing users to more efficiently find relevant properties.

Finally, further usability testing with a larger and more diverse group of users is recommended to refine system performance and improve user experience.

7.5 Future work

Future work for this study can focus on enhancing both the technical capabilities and practical applicability of the developed system. One potential area for improvement is the adoption of more advanced machine learning techniques, such as deep learning models, to further improve the accuracy and robustness of rental price predictions. Incorporating additional data sources, including real-time market data and socio-economic factors, could also enhance the predictive performance of the model.

Another important direction for future work is the enhancement of the augmented reality component. This includes the integration of full three-dimensional (3D) property models and the development of more advanced visualisation features to support realistic virtual walkthroughs. Such improvements would provide users with a more immersive and detailed understanding of property layouts and features.

Future research could also explore the integration of image-based analysis within the prediction process. For instance, incorporating computer vision techniques to analyse property images could enable the system to consider visual attributes such as interior quality and design, which are not fully captured in structured datasets.

In addition, the system could be extended to include intelligent recommendation mechanisms that suggest properties based on user interactions and preferences. This would further enhance user experience by providing more personalised and relevant property options.

Finally, future work may focus on deploying the system at a larger scale and evaluating its performance in real-world environments with a broader user base. This would provide deeper insights into system usability, scalability, and long-term effectiveness in supporting rental housing decision-making.

Overall, these future directions provide a pathway for transforming the proposed system into a more advanced, scalable, and intelligent platform for rental housing prediction and visualisation.

References

Abolghasemi, M., & Esmaeilbeigi, R. (2021). *State-of-the-art predictive and prescriptive analytics for IEEE CIS 3rd Technical Challenge*. http://arxiv.org/abs/2112.03595

Adzanoukpe, P. (2025). *Predicting House Rental Prices in Ghana Using Machine Learning*. https://doi.org/10.20944/preprints202412.1927.v1

Aher, S., Avhad, R., Bhavsar, E., Pagare, O., & Malode, M. V. (2024). *Augmented Reality in Real Estate: Enhancing Property Visualization and Customer Experience*. www.ijnrd.org

Alam, S. S., Susmit, S., Lin, C. Y., Masukujjaman, M., & Ho, Y. H. (2021). Factors affecting augmented reality adoption in the retail industry. *Journal of Open Innovation: Technology, Market, and Complexity*, *7*(2). https://doi.org/10.3390/joitmc7020142

Asaad & Avksentieva. (2023). A SURVEY ON MACHINE LEARNING TECHNIQUES FOR SOFTWARE ENGINEERING. *ИНФОРМАЦИОННЫЕ ТЕХНОЛОГИИ И ВЫЧИСЛИТЕЛЬНЫЕ СИСТЕМЫ*. https://doi.org/10.14357/20718632230408

Baraka Mwau. (2020). *About the project*. https://www.

Billinghurst, M. (2021). Grand Challenges for Augmented Reality. *Frontiers in Virtual Reality*, *2*. https://doi.org/10.3389/frvir.2021.578080

Brasoveanu, A., Moodie, M., & Agrawal, R. (2020). Textual evidence for the perfunctoriness of independent medical reviews. *CEUR Workshop Proceedings*, *2657*, 1–9. https://doi.org/10.1145/nnnnnnn.nnnnnnn

Cacciamani, C., Conso, A., Zaltron, S., & Dinicolamaria, D. (2024). Perspective Chapter: Artificial Intelligence in the Real Estate Industry. In *Complex Systems with Artificial Intelligence - Sustainability and Self-Constitution*. IntechOpen. https://doi.org/10.5772/intechopen.1007906

Chen, T., & Si, S. (2024). *Predicting Rental Price of Lane Houses in Shanghai with Machine Learning Methods and Large Language Models*. http://arxiv.org/abs/2405.17505

Communications Authority of Kenya. (2025). *Mobile, data, and digital services on the rise, CA report shows*. Communications Authority of Kenya. https://www.ca.go.ke/mobile-data-and-digital-services-rise-ca-report-shows

Cytonn Investments. (2023). *nairobi-metropolitan-area-residential-report-2023-cytonn-weekly-182023*.

Devraj, K., Jones, L., Higgins, B., Thomas, P. B. M., & Moosajee, M. (2024). User-Centred Design and Development of a Smartphone Application (OverSight) for Digital Phenotyping in Ophthalmology. *Healthcare (Switzerland)*, *12*(24). https://doi.org/10.3390/healthcare12242550

Ekstrand, O., & Lundqvist, S. (2020). *Evaluating an ARCore application to get an image of the state of AR technology today*. http://www.hitl.washington.edu/artoolkit/

Eren, L. (2023). *Comparative Analysis of Machine Learning Methods for Predicting Property Prices and Sale Velocities in the Real Estate Industry; Comparative Analysis of Machine Learning Methods for Predicting Property Prices and Sale Velocities in the Real Estate Industry; Jämförande analys av maskininlärningsmetoder för att förutsäga fastighetspriser och försäljningshastigheter inom fastighetsbranschen: TRITA-EECS-EX*.

European Morgage Federation. (2025). *Joint Paper on the Use of Automated Valuation Models in Europe*.

Felderer, M., & Ramler, R. (2021). *Quality Assurance for AI-based Systems: Overview and Challenges*. https://doi.org/10.1007/978-3-030-65854-0_3

Fombona-Pascual, A., Fombona, J., & Vicente, R. (2022). Augmented Reality, a Review of a Way to Represent and Manipulate 3D Chemical Structures. *Journal of Chemical Information and Modeling*, *62*(8), 1863–1872. https://doi.org/10.1021/acs.jcim.1c01255

Fred, N., & Ru, I. M. O. (2024). *A Systematic Literature Review on the Use of Machine Learning in Software Engineering*.

Fu Ginger Zhe Jin Meng Liu, R., Fu, R., Zhe Jin, G., & Liu, M. (2022). *Human-Algorithm Interactions: Evidence from Zillow.com*. http://www.nber.org/papers/w29880

Goldwaser, A., & Ge, H. (2023). *Understanding Sparse Feature Updates in Deep Networks using Iterative Linearisation*. http://arxiv.org/abs/2211.12345

Hasan, M. H., Jahan, M. A., Ali, M. E., Li, Y.-F., & Sellis, T. (2024). *A Multi-Modal Deep Learning Based Approach for House Price Prediction*. http://arxiv.org/abs/2409.05335

He, F., Ong, S. K., & Nee, A. Y. C. (2021). An integrated mobile augmented reality digital twin monitoring system. *Computers*, *10*(8). https://doi.org/10.3390/computers10080099

Huang, C., Nourian, A., & Griest, K. (2021). *Hidden Technical Debts for Fair Machine Learning in Financial Services*. http://arxiv.org/abs/2103.10510

International Organization for Standardization. (n.d.). *SIST-EN-ISO-9241-210-2020*.

International Organization for Standardization & International Electrotechnical Commission. (2023). *Systems and software engineering-Systems and software Quality Requirements and Evaluation (SQuaRE)-Product quality model*.

Jarrahi, M. H., Memariani, A., & Guha, S. (2023). The Principles of Data-Centric AI. In *Communications of the ACM* (Vol. 66, Number 8, pp. 84–92). Association for Computing Machinery. https://doi.org/10.1145/3571724

Kapoor, S., & Narayanan, A. (2022). *Leakage and the Reproducibility Crisis in ML-based Science*. http://arxiv.org/abs/2207.07048

Kenya Mortgage Refinance Company. (2023). *9bb9b0e7--kmrc-annual-report-2023*.

Kenya National Bureau of Statistics. (2020). *2019-Kenya-population-and-Housing-Census-Volume-4-Distribution-of-Population-by-Socio-Economic-Characteristics*.

Kenya National Bureau of Statistics. (2023). *KENYA HOUSING SURVEY BASIC REPORT*.

Kenya National Bureau of Statistics. (2024). *REAL ESTATE SURVEY REPORT*.

Kiratiratanapruk, K., Temniranrat, P., Sinthupinyo, W., Marukatat, S., & Patarapuwadol, S. (2022). *Automatic Detection of Rice Disease in Images of Various Leaf Sizes*. http://arxiv.org/abs/2206.07344

Kundisch, D., Muntermann, J., Oberländer, A. M., Rau, D., Röglinger, M., Schoormann, T., & Szopinski, D. (2022). An Update for Taxonomy Designers: Methodological Guidance from Information Systems Research. *Business and Information Systems Engineering*, *64*(4), 421–439. https://doi.org/10.1007/s12599-021-00723-x

Liu, C., Zhao, Y., & Braud, T. (2023). *Robust Localization with Visual-Inertial Odometry Constraints for Markerless Mobile AR*. http://arxiv.org/abs/2308.05394

Liu, C., Zhao, Y., & Braud, T. (2024). *MobileARLoc: On-device Robust Absolute Localisation for Pervasive Markerless Mobile AR*. http://arxiv.org/abs/2401.11511

Macchiaroli, M., Dolores, L., & De Mare, G. (2025). Extended reality (XR) impacts on real estate market and economic-financial project management of residential built environments: systematic review and analysis. *Smart and Sustainable Built Environment*. https://doi.org/10.1108/SASBE-07-2024-0258

MagicBricks. (2021). *COVID-19 Contact–less Site Visit Report*.

Marino, E., Bruno, F., Barbieri, L., & Lagudi, A. (2022). Benchmarking Built-In Tracking Systems for Indoor AR Applications on Popular Mobile Devices. *Sensors*, *22*(14). https://doi.org/10.3390/s22145382

McKinsey & Company. (2020). *The State of AI in 2020*. https://www.mckinsey.com/

Mora-Garcia, R. T., Cespedes-Lopez, M. F., & Perez-Sanchez, V. R. (2022). Housing Price Prediction Using Machine Learning Algorithms in COVID-19 Times. *Land*, *11*(11). https://doi.org/10.3390/land11112100

Mwau, B. &, & Sverdlik, A. (2020). High rises and low-quality shelter: rental housing dynamics in Mathare Valley, Nairobi. *Environment and Urbanization*, *32*(2), 481–502. https://doi.org/10.1177/0956247820942166

Olaolu, A. (2022). Digital Banking Adoption in Nigeria: The Place of Technology Acceptance Model. *Asian Journal of Economics, Business and Accounting*, 59–72. https://doi.org/10.9734/ajeba/2022/v22i730579

Onyango, G., & Ondiek, J. O. (2021). Digitalization and Integration of Sustainable Development Goals (SGDs) in Public Organizations in Kenya. *Public Organization Review*, *21*(3), 511–526. https://doi.org/10.1007/s11115-020-00504-2

Pamungkas & Astrianty. (2024). *Development of Mobile Augmented Reality-Based Boarding House Search Application*.

Patil, R., Kumar Sahay, M., Singh Rathore, R., Pandagale, A., Year Student, F. U., & Professor, A. (2022). *House Price Prediction Using Neural Network*. https://www.ijasrw.com

Priyanka Das. (2020). Advancing Simultaneous Localization and Mapping (SLAM) for Robots in Unstructured Terrain. *International Journal For Multidisciplinary Research*, *2*(6). https://doi.org/10.36948/ijfmr.2020.v02i06.25432

Rahaman, Md. J. (2024). A Comprehensive Review to Understand the Definitions, Advantages, Disadvantages and Applications of Machine Learning Algorithms. *International Journal of Computer Applications*, *186*(31), 43–47. https://doi.org/10.5120/ijca2024923868

Raju, R., Neyaz, A., Ahmed, A., & Singh, A. (2023). Machine Learning for Rental Price Prediction: Regression Techniques and Random Forest Model. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.4587725

Sarker, I. H. (2021). Machine Learning: Algorithms, Real-World Applications and Research Directions. In *SN Computer Science* (Vol. 2, Number 3). Springer. https://doi.org/10.1007/s42979-021-00592-x

scikit-learn. (2020). *scikit-learn user guide Release 0.22.2 scikit-learn developers*.

Silpa Chaitanya, P., Devi, V. V., Siva, Y., Jyothi, N., Durga, T., Malleswari, J., & Hema, P. (2024). *Issue 11 www.jetir.org (ISSN-2349-5162) JETIRGO06048 Journal of Emerging Technologies and Innovative Research (JETIR) www.jetir* (Vol. 11). www.jetir.org

Silva, R., Jesus, R., & Jorge, P. (2023). Development and Evaluation of a Mobile Application with Augmented Reality for Guiding Visitors on Hiking Trails. *Multimodal Technologies and Interaction*, *7*(6). https://doi.org/10.3390/mti7060058

Syed, T. A., Siddiqui, M. S., Abdullah, H. B., Jan, S., Namoun, A., Alzahrani, A., Nadeem, A., & Alkhodre, A. B. (2023). In-Depth Review of Augmented Reality: Tracking Technologies, Development Tools, AR Displays, Collaborative AR, and Security Concerns. In *Sensors* (Vol. 23, Number 1). MDPI. https://doi.org/10.3390/s23010146

UN-Habitat. (2022). *Envisaging the Future of Cities*.

Vindigni, G. (2024). Enhancing Human-Computer Interaction in Socially Inclusive Contexts: Flow Heuristics and AI Systems in Compliance with DIN EN ISO 9241 Standards. *European Journal of Contemporary Education and E-Learning*, *2*(4), 115–139. https://doi.org/10.59324/ejceel.2024.2(4).10

Weichbroth, P. (2024). Usability Testing of Mobile Applications: A Methodological Framework. *Applied Sciences (Switzerland)*, *14*(5). https://doi.org/10.3390/app14051792

Weichbroth, P. (2025). *Usability Issues With Mobile Applications: Insights From Practitioners and Future Research Directions*. https://doi.org/10.1109/ACCESS.2025.3573503

Winnie, L., & Otieno, A. (2022). *Title: ANALYSING THE USE OF DIGITAL TECHNOLOGIES IN EMERGING REAL ESTATE MARKETS: A CASE STUDY OF DIGITALIZATION IN THE KENYAN HOUSING SECTOR*. https://doi.org/10.13140/RG.2.2.34923.82726

World Bank. (2023). *KENYA CCDR Digital sector background note*. www.worldbank.org

Yang, X. (2025). Research on House Price Prediction based on Machine Learning. *ITM Web of Conferences*, *70*, 02018. https://doi.org/10.1051/itmconf/20257002018

Yeboah, E., Boateng, R., Owusu, A., Afful-Dadzie, E., & Ofori-Amanfo, J. (2020). Assessing the Role of Trust in Merchant Adoption of Mobile Payments in Ghana. *Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, *12066 LNCS*, 204–215. https://doi.org/10.1007/978-3-030-44999-5_17

Żbikowski & Antosiuk. (2021). A machine learning, bias-free approach for predicting business success using Crunchbase data. *Information Processing and Management*, *58*(4). https://doi.org/10.1016/j.ipm.2021.102555

Zhang, L. (2023). Housing Price Prediction Using Machine Learning Algorithm. *Journal of World Economy*, *2*(3), 18–26. https://doi.org/10.56397/jwe.2023.09.03

Appendices

Appendix A: Similarity Report

turnitin Page 1 of 174 - Cover Page Submission ID trn:oid:::2945:369267010

Baraka Mitigoa

A Machine Learning-based Mobile Application with Augmented Reality for Rental Housing Prediction and Visualisation in Nair...

Strathmore University (Main Account)

Document Details

Submission ID
trn:oid:::2945:369267010

Submission Date
Mar 23, 2026, 7:19 PM GMT+3

Download Date
Mar 23, 2026, 7:43 PM GMT+3

File Name
A Machine Learning-based Mobile Application with Augmented Reality for Rental Housing Predi....docx

File Size
7.5 MB

146 Pages

31,038 Words

193,620 Characters

turnitin Page 1 of 174 - Cover Page Submission ID trn:oid:::2945:369267010

20% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text

Match Groups

580 Not Cited or Quoted 19%
Matches with neither in-text citation nor quotation marks

44 Missing Quotations 1%
Matches that are still very similar to source material

0 Missing Citation 0%
Matches that have quotation marks, but no in-text citation

0 Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

11% Internet sources
8% Publications
18% Submitted works (Student Papers)

Match Groups

- 580 Not Cited or Quoted 19%
  Matches with neither in-text citation nor quotation marks
- 44 Missing Quotations 1%
  Matches that are still very similar to source material
- 0 Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- 0 Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

11%	Internet sources
8%	Publications
18%	Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Internet	www.mdpi.com	<1%
2	Submitted works	University of Wales Institute, Cardiff on 2026-03-09	<1%
3	Internet	su-plus.strathmore.edu	<1%
4	Submitted works	Strathmore University (Main Account) on 2025-12-09	<1%
5	Submitted works	Strathmore University (Main Account) on 2026-03-13	<1%
6	Submitted works	Asia Pacific Instutute of Information Technology on 2026-02-12	<1%
7	Publication	Bal S. Virdee, Tanweer Ali, Jaume Anguera, Suman Lata Tripathy. "Connecting Int...	<1%
8	Submitted works	Strathmore University (Main Account) on 2026-03-18	<1%
9	Publication	Mutlu Yuksel, Yigit Aydede. "Causal Inference and Machine Learning - In Economi...	<1%
10	Submitted works	Liverpool John Moores University on 2026-01-30	<1%

Appendix B: Ethical Clearance Confirmation

Strathmore
UNIVERSITY

11th November 2025

Mr Mitigoa Baraka,
baraka.mitigoa@strathmore.edu

Dear Mr Mitigoa,

RE: A Machine Learning-based Mobile Application with Augmented Reality for Rental Housing Prediction and Visualisation in Nairobi

This is to inform you that SU-ISERC has reviewed and **approved** your above **SU-masters** research proposal. Your application reference number is **SU-ISERC3112/25.** The approval period is from **11th November 2025 to 10th November 2026.**

This approval is subject to compliance with the following requirements:

i. Only approved documents including (informed consents, study instruments, and MTA) will be used
ii. All changes including (amendments, deviations, and violations) are submitted for review and approval by SU-ISERC.
iii. Death and life-threatening problems and serious adverse events or unexpected adverse events whether related or unrelated to the study must be reported to SU-ISERC within 72 hours of notification
iv. Any changes, anticipated or otherwise, that may increase the risks or affect the safety or welfare of study participants and others or affect the integrity of the research must be reported to SU-ISERC within 72 hours
v. Clearance for the export of biological specimens must be obtained from relevant institutions.
vi. Submission of a request for renewal of approval at least 60 days prior to the expiry of the approval period. Attach a comprehensive progress report to support the renewal.
vii. Submission of an executive summary report within 90 days of completion of the study to SU-ISERC.

Before commencing your study, you will be expected to obtain a research license from National Commission for Science, Technology, and Innovation (NACOSTI) https://research-portal.nacosti.go.ke/ and obtain other clearances needed.

Yours sincerely,

Mr Ambrose Rachier,
Chairperson; SU-ISERC

Ole Sangale Rd, Madaraka Estate. PO Box 59857-00200, Nairobi, Kenya. Tel +254 (0)703 034000
Email admissions@strathmore.edu www.strathmore.edu

Appendix C: System Usability Scale (SUS) Questionnaire

System Usability Scale

Guidelines
All items should be checked. If you feel that you cannot respond to a particular item, mark the center point of the scale
1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree

	Strongly disagree 1	2	3	4	Strongly agree 5
1. I think that I would like to use this system frequently					
2. I found the system unnecessarily complex					
3. I thought the system was easy to use					
4. I think that I would need the support of a technical person to be able to use this system					
5. I found the various functions in this system were well integrated					
6. I thought there was too much inconsistency in this system					
7. I would imagine that most people would learn to use this system very quickly					
8. I found the system very cumbersome to use					
9. I felt very confident using the system					
10. I needed to learn a lot of things before I could get going with this system					

Appendix D: Research Budget

Category	Item Description	Justification / Purpose	Estimated Cost (KSh)
Personnel / Human Resources	Transport Allowance for SUS Data Collection	Covers transport costs within Nairobi to administer and collect SUS questionnaire responses from participants.	8,000
Equipment and Software	Mobile Phone for Prototype Testing	Purchase of a smartphone compatible with AR features to test and demonstrate the developed application.	25,000
Data Collection and Processing	Printing of SUS Questionnaires and Consent Forms	Printing hard copies of questionnaires and consent forms for participants during usability evaluation.	3,000
Documentation and Report Preparation	Printing and Binding	Printing and binding the final report and appendices for submission.	4,000
	Presentation Materials (Slides and Posters)	Preparation of PowerPoint slides and visual materials for proposal or project defense.	3,000
Contingency (10%)	Miscellaneous and Unforeseen Costs	To cater for any unplanned expenses or price variations during the project period.	4,300
TOTAL ESTIMATED COST			**KSh 47,300**

Appendix E: Research Implementation Timeline (Gantt Chart)